

Block

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Parameter Estimation and Hypothesis Testing

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BLOCK-3 INTRODUCTION

In the previous block, the basic rules of probability and various probability distributions such as Bernoulli, Binomial, Poisson, Uniform, Normal, etc. were discussed. This block uses concepts of probability theory and probability distributions for drawing inferences about the population parameters. Generally, population parameters are unknown, and when the population is too large, or there are limited resources and manpower available, then it is not practically possible to examine each and every unit of the population to obtain the population parameters. In such situations, one can draw a sample from the population under study and utilise sample observations to draw reliable conclusions about the population parameters. The results obtained from the sample are projected in such a way that they are valid for the entire population. This technique is known as statistical inference. The statistical inference may be divided into two areas or parts:

- i) The population parameters are unknown, and we may want to guess the true value of the unknown parameters on the basis of a random sample drawn from the population. This type of problem is known as “Estimation”.
- ii) Some information is available about the population, and we may like to verify whether the information is true or not on the basis of a random sample drawn from the population. This type of problem is known as “Testing of hypothesis”.

This block explains these concepts in detail. The block is divided into three units:

Unit 8 explains the concepts related to Sampling Distribution. This unit discusses the Central Limit Theorem, the Law of Large Numbers, the Sampling Distribution of the Sample Mean and the Sampling Distribution of the Difference of Two Sample Means.

Unit 9 explains the concepts of Point Estimation, Method of Maximum Likelihood, and Interval Estimation. Further, the unit discusses the Confidence Interval for the Population Mean. In addition, the unit discusses the Confidence Interval for a Population Proportion and the concept of Determining Sample Size.

Unit 10 describes the concept of hypothesis and hypothesis testing. It discusses the key related concepts like Critical Region, Type-I and Type-II Errors, Level of Significance, One-Tailed and Two-Tailed Tests, and the Concept of p-Value. Further, the unit discusses The Hypothesis for Population Mean, and the difference of two population means using the Z-Test and the t-Test.

The content of each unit is adopted from several units, listed in the reference section of each unit.

UNIT 8 SAMPLING DISTRIBUTION

Structure

- 8.0 Introduction
- 8.1 Objectives
- 8.2 Basic Terminology
- 8.3 Introduction to Sampling Distribution
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- 8.7 Sampling Distribution of Sample Mean
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8.0 INTRODUCTION

In many situations, if we extract some information from all the elements or items of a large population in general, it may be time consuming and expensive. Also, if the elements or items of a population are destroyed under investigation, then the information gathering from all the elements does not make sense because all the elements will be destroyed. Therefore, in most of the cases in daily life, business and industry, the information is gathered through **sampling**. The results of a properly taken sample enable the investigator to arrive at generalisations that are valid for the entire population. The process of generalising sample results to the population is called **Statistical Inference**.

For drawing inferences about the population parameters, we draw all possible samples of the same size and determine a function of sample values, which is called a **statistic**, for each sample. The values of statistic are generally varied from one sample to another sample. Therefore, the sample statistic is a random variable and follows a distribution. The distribution of the statistic is called the **sampling distribution of the statistic**.

This unit introduces the basic terminology and sampling distribution used in statistical inference. One of the important concepts in Statistics is the standard error. It has been discussed in this unit. In addition, this unit discusses the concept of the law of large numbers and sample size.

One of the most important sample statistics that is used to draw a conclusion about the population mean is the sample mean. This unit discusses the sampling distribution of the mean and the sampling distribution of the difference of two sample means. The unit ends by providing a summary of what we have discussed in this unit and the solution to the check your progress questions.

8.0 OBJECTIVES

After studying this unit, you should be able to:

- define statistical inference;
- define the basic terms as population, sample, parameter, statistic, estimator, estimate, etc. used in statistical inference;

- explain the concept and need of sampling distribution;
- explain the concept of standard error;
- describe the most important theorem of Statistics, “Central Limit Theorem”;
- explain the concept of the law of large numbers and adequate sample size; and
- explain the sampling distribution of sample mean and difference of two sample means.

8.2 BASIC TERMINOLOGY

Before discussing the sampling distribution of a statistic, we shall be discussing basic definitions of some of the important terms which are very helpful to understand the fundamentals of statistical inference.

Population

In Statistics, the term population has a broader meaning. A population is the collection or group of individuals /items /units/ observations under study. For example, the collection of books in a library, the particles in a salt bag, the rivers in India, the students in a classroom, etc., are considered as populations in Statistics.

The total number of elements/items/units/observations in a population is known as population size and denoted by N . The characteristic under study may be denoted by X or Y .

We may classify the population into two types as:

(1) Finite and Infinite Population

If a population contains a finite number of units or observations, it is called a **finite population**. For example, the population of students in a class, the population of bolts produced in a factory in a day, the population of electric bulbs in a lot, the population of books in a library, etc.

If a population contains an infinite (uncountable) number of units or observations, it is called an **infinite population**. For example, the population of particles in a salt bag, the population of stars in the sky, the population of hairs on a human body, etc. Theoretically, sometimes, populations that are too large are assumed to be infinite.

(2) Real and Hypothetical Population

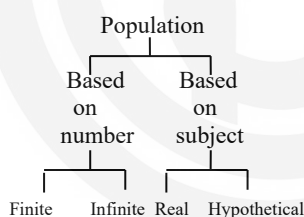
The population can also be classified as real and hypothetical. A population comprising the items or units which are all physically present is known as **real population**. All of the examples given for finite and infinite population are examples of a real population.

If a population consists of items or units which are not physically present but the existence of them can only be imagined or conceptualised then it is known as a **hypothetical population**. For example, the population of heads or tails in successive tosses of a coin a large number of times is considered as a hypothetical population.

Sample

To extract the information from all the elements or units or items of a large population may be, in general, time-consuming, expensive and difficult. And

A group of individuals / items / units/ observations under study is known as population.



also, if the elements or units of a population are destroyed under investigation, then gathering the information from all the units does not make sense. For example, to test the strength of the chalk, to test the quality of crackers, etc., the chalks and crackers are destroyed under the testing procedure. In such situations, a small part of the population is selected from the population, which is called a **sample**. Thus, the sample can be defined as given below:

“A sample is a part/fraction/subset of the population.”

The procedure for drawing a sample from the population is called **sampling**. A detailed discussion on sampling is beyond this course. You may learn about sampling from the references given in Section 8.11. The number of units selected in the sample is known as sample size and it is denoted by n .

Complete Survey and Sample Survey

Statistical data may be collected by two methods:

- (1) Complete Survey or Complete Enumeration or Census
- (2) Sample Survey or Sample Enumeration

(1) Complete Survey or Complete Enumeration or Census

When every element or unit of the population is investigated or studied for the characteristics under study then we call it **complete survey** or **census**. For example, suppose we want to find out the average height of the students of a study centre then if we measure the height of each student of this study centre to find the average height of the students then such type of survey is called a complete survey.

(2) Sample Survey or Sample Enumeration

When only a part or a small number of elements or units (i.e. sample) of the population are investigated or studied for the characteristics under study then we call it **sample survey** or **sample enumeration**. In the above example, if we select some students of only one study centre and measure the height to find the average height of the students then such type of survey is called a sample survey.

Simple Random Sampling or Random Sampling

The simplest and most common method of sampling is simple random sampling. In simple random sampling, the sample is drawn in such a way that each element or unit of the population has an equal and independent chance of being included in the sample. If a sample is drawn by this method, then it is known as a **simple random sample** or a **random sample**. The random sample of size n is denoted by $X_1, X_2, \dots, X_n$ or $Y_1, Y_2, \dots, Y_n$ and the observed value of this sample is denoted by $x_1, x_2, \dots, x_n$ or $y_1, y_2, \dots, y_n$. Some author use $x_1, x_2, \dots, x_n$ to represent the random sample instead of $X_1, X_2, \dots, X_n$. But throughout the course, we use capital letters to represent the random sample, that is, $X_1, X_2, \dots, X_n$. In addition, $X_1, X_2, \dots, X_n$ simply be assumed as the independent identically distributed (iid) random variables.

In simple random sampling, elements or units are selected or drawn one by one therefore, it may be classified into two types as:

(1) Simple Random Sampling without Replacement (SRSWOR)

In simple random sampling, if the elements or units are selected or drawn one by one in such a way that an element or unit drawn at a time is not replaced back to

Parameter Estimation and Hypothesis Testing

In SRSWR each of the 1st, 2nd, ..., nth draw the elements are remain same N due to replacement so by rule of multiplication total number of possible samples is $N \times N \times \dots n \text{ times} = N^n$.

the population before the subsequent draws, is called **SRSWOR**. If we draw a sample of size n from a population of size N without replacement then the total number of possible samples is ${}^N C_n$. For example, consider a population that consists of three elements, A, B and C. Suppose we wish to draw a random sample of two elements then $N = 3$ and $n = 2$. The total number of possible random samples without replacement is ${}^N C_n = {}^3 C_2 = 3$. The possible sample would be: (A, B), or (A, C) and (B, C).

(2) Simple Random Sampling with Replacement (SRSWR)

In simple random sampling, if the elements or units are selected or drawn one by one in such a way that a unit drawn at a time is replaced back to the population before the subsequent draw is called **SRSWR**. In this method, the same element or unit can appear more than once in the sample and the probability of selection of a unit at each draw remains the same i.e. $1/N$. In this method, the total number of possible samples is N^n . In above example, the total number of possible random samples with replacement is $N^n = 3^2 = 9$ as (A, A), (A, B), (A, C), (B, A), (B, B), (B, C), (C, A), (C, B) and (C, C).

Note 1: The statistical inference is based on the theoretical distribution so, in whole statistical inference, we assume that the random sample is selected from an infinite population or a finite population with replacement so that removal of the sample unit has no appreciable effect on the composition of the population. That is, we draw the random sample such that:

- (a) The n successive sample observations are independent and
- (b) The population composition remains constant from draw to draw.

For example, if we draw a random sample $X_1, X_2, \dots, X_n$ from a normal population with mean μ and variance σ^2 then X_i 's are independent and follow the same normal distribution.

Parameter

The characteristics of a population can be described with some measures such as total numbers of items, average, variance, etc. These measures are known as parameters of the population. Thus, we can define a parameter as:

A parameter is a function of population values, which is used to represent the certain characteristic of the population. For example, population mean, population variance, population coefficient of variation, population correlation coefficient, etc. are all parameters. Population parameter mean is usually denoted by μ and population variance is denoted by σ^2 .

We know that the population can be described with the help of distribution and the distribution is fully determined with the help of its constants such as, in case of a normal distribution, we need to know μ and σ^2 to determine the normal distribution, in case of Poisson distribution, we need to know λ , etc. These constants are known as the parameters.

Sample Mean and Sample Variance

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population whose probability density(mass) function $f(x, \theta)$ then the sample mean is defined as

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i$$

A parameter is a function of population values which is used to represent the certain characteristic of the population.

And sample variance is defined as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Here, we divide $\sum_{i=1}^n (X_i - \bar{X})^2$ by $(n-1)$ rather than n as our definition of the variance described earlier. The reason for taking $(n-1)$ in place of n will be discussed later in the unit.

Statistic

Any quantity calculated from sample values and does not contain any unknown parameter is known as a **statistic**. For example, if $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population with mean μ and variance σ^2 (both are unknown) then the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is a statistic whereas $\bar{X} - \mu$ and $\bar{X}/\sigma$ are not statistics because both are functions of unknown parameters.

Estimator and Estimate

Generally, population parameters are unknown and the whole population is too large to find out the parameters. Since the sample drawn from a population always contains some or more information about the population, therefore in such situations, we guess or estimate the value of the parameter under study based on a random sample drawn from that population.

So if a statistic is used to estimate an unknown population parameter then it is known as an **estimator** and the value of the estimator based on the observed value of the sample is known as an **estimate** of the parameter. For example, suppose the parameter λ of the Poisson population $f(x, \lambda)$ is unknown so we draw a random sample $X_1, X_2, \dots, X_n$ of size n from this Poisson population to

estimate λ . If we use the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ to estimate λ then $\bar{X}$ is

called estimator and any particular value of this estimator, say, $\bar{x}$ is called the estimate of unknown parameter λ . Consider another example, if we want to estimate the average height (μ) of students in a college with the help of sample mean $\bar{X}$ then $\bar{X}$ is the estimator and its particular value, say, 165 cm is the estimate of the population average height (μ).

In this course, we use capital letters for the estimator and small letters for the estimated value. In general, the estimator is denoted by $T_n = t(X_1, X_2, \dots, X_n)$ where 'n' denotes the sample size and the estimator T is a function of the random sample $X_1, X_2, \dots, X_n$.

Now, it is the right place to discuss the sampling distribution.

Any statistic used to estimate an unknown population parameter is known as **estimator** and the particular value of the estimator is known as **estimate** of parameter. The estimated value of sample mean and sample variance are denoted by $\bar{x}$ and s^2 respectively.

8.3 INTRODUCTION TO SAMPLING DISTRIBUTION

Consider an example, if we want to know the average income of persons living in a big city, then collecting the information from each person is very much time

and manpower consuming. In such situations, one can draw a sample from the population under study and utilise sample observations to extract the necessary information about the population. **The results obtained from the sample are projected in such a way that they are valid for the entire population.** Therefore, the sample works like a “**Vehicle**” to reach (drawing) at valid conclusions about the population. Thus, statistical inference can be defined as:

“The process of projecting the sample results for the whole population is known as **statistical inference**.”

For drawing the inference about the population, we analyse the sample data, that is, we calculate the value of a sample statistic such as the sample mean, sample proportion, sample variance, etc. Generally, if we want to draw the inference about the population mean we use sample mean, about population variance we use sample variance, etc. However, if you take two independent samples from the population, then it is possible that mean of sample-I is different from mean of sample-II. Thus, if we are using this sample mean to estimate the population mean then we get a different estimate of the mean of population. Similarly, we can take many samples of the same size and for each sample, we get sample mean which may or may not be distinct. From all of the sample means, we try to estimate the mean of the whole population.

For a better understanding of the process of generalising these sample results to the whole population, we consider the example in which the size of the population is very small.

Consider a population comprising four typists who type the sample page of a manuscript. The number of errors made by each typist is shown in Table 8.1, given below:

Table 8.1: Number of Errors per Typist

Typist	Number of Errors
A	4
B	2
C	3
D	1

The population mean (average number of errors) can be obtained as

$$\mu = \frac{4 + 2 + 3 + 1}{4} = 2.5$$

Now, let us assume that we do not know the average number of errors made by typists. So we decide to estimate the population mean on the basis of a sample of size $n = 2$. There are $N^n = 4^2 = 16$ possible simple random samples with replacement of size 2. All possible samples of size $n = 2$ are given below and for each sample the sample mean is calculated as shown in Table 8.2.

From table 8.2, we can see that the value of the sample statistic (sample mean) is varying from sample to sample and out of 16 samples only 4 samples (4, 7, 10 and 13) have their means equal to the population mean whereas other 12 samples result in some error in the estimation process. The error is the difference between the population mean and the sample mean used for the estimate.

Table 8.2: Calculation of Sample Mean

Sample Number	Sample in Term of Typist	Sample Observation	Sample Mean ($\bar{X}$)
1	(A, A)	(4, 4)	4.0
2	(A, B)	(4, 2)	3.0
3	(A, C)	(4, 3)	3.5
4	(A, D)	(4, 1)	2.5
5	(B, A)	(2, 4)	3.0
6	(B, B)	(2, 2)	2.0
7	(B, C)	(2, 3)	2.5
8	(B, D)	(2, 1)	1.5
9	(C, A)	(3, 4)	3.5
10	(C, B)	(3, 2)	2.5
11	(C, C)	(3, 3)	3.0
12	(C, D)	(3, 1)	2.0
13	(D, A)	(1, 4)	2.5
14	(D, B)	(1, 2)	1.5
15	(D, C)	(1, 3)	2.0
16	(D, D)	(1, 1)	1.0

Now, consider all possible values of the sample mean and calculate their probabilities of occurrence. Then we arrange every possible value of the sample mean with their respective probabilities in the following Table 8.3 given below:

Table 8.3: Sampling Distribution of Sample Means

S. No.	$\bar{X}$	Frequency(f)	Probability(p)
1	1.0	1	$1/16 = 0.0625$
2	1.5	2	$2/16 = 0.1250$
3	2.0	3	$3/16 = 0.1875$
4	2.5	4	$4/16 = 0.2500$
5	3.0	3	$3/16 = 0.1875$
6	3.5	2	$2/16 = 0.1250$
7	4.0	1	$1/16 = 0.0625$

So the arrangement of all possible values of the sample mean with their corresponding probabilities is called the sampling distribution of the mean.

Thus, we can define the sampling distribution of a statistic as:

“The probability distribution of all possible values of a sample statistic that would be obtained by drawing all possible samples of the same size from the population is called the sampling distribution of that statistic.”

The sampling distribution of sample means itself has mean, variance, etc. Therefore, the mean of sample means can be obtained by the formula

$$\begin{aligned}\text{Mean of sample means} = \bar{\bar{X}} &= \frac{1}{K} \sum_{i=1}^k \bar{X}_i f_i \quad \text{where, } K = \sum_{i=1}^k f_i \\ &= \frac{1}{16} (1.0 \times 1 + 1.5 \times 2 + \dots + 4.0 \times 1) = 2.5 = \mu\end{aligned}$$

The mean of sample means can also be calculated as

$$E(\bar{X}) = \bar{\bar{X}} = \sum_{i=1}^k \bar{X}_i p_i = 1.0 \times \frac{1}{16} + 1.5 \times \frac{2}{16} + \dots + 4.0 \times \frac{1}{16} = 2.5$$

Thus, we have seen for this population that mean of the sample means is equal to the population mean, that is, $\bar{\bar{X}} = \mu = 2.5$. The fact that these two means are equal is not a chance as it can be shown that $\bar{\bar{X}}$ equals to the population mean for any population and any given sample size.

But if the population is large say having 1,000 elements and we want to draw all samples of size 2 then there are $(1000)^2 = 1000000$ possible simple random samples with replacement. Then the listing of all such samples and finding the sampling distribution would be a difficult task empirically. Therefore, we may consider a theoretical experiment in which we draw all possible sample of a fixed size and obtain a sampling distribution.

Although *in practice only a random sample is actually selected* and the concept of the sampling distribution is used to draw the inference about the population parameters.

8.4 STANDARD ERROR

As we have seen in the previous section that the values of sample statistic may vary from sample to sample and all the sample values are not equal to the population parameter. Now, one can be interested to measure how much the values of sample statistic vary from the population parameter on average. You may use the standard deviation as a measure of variation.

Thus, for measuring the variation in the values of sample statistic around the population parameter we calculate the standard deviation of the sampling distribution. This is known as the standard error of that statistic. Thus, the standard error of a statistic can be defined as:

“The standard deviation of a sampling distribution of a statistic is known as standard error and it is denoted by SE.”

The computation of the standard error is a tedious process. There is a simple formula to compute the standard error of the mean from a single sample as:

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population with mean μ and variance σ^2 then the standard error of the sample mean ($\bar{X}$) is given by

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

The standard error is used to express the accuracy or precision of the estimate of population parameter because the reciprocal of the standard error is the measure of reliability or precision of the statistic. The standard error also determines the probable limits or confidence limits within which the population parameter may

Standard deviation of sampling distribution of a statistic is called standard error.

be expected to lie with a certain level of confidence. The standard error is also applicable in the testing of hypothesis.

The standard errors of some other well-known statistics are given below:

1. The standard error of sample proportion (p) is given by

$$SE(p) = \sqrt{\frac{PQ}{n}}$$

where P is the population proportion and $Q = 1-P$.

2. The standard error of sample median ($\tilde{X}$) is given by

$$SE(\tilde{X}) = \frac{\pi}{2} \frac{\sigma}{\sqrt{n}} = 1.25331 SE(\bar{X})$$

3. The standard error of difference of two sample means is given by

$$SE(\bar{X} - \bar{Y}) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

where, σ_1^2 and σ_2^2 are the population variances and n_1 and n_2 are the sample sizes of two independent samples.

4. The standard error of difference between two sample proportions is given by

$$SE(p_1 - p_2) = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$

where P_1 and P_2 are population proportions of two different populations and $Q_1 = 1-P_1$ and $Q_2 = 1-P_2$.

From all the above formulae, we can understand that standard error is inversely proportional to the sample size. **Therefore, as sample size increases the standard error decreases.**

Note 2: These standard errors discussed above were calculated under the assumption that sampling is done either from an infinite population or from a finite population with replacement. But, in real life sampling problems, most sampling plans do not permit an element to be selected twice in a given sample (i.e. sampling with replacement). Consequently, if the population is not large in the comparison to the size of the sample and sampling is done without replacement then we multiply the above standard errors by the correction factor

$$\sqrt{\frac{N-n}{N-1}}.$$

where N is the population size. This correction factor is known as a finite population correction factor.

Therefore, in this case, the standard error of the sample mean is given by

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

Similarly,

$$SE(p) = \sqrt{\frac{N-n}{N-1} \frac{PQ}{n}}$$

In practice, if the sample size n is less than or equal to 10% of the population size N , this correction factor may be ignored for all practical purposes.

Let us do one example relating to standard error.

Example 1: Diameter of a steel ball bearing produced by a semi-automatic machine is known to be distributed normally with mean 12 cm and standard deviation 0.1 cm. If we take a random sample of size 10 with replacement then find the standard error of the sample mean for estimating the population mean of the diameter of steel ball bearing for the whole population.

Solution: Here, we are given that

$$\mu = 12, \sigma = 0.1, n = 10$$

Since the sampling is done with replacement, therefore, the standard error of the sample mean for estimating the population mean is given by

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{0.1}{\sqrt{10}} = 0.03$$

Check Your Progress 1

- 1) If the lives of 3 Televisions of a certain company are 8, 6 and 10 years then construct the sampling distribution of average life of Televisions by taking all samples of size 2.

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- 2) The average weight of a certain type of tyres is 200 pounds and the standard deviation is 4 pounds. A sample of 50 tyres is selected. Obtain the standard error of the sample mean.

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- 3) A machine produces a large number of items of which 15% are found to be defective. If a random sample of 200 items is taken from the population, then find the standard error of the sampling distribution of proportion.

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8.5 CENTRAL LIMIT THEOREM

The central limit theorem is the most important theorem of Statistics. It was first introduced by De Moivre in the early eighteenth century. According to the central limit theorem, if $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population with mean μ and variance σ^2 then the sampling distribution of the

sample mean tends to a normal distribution with mean μ and variance σ^2/n as the sample size tends to be large ($n > 30$), whatever may be the form of the parent population, that is:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

and the variate

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

follows a normal distribution with mean 0 and variance unity, that is, the variate Z follows a standard normal distribution.

We do not intend to prove this theorem here, but merely show graphical evidence of its validity in Fig. 8.1. Here, we will also try to show how large must the sample size be for which we can assume that the central limit theorem applies?

In Fig. 8.1, we are trying to understand the sampling distribution of sample mean $\bar{X}$ for different populations and for varying sample sizes. We divide this figure into four parts A, B, C and D. The part 'A' of this figure shows four different populations as normal, uniform, binomial and exponential. The rest parts B, C and D represent the shape of the sampling distribution of mean of sizes $n = 2$, $n = 5$ and $n = 30$ respectively drawn from the populations shown in first row (Part-A). From the first column of this figure, we observed that when the parent population is normal then all the sampling distributions for varying sample sizes are also normal, having the same mean but their variances decrease as n increases.

The second column of this figure represents the uniform population. Here, we observe that the sampling distribution of $\bar{X}$ is symmetrical when $n = 2$ and tends to normal when $n = 30$.

However, the third column of this figure represents the binomial population (discrete). Again when $n = 2$, the sampling distribution of $\bar{X}$ is symmetrical and for $n = 5$ it is quite bell-shaped and tends to normal when $n = 30$. The last column of this figure represents the exponential population which is highly skewed. Here, we observe that for $n = 30$, the distribution of $\bar{X}$ is symmetrical bell-shaped normal.

Here, we conclude that if we draw a random sample of large size $n > 30$ from the population then the sampling distribution of $\bar{X}$ can be approximated by a normal probability distribution, whatever the form of the parent population.

Note 3: The central limit theorem can also be applicable in the same way for the sampling distribution of sample proportion, sample standard deviation, difference of two sample means, difference of two sample proportions, etc. that is, if we take a random sample of large size ($n = 36 > 30$) from the population then the sampling distribution of sample proportion, sample standard deviation, difference of two sample means, difference of two sample proportions, etc. approximated by a normal probability distribution, whatever the form of the parent population. The mean and variance of these sampling distributions shall be discussed later.

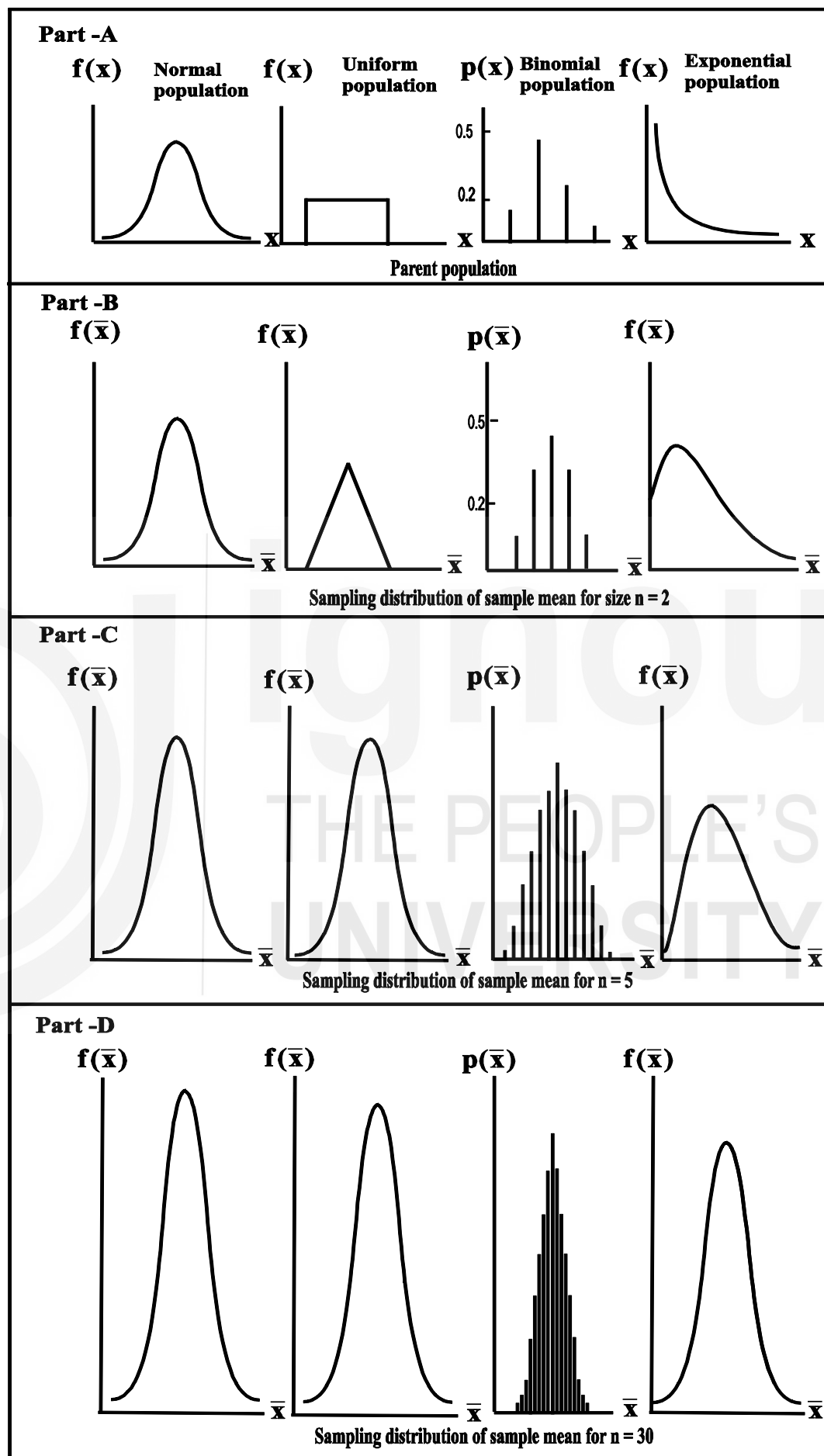


Fig. 8.1: Sampling distribution of sample means for various populations
when $n = 2$, $n = 5$ and $n = 30$

8.6 LAW OF LARGE NUMBERS AND SAMPLE SIZE

We have already discussed in the previous sections that the population parameters are generally unknown and for estimating parameters, we draw all possible random samples of the same size from the population and calculate the values of sample statistic such as sample mean, sample proportion, sample variance, etc. for all samples and with the help of these values we form sampling distribution of that statistic. Then we draw inference about the population parameters.

But in real-world the sampling distributions are never really observed. The process of finding sampling distribution would be very tedious because it would involve a very large number of samples. So in real-world problems, we draw a random sample from the population to draw inference about the population parameters. A very crucial question then arises: “Using a random sample of finite size, say n , can we make a reliable inference about population parameter?” The answer is “yes”, reliable inference about population parameter can be made by using only a finite sample and we shall demonstrate this by “**law of large numbers**”. This law of large numbers can be stated in words as:

“A positive integer can be determined such that if a random sample of size n or larger is taken from a population with a parameter, say, population mean (μ), the probability that the sample mean $\bar{X}$ will deviate from the population mean μ can be made to be as close to 1 as desired.”

This law states that for any two arbitrary small numbers ε and η where $\varepsilon > 0$ and $0 < \eta < 1$, there exists an integer n such that if a random sample of size n or larger is drawn from the population and calculate the sample mean $\bar{X}$ for this sample, then the probability that $\bar{X}$ deviates from μ by less than ε is greater than $1 - \eta$ (as close to 1 as desired), that is, $\bar{X}$ arbitrarily close to μ .

Symbolically this can be written as

For any $\varepsilon > 0$ and $0 < \eta < 1$ there exists an integer n such that $n \geq \frac{\sigma^2}{\varepsilon^2 \eta}$, where σ^2 is the finite variance of the population then

$$P[|\bar{X} - \mu| < \varepsilon] \geq 1 - \eta$$

$$\text{or } P[-\varepsilon < \bar{X} - \mu < \varepsilon] \geq 1 - \eta \quad \left[\because |X| < a \Rightarrow -a < X < a \right]$$

The proof of “law of large numbers” is beyond the scope of this Unit.

Here, with an example, we have to try to understand the law of large numbers in action, as we take more observations the sample mean approaches to the population mean.

Suppose that the distribution of the weight of all the young men living in a city is close to a normal distribution with mean 65 kg and standard deviation of 25 kg. To understand the law of large numbers, we calculate the sample mean weight ($\bar{X}$) for varying sample size $n = 1, 2, 3, \dots$. Fig. 8.2 shows the behaviour of the sample mean weight $\bar{X}$ of men chosen at random from this city. The graph plots the values of $\bar{X}$ (along the vertical axis and sample size along the horizontal axis) as sample size varying from $n = 1, 2, 3, \dots$

First, we start a sample of size $n = 1$, that is, we select a man randomly from the city. Suppose a selected man had weight 70 kg, therefore, the line of the graph starts from this point. Now, select the second man randomly and suppose his weight is 55 kg. So for $n = 2$ the sample mean is

$$\bar{X} = \frac{70 + 55}{2} = 62.5 \text{ kg}$$

This is the second point on the graph. Now, select the third man randomly from the city and suppose his weight is 83 kg. Therefore, for $n = 3$ the sample mean is

$$\bar{X} = \frac{70 + 55 + 83}{3} = 69.3 \text{ kg}$$

This is the third point on the graph.

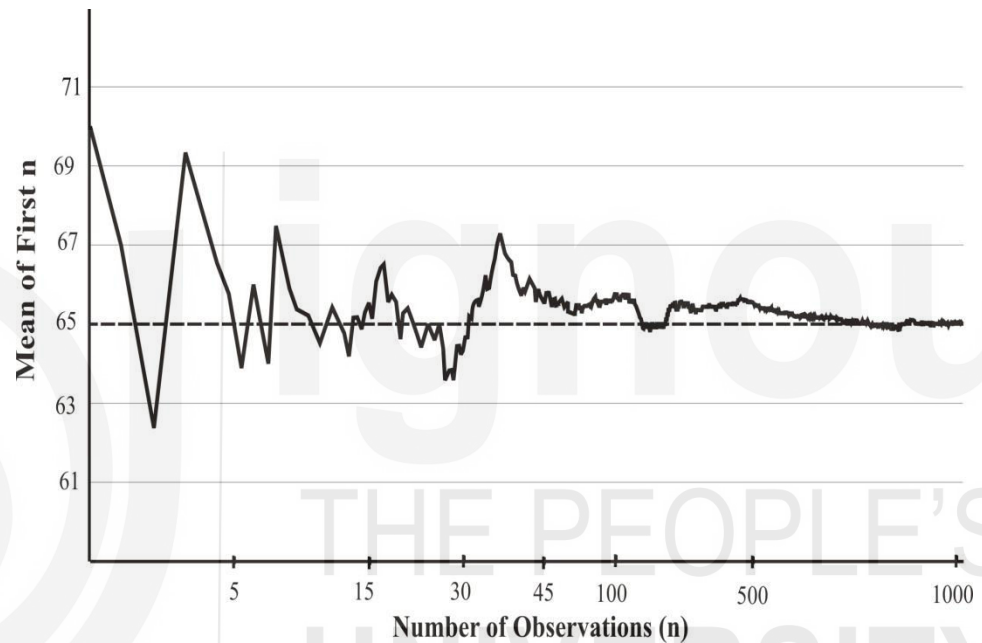


Fig. 8.2: Mean for Varying Sample Size

This process will continue. From the graph, we can see that the mean of the sample changes as we make more observations, eventually, however, the mean of the observations get close and close to the population mean 65 kg as the sample size increases.

Note 4: The law of large numbers can also be applied for the sample proportion, sample standard deviation, difference of two sample means, difference of two sample proportions, etc.

Example 2: A pathologist wants to estimate the mean time required to complete a certain analysis on the basis of sample study so that he may be 99% confident that the mean time may remain with ± 2 days of the mean. As per the available records, the population variance is 5 days. What must be the size of the sample for this study?

Solution: Here, we are given

$$1 - \eta = 0.99 \Rightarrow \eta = 0.01, \quad \varepsilon = 2, \quad \sigma^2 = 5$$

By the law of large numbers, we have

$$n \geq \frac{\sigma^2}{\varepsilon^2 \eta}$$

$$= \frac{5}{2^2 \times 0.01} = 125$$

That is, $n \geq 125$

Hence, at least 125 units must be drawn in a sample.

Example 3: An investigator wishes to estimate the mean of a population using a sample large enough that the probability will be 0.95 that the sample mean will not differ from the population mean by more than 25 per cent of the standard deviation. How large a sample should be taken?

Solution: We have

$$1 - \eta = 0.95 \Rightarrow \eta = 0.05, \varepsilon = 0.25\sigma$$

By the law of large numbers, we have

$$n \geq \frac{\sigma^2}{\varepsilon^2 \eta}$$

$$= \frac{\sigma^2}{(0.25\sigma)^2 \times 0.05} = 320$$

That is, $n \geq 320$

Therefore, at least 320 units must be taken in the sample so that the required probability will be 0.95.

8.7 SAMPLING DISTRIBUTION OF SAMPLE MEAN

One of the most important sample statistics, which is used to draw a conclusion about the population mean, is the sample mean. For example, an investigator may want to estimate the average income of the people living in a particular geographical area, a product manager may want to estimate the average life of electric bulbs manufactured by a company, a pathologist may want to estimate the mean time required to complete a certain analysis, etc. In the above cases, an estimate of the population mean is required, and one may estimate this on the basis of a sample taken from that population. For this, the sampling distribution of sample mean is required.

We have already given you the flavour of the sampling distribution of sample mean with the help of an example in earlier section in which we draw all possible samples of the same size from the population and calculate the sample mean for each sample. After calculating the value of sample mean for each sample we observed that the values of sample mean vary from sample to sample. Then the sample mean is treated as a random variable and a probability distribution is constructed for the values of sample mean. This probability distribution is known as the sampling distribution of sample mean. Therefore, the sampling distribution of sample mean can be defined as:

“The probability distribution of all possible values of the sample mean that would be obtained by drawing all possible samples of the same size from the population is called the sampling distribution of sample mean or simply sampling distribution of mean.”

The sampling distribution of sample means is a theoretical probability distribution of sample means that would be obtained by drawing all possible samples of the same size from the population.

Now, the question may arise in your mind “what is the shape of the sampling distribution of mean? Is it normal, exponential, binomial, Poisson, etc.?”

With the help of Fig. 8.1, we have described the shape of the sampling distribution of mean for different populations and for varying sample sizes. From this figure, we observed that when the parent population is normal then all the sampling distributions for varying sample sizes are also normal whereas when parent population is uniform, binomial, or exponential then the shapes of the sampling distributions of mean are not in the form of specifying distribution when the sample size is small ($n = 2$ or 5).

Generally, when samples are drawn from non-normal populations then it is not possible to specify the shape of the sampling distribution of mean when the sample size is small. Although when sample size is large (> 30) then we observed that sampling distribution of mean converges to normal distribution whatever the form of the population i.e. normal or non-normal.

After knowing the shapes of the sampling distribution of mean in different situations, you may be interested to know the mean and variance of the sampling distribution of mean when samples are drawn from a normal population.

In practice, only **one random sample** is selected and the concept of the sampling distribution is used to draw the inference about the population parameters. If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a normal population with mean μ and variance σ^2 then it has also been established that the sampling distribution of sample mean $\bar{X}$ is also normal. The mean and variance of the sampling distribution of $\bar{X}$ can be obtained as

$$\begin{aligned}\text{Mean of } \bar{X} &= E(\bar{X}) = E\left[\frac{X_1 + X_2 + \dots + X_n}{n}\right] \quad [\text{By definition of } \bar{X}] \\ &= \frac{1}{n}[E(X_1) + E(X_2) + \dots + E(X_n)]\end{aligned}$$

Since $X_1, X_2, \dots, X_n$ are randomly drawn from the same population so they also follow the same distribution as the population. Therefore,

$$E(X_1) = E(X_2) = \dots = E(X_n) = E(X) = \mu$$

and

$$\text{Var}(X_1) = \text{Var}(X_2) = \dots = \text{Var}(X_n) = \text{Var}(X) = \sigma^2$$

Thus,

$$\begin{aligned}E(\bar{X}) &= \frac{1}{n} \left(\underbrace{\mu + \mu + \dots + \mu}_{n\text{-times}} \right) \\ &= \frac{1}{n}(n\mu) = \mu\end{aligned}$$

$$E(\bar{X}) = \mu$$

and variance

$$\begin{aligned}\text{Var}(\bar{X}) &= \text{Var}\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \\ &= \frac{1}{n^2}[\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)]\end{aligned}$$

The statistical inference is based on the theoretical distribution so if we say that a random sample, say, $X_1, X_2, \dots, X_n$ is taken from a population it means that the random sample is selected from the infinite population or from a finite population with replacement such that X_i 's are independent and followed same population distribution.

If X and Y are two independent random variables and a & b are two constant then by the addition theorem of expectation, we have

$$E(aX + bY) = aE(X) + bE(Y)$$

and

$$\begin{aligned}\text{Var}(aX + bY) \\ &= a^2 \text{Var}(X) + b^2 \text{Var}(Y)\end{aligned}$$

$$= \frac{1}{n^2} \left(\underbrace{\sigma^2 + \sigma^2 + \dots + \sigma^2}_{n\text{-times}} \right) = \frac{1}{n^2} (n\sigma^2)$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Hence, we conclude that if the samples are drawn from a normal population with mean μ and variance σ^2 then the sampling distribution of mean $\bar{X}$ is also normal distribution with mean μ and variance σ^2/n , that is,

$$\text{If } X_i \sim N(\mu, \sigma^2) \text{ then } \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

The standard error (SE) of sample mean can be obtained by the definition of SE as

$$\text{SE}(\bar{X}) = \text{SD}(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}}$$

Note 5: The statistical inference is based on the theoretical distribution. Therefore, in whole statistical inference, we assume that the random sample is selected from the infinite population or a finite population with replacement.

If the sampling is done without replacement from a finite normal population of size N then the sample mean is distributed normally with mean

$$E(\bar{X}) = \mu$$

and variance

$$\text{Var}(\bar{X}) = \frac{N-n}{N-1} \frac{\sigma^2}{n}$$

The proof of this formula is beyond the scope of this course.

Therefore, the standard error of sample mean is given by

$$\text{SE}(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}$$

If the sample is drawn from any population other than normal then by central limit theorem, the sampling distribution of $\bar{X}$ tends to normal as the sample size n increases, that means, the sample mean $\bar{X}$ is normally distributed with mean μ and variance σ^2/n when the sample size is large (> 30).

Let us see an application of the sampling distribution of mean with the help of an example.

Example 4: Diameter of a steel ball bearing produced on a semi-automatic machine is known to be distributed normally with mean 12 cm and standard deviation 0.1 cm. If we take a random sample of size 10 then find

- (i) Mean and variance of the sampling distribution of mean.
- (ii) The probability that the sample mean lies between 11.95 cm and 12.05 cm.

Solution: Here, we are given that

$$\mu = 12, \sigma = 0.1, n = 10$$

- (i) Since the population is normally distributed therefore the sampling distribution of the sample mean also follows a normal distribution with mean

$$E(\bar{X}) = \mu = 12$$

and variance

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{(0.1)^2}{10} = 0.001$$

- (ii) The probability that the sample mean lies between 11.95 cm and 12.05 cm is given by

$$P[11.95 < \bar{X} < 12.05]$$

To get the value of this probability, we convert $\bar{X}$ into a standard normal variate Z by the transformation

$$Z = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\bar{X} - 12}{\sqrt{0.001}} = \frac{\bar{X} - 12}{0.03}$$

Therefore, by subtracting 12 from each term and then dividing each term by 0.03 in the above inequality, we get the required probability as

$$\begin{aligned} P\left[\frac{11.95 - 12}{0.03} < \frac{\bar{X} - 12}{0.03} < \frac{12.05 - 12}{0.03}\right] \\ &= P[-1.67 < Z < 1.67] \\ &= P[-1.67 < Z < 0] + P[0 < Z < 1.67] \\ &= P[0 < Z < 1.67] + P[0 < Z < 1.67] \\ &= 2P[0 < Z < 1.67] \\ &= 2 \times 0.4525 \quad \left[\begin{array}{l} \text{Using table for area} \\ \text{under normal curve} \end{array} \right] \\ &= 0.9050 \end{aligned}$$

Therefore, the probability that the sample mean lies between 11.95 and 12.05 cm is 0.9050.

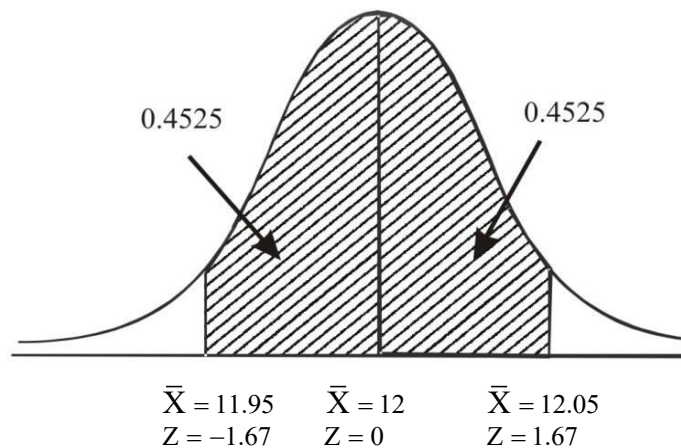


Fig. 8.3: Area Computation in Standard Normal Curve

Now, continuing our discussion about the sampling distribution of mean, we consider another situation.

In the earlier examples, we assume that variance σ^2 of the normal population is known but in general, it is not known and in such a situation the only alternative

left is to estimate the unknown σ^2 . The value of sample variance (S^2) is used to estimate the σ^2 where,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2$$

Thus, in this case, it has been established that the sampling distribution of mean is not normal and the variate $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ is known as t-variate and follows t-distribution with $(n-1)$ df instead of normal distribution. This distribution has major applications in Statistics.

If n is sufficiently large (> 30) then we know that almost all the distributions are very closely approximated by a normal distribution. Thus, in this case, t-distribution is also approximated as a normal distribution. So for large sample size (> 30), the sampling distribution of mean also follows a normal distribution with mean μ and variance S^2/n .

Now, try some exercises for your practice

Check Your Progress 2

- 1) The mean of a population is unknown and having a variance equal to 2. Find out that how large a sample must be taken, so that the probability will be at least 0.95 that the sample mean will lie within the range of 0.5 of the population mean?

.....

- 2) If parent population is normal with mean μ and variance σ^2 then, what is sampling distribution of sample means?

.....

- 3) The weight of a certain type of a truck tyre is known to be distributed normally with mean 200 pounds and standard deviation 4 pounds. A random sample of 10 tyres is selected. What is the sampling distribution of sample mean? Also, obtain the mean and variance of this distribution.

.....

- 4) The mean life of electric bulbs produced by a company is 2550 hours. A random sample of 100 bulbs is selected and the standard deviation is found to be 54 hours. Find the mean and variance of the sampling distribution of mean.

.....

8.8 SAMPLING DISTRIBUTION OF DIFFERENCE OF TWO SAMPLE MEANS

The sampling distribution of difference of two sample means is the theoretical sampling distribution of difference of sample means $(\bar{x} - \bar{y})$ that would be obtained by drawing all possible samples from both the populations.

There are so many problems where someone may be interested to draw the inference about the difference of two population means. For example, two manufacturing companies of bulbs are produced the same type of bulbs and one may be interested to know which one is better than the other, an investigator may want to know the difference of average income of the peoples living in two cities, say, A and B, two different types of drugs, were tried on a certain number of patients for controlling blood pressure and one may be interested to know which one has a better effect on controlling blood pressure, etc. Therefore, in such situations, to draw the inference we require the sampling distribution of difference of two sample means.

Let the same characteristic measures from two populations be represented by X and Y variables and the variation in the values of these constitute two populations, say, population-I for variation in X and population-II for variation in Y. Suppose population-I is having mean μ_1 and variance σ_1^2 and population-II is having mean μ_2 and variance σ_2^2 . Then we take all possible samples of same size n_1 from population-I and then the sample mean, say, $\bar{X}$ is calculated for each sample. Similarly, all possible samples of same size n_2 are taken from the population-II and the sample mean, say, $\bar{Y}$ is calculated for each sample. Then we consider all possible differences of means $\bar{X}$ and $\bar{Y}$. The difference of these means may or may not differ from sample to sample and so we construct the probability distribution of these differences. The probability distribution thus obtained is known as the sampling distribution of the difference of sample means. Therefore, the sampling distribution of difference of sample means can be defined as:

“The probability distribution of all values of the difference of two sample means would be obtained by drawing all possible samples from both the populations is called the sampling distribution of difference of two sample means.”

If both the parent populations are normal, that is,

$$X \sim N(\mu_1, \sigma_1^2) \text{ and } Y \sim N(\mu_2, \sigma_2^2)$$

then as we discussed in the previous section that

$$\bar{X} \sim N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \text{ and } \bar{Y} \sim N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)$$

Thus, as we have discussed in earlier unit that if two independent random variables X and Y are normally distributed then the difference $(X - Y)$ of these also normally distributed. Therefore, in our case, the sampling distribution of difference of two sample means $(\bar{X} - \bar{Y})$ also follows a normal distribution with mean

$$E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2$$

and variance

$$\text{Var}(\bar{X} - \bar{Y}) = \text{Var}(\bar{X}) + \text{Var}(\bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

If X and Y are independent random variables then
 $E(X - Y) = E(X) - E(Y)$
 and
 $\text{Var}(X - Y)$
 $= \text{Var}(X) + \text{Var}(Y)$

Therefore, the standard error of difference of two sample means is given by

$$SE(\bar{X} - \bar{Y}) = \sqrt{\text{Var}(\bar{X} - \bar{Y})} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

If the parent populations are not normal but n_1 and n_2 are large (> 30), then by central limit theorem, the sampling distribution of $(\bar{X} - \bar{Y})$ is very closely normally distributed with mean $(\mu_1 - \mu_2)$ and variance $\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$.

Let us see an application of the sampling distribution of difference of two sample means with the help of an example.

Example 5: Electric CFLs manufactured by company A have mean lifetime of 2400 hours with standard deviation 200 hours, while CFLs manufactured by company B have mean lifetime of 2200 hours with the standard deviation of 100 hours. If random samples of 125 electric CFL of each company are tested, find

- The mean and standard error of the sampling distribution of the difference of mean lifetime of electric CFLs.
- The probability that the CFLs of company A will have a mean lifetime at least 160 hours more than the mean lifetime of the CFLs of company B.

Solution: Here, we are given that

$$\mu_1 = 2400, \sigma_1 = 200, \mu_2 = 2200, \sigma_2 = 100$$

$$n_1 = n_2 = 125$$

- Let $\bar{X}$ and $\bar{Y}$ denote the mean lifetime of CFLs taken from companies A and B, respectively. Since n_1 and n_2 are large ($n_1, n_2 > 30$) therefore, by the central limit theorem, the sampling distribution of $(\bar{X} - \bar{Y})$ follows a normal distribution with mean

$$E(\bar{X} - \bar{Y}) = \mu_1 - \mu_2 = 2400 - 2200 = 200$$

and variance

$$\begin{aligned} \text{Var}(\bar{X} - \bar{Y}) &= \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \\ &= \frac{(200)^2}{125} + \frac{(100)^2}{125} = 400 \end{aligned}$$

Therefore, the standard error is given by

$$SE(\bar{X} - \bar{Y}) = \sqrt{\text{Var}(\bar{X} - \bar{Y})} = \sqrt{400} = 20$$

- Here, we want to find out the probability which is given by

$$P[\bar{X} \geq 160 + \bar{Y}] = P[(\bar{X} - \bar{Y}) \geq 160]$$

To get the value of this probability, we convert variate $(\bar{X} - \bar{Y})$ into a standard normal variate Z by the transformation

$$Z = \frac{(\bar{X} - \bar{Y}) - E(\bar{X} - \bar{Y})}{\sqrt{\text{Var}(\bar{X} - \bar{Y})}} = \frac{(\bar{X} - \bar{Y}) - 200}{\sqrt{400}} = \frac{(\bar{X} - \bar{Y}) - 200}{20}$$

Therefore, by subtracting 200 from each term and then dividing each term by 20 in the above inequality, we get the required probability

$$\begin{aligned}
 P\left[\frac{(\bar{X} - \bar{Y}) - 200}{20} \geq \frac{160 - 200}{20}\right] &= P[Z \geq -2] \\
 &= P[-2 < Z < 0] + P[0 < Z < \infty] \\
 &= P[0 < Z < 2] + P[0 < Z < \infty] \\
 &= 0.4772 + 0.5 = 0.9772
 \end{aligned}$$

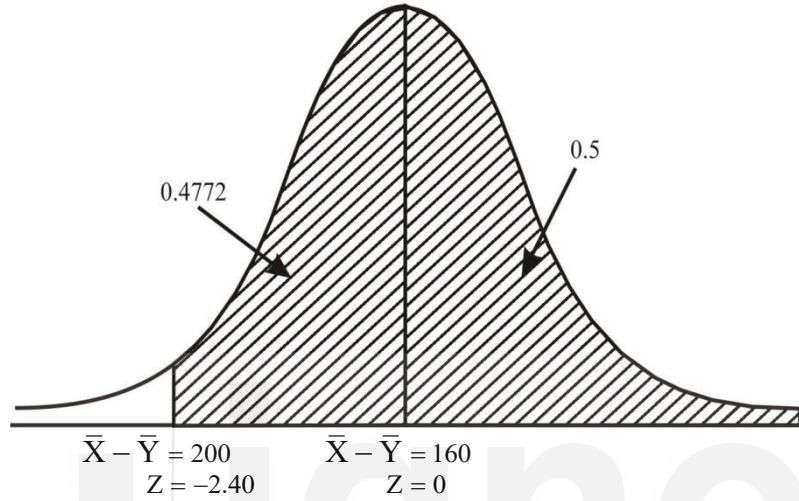


Fig. 8.4: Computation of Probability

Now, continuing our discussion about the sampling distribution of difference of two means, we consider another situation.

If population variances σ_1^2 & σ_2^2 are unknown then we estimate σ_1^2 & σ_2^2 by the values of the sample variances S_1^2 & S_2^2 of the samples taken from the first and second population respectively. And for large sample sizes n_1 and n_2 (> 30), the sampling distribution of $(\bar{X} - \bar{Y})$ is very closely normally distributed with

mean $(\mu_1 - \mu_2)$ and variance $\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)$.

If population variances σ_1^2 & σ_2^2 are unknown and $\sigma_1^2 = \sigma_2^2 = \sigma^2$ then σ^2 is estimated by pooled sample variance S_p^2 where,

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} [n_1 S_1^2 + n_2 S_2^2]$$

and variate

$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1 + n_2 - 2)}$$

follows t-distribution with $(n_1 + n_2 - 2)$ degrees of freedom. As similar as the sampling distribution of mean, for large sample sizes n_1 and n_2 (> 30) the sampling distribution of $(\bar{X} - \bar{Y})$ is very closely normally distributed with mean

$(\mu_1 - \mu_2)$ and variance $S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)$.

The sampling distributions are also defined for proportions also, but they are beyond the scope of this Unit. You may refer to references for more details on that topic.

Check Your Progress 3

- 1) The average height of all the workers in a hospital is known to be 68 inches with a standard deviation of 2.3 inches, whereas the average height of all the workers in a company is known to be 65 inches with a standard deviation of 2.5 inches. If a sample of 35 hospital workers and a sample of 50 company workers are selected at random, what is the probability that the sample mean of the height of hospital workers is at least 2 inch greater than that of company workers?

.....

8.9 SUMMARY

In this unit, we have covered the following points:

1. The statistical procedure which is used for drawing conclusions about the population parameter on the basis of the sample data is called “**statistical inference**”.
2. The group of units or items under study is known as “**Population**” whereas a part or a fraction of population is known as “**sample**”.
3. A “**parameter**” is a function of population values which is used to represent the certain characteristic of the population and any quantity calculated from sample values and does not contain any unknown population parameter is known as “**statistic**”.
4. Any statistic used to estimate an unknown population parameter is known as “**estimator**” and the particular value of the estimator is known as “**estimate**” of the parameter.
5. The probability distribution of all possible values of a sample statistic that would be obtained by drawing all possible samples of the same size from the population is called “**sampling distribution**” of that statistic.
6. The standard deviation of the sampling distribution of a statistic is known as “**standard error**”.
7. The most important theorem of Statistics is “**central limit theorem**” which state that the sampling distribution of the sample means tends to a normal distribution as the sample size tends to large ($n > 30$).
8. According to the law of large numbers, a positive integer can be determined such that if a random sample of size n or larger is taken from a population with a parameter, say, population mean (μ), the probability that the sample mean will be very closed to the population mean, can be made as close to 1 as desired.
9. The sampling distribution of sample mean and the sampling distribution of difference between two sample means are normally distributed.

8.10 SOLUTIONS / ANSWERS

Check Your Progress 1

- 1) Here, we are given that
 $N = 3, n = 2$

Since we have to estimate the average life of Televisions on the basis of samples of size $n = 2$ therefore, all possible samples (with replacement) are $N^n = 3^2 = 9$ and for each sample, we calculate the sample mean as shown in Table 8.4 given below:

Table 8.4: Calculation of Sample Mean

Sample Number	Sample Observation	Sample Mean ($\bar{X}$)
1	8, 8	8
2	8, 6	7
3	8, 10	9
4	6, 8	7
5	6, 6	6
6	6, 10	8
7	10, 8	9
8	10, 6	8
9	10, 10	10

Since the arrangement of all possible values of the sample mean with their corresponding probabilities is called the sampling distribution of mean thus, we arrange every possible value of the sample mean with their respective probabilities in the following Table 8.5 given below:

Table 8.5: Sampling distribution of sample means

S.No.	Sample Mean ($\bar{X}$)	Frequency	Probability
1	6	1	1/9
2	7	2	2/9
3	8	3	3/9
4	9	2	2/9
5	10	1	1/9

2) Here, we are given that

$$\mu = 200, \sigma = 4, n = 50$$

Assuming that the sampling is done with replacement therefore we know that the standard error of sample mean for estimating population mean is given by

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{50}} = 0.57$$

3) Here, we are given that

$$P = \frac{15}{100} = 0.15, n = 200$$

Therefore, we know that the standard error of sample proportion is given by

$$SE(p) = \sqrt{\frac{PQ}{n}} = \sqrt{\frac{0.15 \times (1 - 0.15)}{200}} = 0.025$$

Check Your Progress 2

1) Here, we are given that

$$\sigma^2 = 2, \varepsilon = 0.5, 1 - \eta = 0.95 \Rightarrow \eta = 0.05$$

By the law of large numbers, we have

$$n \geq \frac{\sigma^2}{\varepsilon^2 \eta}$$

$$= \frac{2}{(0.5)^2 \times (0.05)} = 160$$

That is, $n \geq 160$

Therefore, at least 160 units must be taken in the sample so that the probability will be at least 0.95 that the sample mean will lie within 0.5 of the population mean.

- 2) Since parent population is normal so sampling distribution of sample means is also normal with mean μ and variance σ^2/n .

- 3) Here, we are given that

$$\mu = 200, \sigma = 4, n = 10$$

Since parent population is normal so the sampling distribution of sample means is also normal. Therefore, the mean of this distribution is given by

$$E(\bar{X}) = \mu = 200$$

and variance

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{(4)^2}{10} = \frac{16}{10} = 1.6$$

- 4) Here, we are given that

$$\mu = 2550, n = 100, S = 54$$

First of all, we find the sampling distribution of sample mean. Since the sample size is large ($n = 100 > 30$) therefore, by the central limit theorem, the sampling distribution of sample mean follows a normal distribution. Therefore, the mean of this distribution is given by

$$E(\bar{X}) = \mu = 2550$$

and variance

$$\text{Var}(\bar{X}) = \frac{S^2}{n} = \frac{(54)^2}{100} = \frac{2916}{100} = 29.16$$

Check Your Progress 3

- 1) Here, we are given that

$$\mu_1 = 68, \sigma_1 = 2.3, n_1 = 35$$

$$\mu_2 = 65, \sigma_2 = 2.5, n_2 = 50$$

To find the required probability, first of all, we find the sampling distribution of difference of two sample means. Let $\bar{X}$ and $\bar{Y}$ denote the mean height of workers taken from hospital and company respectively. Since n_1 and n_2 are large ($n_1, n_2 > 30$) therefore, by the central limit theorem, the sampling distribution of $(\bar{X} - \bar{Y})$ follows a normal distribution with mean

$$E(\bar{X} - \bar{Y}) = \mu_1 - \mu_2 = 68 - 65 = 3$$

and variance

$$\text{Var}(\bar{X} - \bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{(2.3)^2}{35} + \frac{(2.5)^2}{50}$$

$$= 0.1511 + 0.1250 = 0.2761$$

The probability that the sample mean of the height of hospital workers is 2 inches less than that of company workers is given by

$$P[(\bar{X} - \bar{Y}) < 2]$$

To get the value of this probability, we convert variate $(\bar{X} - \bar{Y})$ into a standard normal variate Z by the transformation

$$Z = \frac{(\bar{X} - \bar{Y}) - E(\bar{X} - \bar{Y})}{\sqrt{\text{Var}(\bar{X} - \bar{Y})}} = \frac{(\bar{X} - \bar{Y}) - 3}{\sqrt{0.2761}} = \frac{(\bar{X} - \bar{Y}) - 3}{0.53}$$

Therefore, by subtracting 3 from each term and then dividing each term by 0.53 in the above inequality, we get the required probability

$$\begin{aligned} P\left[\frac{(\bar{X} - \bar{Y}) - 3}{0.53} < \frac{2 - 3}{0.53}\right] &= P[Z < -1.89] = 0.5 - P[-1.89 \leq Z \leq 0] \\ &= 0.5 - P[0 \leq Z \leq 1.89] \\ &= 0.5 - 0.4706 = 0.0294 \end{aligned}$$

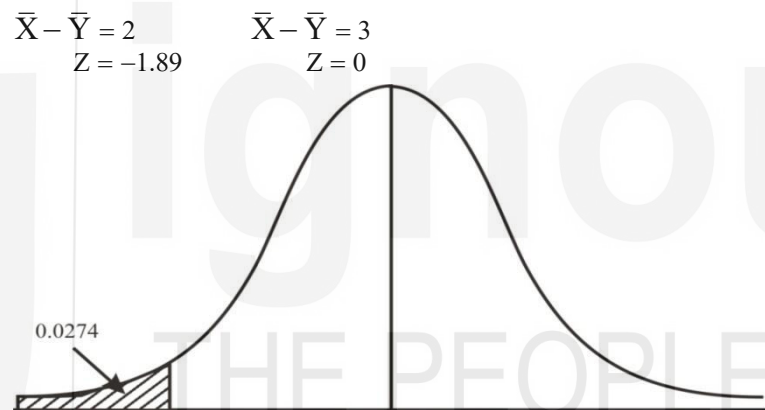


Fig. 8.5: The Distribution of Difference of Two Normal Variables

8.11 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 1: Sampling Distributions, UNIT 1: Introduction to Sampling Distribution (for Sections 8.2 to 8.6 and related portions in Sections 8.0 , 8.1, 8.9, 8.10 and Check Your Progress questions)
2. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 1: Sampling Distributions, UNIT 2: Sampling Distribution(s) of Statistic(s) (for Sections 8.6 to 8.8 and related portions in Sections 8.0 , 8.1, 8.9, 8.10 and Check Your Progress questions)

Reference Book

3. "Introduction to Probability and Statistics for Engineers and Scientists", 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004

UNIT 9 ESTIMATING PARAMETERS

Structure

- 9.0 Introduction
- 9.1 Objectives
- 9.2 Basic Terminology
- 9.3 Characteristics of Estimators
- 9.4 Point Estimation
- 9.5 Method of Maximum Likelihood
- 9.6 Interval Estimation
- 9.7 Confidence Interval for Population Mean
 - 9.7.1 Confidence Interval for Population Mean when Population Variance is Known
 - 9.7.2 Confidence Interval for Population Mean when Population Variance is Unknown
 - 9.7.3 Confidence Interval for Difference of Two Population Means when Samples are Independent and Population Variances are known
 - 9.7.4 Confidence Interval for Difference of Two Population Means when Population Variances are Unknown.
- 9.8 Confidence Interval for Population Proportion
- 9.9 Determination of Sample Size
- 9.10 Summary
- 9.11 Solutions/ Answers
- 9.12 References

9.0 INTRODUCTION

In many real-life problems, population parameter(s) is (are) unknown and someone is interested to obtain the value(s) of the parameter(s). But if the whole population is too large to study or the units of the population are destructive in nature or there are limited resources and manpower available, then it is not practically convenient to examine each unit of the population to find the value(s) of the parameter(s). In such situations, one can draw a sample from the population under study and utilise sample observations to estimate the parameter(s). The technique of finding an estimator to produce an estimate of the unknown parameter on the basis of a sample is called estimation. There are two methods of estimation:

1. Point Estimation
2. Interval Estimation

In point estimation, we determine an appropriate single statistic whose value is used to estimate the unknown parameter, whereas in interval estimation, we determine an interval that contains the true value of the unknown parameter with a certain confidence. The unit also elaborates on confidence intervals for population means for one and two variables under different conditions and the confidence interval for population proportions.

9.1 OBJECTIVES

After studying this unit, you should be able to:

- define the parameter space and estimator;

- explain the properties of an estimator;
- explore the different methods of point estimation;
- explain the method of maximum likelihood;
- describe the properties of maximum likelihood estimators;
- define the interval estimation;
- obtain the confidence interval for population mean of a normal population when population variance is known and unknown;
- describe the method of obtaining the confidence interval for difference of means of two normal populations when variances are known and unknown;
- obtain the confidence interval for population proportion; and
- determination of sample size.

9.2 BASIC TERMINOLOGY

Before discussing the properties of a good estimator, we discuss basic definitions of some important terms. These terms are very useful in understanding the fundamentals of the theory of estimation discussed in this block.

Discrete and Continuous Distributions

In the earlier Units, we have discussed standard discrete and continuous distributions as binomial, Poisson, normal, etc. We know that the populations can be described with the help of distributions, therefore, standard discrete and continuous distributions are used in statistical inference. Here, we present some standard discrete and continuous distributions in brief as in tabular form:

S. No.	Distribution	Parameter (s)	Mean	Variance
1	Bernoulli (discrete) $P[X = x] = p^x (1 - p)^{1-x}; x = 0, 1$	p	p	pq
2	Binomial (discrete) $P[X = x] = {}^n C_x p^x q^{n-x}; x = 0, 1, \dots, n$	$n \text{ \& } P$	np	npq
3	Poisson (discrete) $P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, \dots \text{ \& } \lambda > 0$	λ	λ	λ
4	Uniform (discrete) $P[X = x] = \frac{1}{n}; x = 0, 1, \dots, n$	n	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$
5	Hypergeometric (discrete) $P[X = x] = \frac{{}^M C_x {}^{N-M} C_{n-x}}{{}^N C_n}; x = 0, 1, \dots, \min\{M, n\}$	$N, M \text{ \& } n$	$\frac{nM}{N}$	$\frac{NM(N-M)(N-n)}{N^2(N-1)}$

6	Normal(continuous) $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; -\infty < x < \infty$ $\& \sigma > 0, -\infty < \mu < \infty$	$\mu \& \sigma^2$	μ	σ^2
7	Standard Normal(continuous) $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}; -\infty < x < \infty$	--	0	1
8	Uniform (continuous) $f(x) = \frac{1}{b-a}; a < x < b, b > a$	a & b	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$

Parameter Space

The set of all possible values that the parameter θ or parameters $\theta_1, \theta_2, \dots, \theta_k$ can assume is called the parameter space. It is denoted by Θ and is read as “**big theta**”. For example, if parameter θ represents the average life of electric bulbs manufactured by a company, then parameter space of θ is $\Theta = \{\theta : \theta \geq 0\}$, that is, the parameter average life θ can take all possible values greater than or equal to 0. Similarly, in a normal distribution (μ, σ^2) , the parameter space of parameters μ and σ^2 is $\Theta = \{(\mu, \sigma^2) : -\infty < \mu < \infty; 0 < \sigma < \infty\}$.

9.3 CHARACTERISTICS OF ESTIMATORS

It is to be noted that a large number of estimators can be proposed for an unknown parameter. For example, if we want to estimate the average income of the persons living in a city then the sample mean, sample median, sample mode, etc. can be used to estimate the average income. Now, the question arises, “Are some of the possible estimators better, in some sense, than the others?” Generally, an estimator can be called good for two different situations:

- (i) **When the true value of the parameter is being estimated is known**– An estimator might be called good if its value is close to the true value of the parameter to be estimated. In other words, the estimator whose sampling distribution concentrates as closely as possible near the true value of the parameter may be regarded as the good estimator.
- (ii) **When the true value of the parameter is unknown**– An estimator may be called good if the data give good reason to believe that the estimate will be close to the true value.

In the whole estimation, we estimate the parameter when the true value of the parameter is unknown. Hence, we must choose estimates not because they are certainly close to the true value, but because there is a good reason to believe that the estimated value will be close to the true value of the parameter. Let us describe certain properties, which help us in deciding whether an estimator is better than others.

Prof. Ronald A. Fisher was the man who pushed ahead the theory of estimation and introduced these concepts and gave some properties of a good estimator as follows:

1. Unbiasedness
2. Consistency



*Prof. Ronald A. Fisher,
a Great English
Mathematical
Statistician.
(1890-1962)*

3. Efficiency
4. Sufficiency

We shall discuss these properties one by one in this section.

1. Unbiasedness

Generally, population parameter(s) is (are) unknown and if the whole population is too large to study to find the value of an unknown parameter(s) then one can estimate the population parameter(s) with the help of estimator(s) which is (are) always a function of sample values.

An estimator is said to be unbiased if the expected value of the estimator is equal to the true value of the parameter being estimated.

An estimator is said to be unbiased for the population parameter such as population mean, population variance, population proportion, etc. if and only if the average or mean of the sampling distribution of the estimator is equal to the true value of the parameter.

Now, we explain the procedure to show that a statistic is unbiased or not for a parameter with the help of some examples:

Example 1: A random sample of 10 cadets of a centre is selected and their weights (in kg) are measured, which are given below:

48, 50, 62, 75, 80, 60, 70, 56, 52, 78

Determine an unbiased estimate of the average weight of cadets of the centre.

Solution: We know that sample mean ($\bar{X}$) is an unbiased estimator of the population mean and its particular value is the unbiased estimate of the population mean, therefore,

$$\begin{aligned}\bar{X} &= \frac{1}{n} \sum_{i=1}^n X_i = \frac{X_1 + X_2 + \dots + X_n}{n} \\ &= \frac{48 + 50 + 62 + 75 + 80 + 60 + 70 + 56 + 52 + 78}{10} = 63.10\end{aligned}$$

Hence, an unbiased estimate of the average weight of cadets of the centre is 63.10 kg.

Note 1: If $X_1, X_2, \dots, X_n$ is a random sample taken from a population with mean μ and variance σ^2 , then

$$S'^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ is a biased estimator of } \sigma^2.$$

whereas, $S^2 = \frac{n}{n-1} S'^2$ i.e. $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of σ^2 .

The proof of the above result is beyond the scope of this course.

Remark:

1. Unbiased estimators may not be unique. For example, the sample mean and sample median are unbiased estimators of the population mean of a normal population.
2. Unbiased estimators do not always exist for all the parameters. For example, for a Bernoulli distribution (θ), there is no unbiased estimator for θ^2 .

Similarly, for a Poisson distribution (λ), there exists no unbiased estimator for $1/\lambda$.

3. If an estimator is unbiased for all types of distribution, then it is called an absolutely unbiased estimator. For example, sample mean is an absolutely unbiased estimator of the population mean, if population mean exists.

One weakness of unbiasedness is that it requires only the average value of the estimator equals the true value of the population parameter. It does not require those values of the estimator to be reasonably close to the population parameter. For this reason, we require some other properties of a good estimator - consistency, efficiency and sufficiency which are described next.

2. Consistency

As defined earlier, an estimator T is said to be an unbiased estimator of a parameter, say, θ if the mean of the sampling distribution of estimator T is equal to the true value of the parameter θ . This concept was defined for a fixed sample size. Now, we will learn about consistency which is defined for increasing sample size.

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population whose probability density (mass) function is $f(x, \theta)$ where, θ is the population parameter then consider a sequence of estimators, say, $T_1 = t_1(X_1)$, $T_2 = t_2(X_1, X_2)$, $T_3 = t_3(X_1, X_2, X_3), \dots, T_n = t_n(X_1, X_2, \dots, X_n)$. A sequence of estimators is said to be consistent for a parameter θ if the deviation of the values of an estimator from the parameter tends to zero as the sample size increases. That means values of the estimator tend to get closer to the parameter θ as sample size increases.

Remark:

1. Consistent estimators may not be unique. For example, sample mean and sample median both are consistent estimators of the population mean of a normal population.
2. An unbiased estimator may or may not be consistent.
3. A consistent estimator may or may not be unbiased.

3. Efficiency

In some situations, we see that there are more than one estimators of a parameter which are unbiased as well as consistent. For example, sample mean and sample median both are unbiased and consistent for the parameter μ when sampling is done from a normal population with mean μ and known variance σ^2 . In such situations, there arises a necessity of some other criterion which will help us to choose 'best estimator' among them. A criterion which is based on the concept of the variance of the sampling distribution of the estimator is termed as efficiency.

If T_1 and T_2 are two estimators of a parameter θ , then T_1 is said to be more efficient than T_2 for all sample sizes if

$$\text{Var}(T_1) < \text{Var}(T_2) \quad \text{for all } n$$

4. Sufficiency

In statistical inference, the aim of the investigator or statistician may be to decide the value of the unknown parameter (θ). The information that guides the investigator in making a decision is supplied by the random sample

$X_1, X_2, \dots, X_n$. However, in most of the cases, the observations would be too numerous and too complicated. Direct use of these observations is complicated or cumbersome, therefore, a simplification or condensation would be desirable. The technique of condensing or reducing the random sample $X_1, X_2, \dots, X_n$ into a statistic such that it contains all the information about parameter θ that is contained in the sample is known as sufficiency. So before continuing our search of finding the best estimator, we introduce the concept of sufficiency.

A sufficient statistic is a particular kind of statistic that condenses random sample $X_1, X_2, \dots, X_n$ in a statistic $T = t(X_1, X_2, \dots, X_n)$ in such a way that no information about parameter θ is lost. That means, it contains all the information about θ that is contained in the sample and if we know the value of sufficient statistic, then the sample values themselves are not needed and can tell you nothing more about θ . In other words:

A statistic T is said to be sufficient statistic for estimating a parameter θ if it contains all the information about θ which are available in the sample.

This property of an estimator is called sufficiency.

9.4 POINT ESTIMATION

The technique of estimating the unknown parameter with a single value is known as point estimation.

There are so many situations in our day to day life where we need to estimate some unknown parameter(s) of the population on the basis of the sample observations. For example, a housewife may want to estimate the monthly expenditure, a sweet shopkeeper may want to estimate the sales of sweets on a day, a student may want to estimate the study hours for the reading of a particular unit of this course, etc. This need is fulfilled by the technique of estimation. So the technique of finding an estimator to produce an estimate of the unknown parameter is called estimation.

Estimation is broadly divided into two categories namely:

- Point estimation and
- Interval estimation

If we find a single value with the help of sample observations which is taken as the estimated value of unknown parameter then this value is known as a point estimate and the technique of estimating the unknown parameter with a single value is known as “**point estimation**”.

If instead of finding a single value to estimate the unknown parameter if we find two values between which the parameter may be considered to lie with a certain probability(confidence) is known as interval estimate of the parameter and this technique of estimating is known as “**interval estimation**”.

For example, if we estimate the average weight of men living in a colony on the basis of sample mean, say, 62 kg then 62 kg is called point estimate of the average weight of men in the colony and this procedure is called as point estimation. If we estimate the average weight of men by an interval, say, [40,110] with 90% confidence that true value of the weight lies in this interval then this interval is called interval estimate and this procedure is called as interval estimation.

Now, the question may arise in your mind that “how the point and interval estimates are obtained?” So we will describe one of the important and frequently used methods of point estimation, next.

Methods of Point Estimation

Some of the important and frequently used methods of point estimation are:

1. Method of maximum likelihood
2. Method of moments
3. Method of least squares
4. Method of minimum chi-square
5. Method of minimum variance

In the next section, we will explain the method of maximum likelihood. Other methods are beyond the scope of this course.

Check Your Progress 1

- 1) Write the four properties of a good estimator.

.....

- 2) Find which technique of estimation (point estimation or interval estimation) is used in each case given below:

- (i) An investigator estimates average income Rs. 1.5 lakh per annum of the people living in a particular geographical area, on the basis of a sample of 50 people taken from that geographical area.
- (ii) A product manager of a company estimates the average life of electric bulbs in the range of 800 hours and 1000 hours, with a certain confidence, on the basis of a sample of 20 bulbs.
- (iii) A pathologist estimates the mean time required to complete a certain analysis in the range 30 minutes to 45 minutes, with a certain confidence, on the basis of a random sample of 25.

.....

- 3) List any two methods of point estimation.

.....

.....

9.5 METHOD OF MAXIMUM LIKELIHOOD

For describing the method of maximum likelihood, first, we have to define likelihood function.

Likelihood Function

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population with joint probability density (mass) function $f(x_1, x_2, \dots, x_n, \theta)$ of sample values then likelihood function is denoted by $L(\theta)$ and is defined as follows:

$$L(\theta) = f(x_1, x_2, \dots, x_n, \theta)$$

For discrete case,

$$L(\theta) = P[X_1 = x_1]P[X_2 = x_2] \dots P[X_n = x_n]$$

For continuous case,

$$L(\theta) = f(x_1, \theta) \cdot f(x_2, \theta) \dots f(x_n, \theta)$$

From the theoretical point of view, one of the most important methods of point estimation is method of maximum likelihood because it generally gives very good estimators as judged from various criteria. It was initially given by Prof. C.F. Gauss but later on, it was used as a general method of estimation by Prof. Ronald A. Fisher in 1912. The principle of maximum likelihood estimation is to find /estimate /choose the value of the unknown parameter which would most likely generate the observed data. We know that the likelihood function gives the relative likelihoods for different values of the parameters for the observed data. Therefore, we search the value of the unknown parameter for which the likelihood function is maximum corresponding to the observed data. The concept of maximum likelihood estimation is explained with a simple example given below:

Suppose we toss a coin 5 times and we observe 3 heads and 2 tails. Instead of assuming that the probability of getting head is 0.5, we want to find/ estimate the value of p that makes the observed data most likely. Since the number of heads follows the binomial distribution, therefore, the probability (likelihood function) of getting 3 heads in 5 tosses is given by

$$P[X = 3] = {}^5C_3(p)^3(1-p)^2$$

Imagine that p was 0.1 then

$$P[X = 3] = {}^5C_3(0.1)^3(0.1)^2 = 0.0081$$

Similarly, for different values of p the probability of getting 3 heads in 5 tosses is given in Table 9.1 given below:

Table 9.1: Probability/Likelihood Function Corresponding to Different Values of p

S. No.	p	Probability/ Likelihood Function
1	0.1	0.0081
2	0.2	0.0512
3	0.3	0.1323
4	0.4	0.2304
5	0.5	0.3125
6	0.6	0.3456
7	0.7	0.3087
8	0.8	0.2048
9	0.9	0.0729

From Table 9.1, we can conclude that p is more likely to be 0.6 because at $p = 0.6$ the probability is maximum or the likelihood function is maximum.

Therefore, the principle of maximum likelihood (ML) consists in finding an estimate for the unknown parameter θ within the admissible range of θ , i.e.

within parameter space Θ , which makes the likelihood function as large as possible, that is, maximise the likelihood function. Such an estimate is known as maximum likelihood estimate for unknown parameter θ . Thus, if there exists an estimate, say, $\hat{\theta}(x_1, x_2, \dots, x_n)$ of the sample values which maximises likelihood function $L(\theta)$, is known as “**maximum likelihood estimate**”. That is,

$$L(\hat{\theta}) = \sup_{\theta \in \Theta} L(\theta)$$

Example 2: If the number of weekly accidents occurring on a mile stretch of a particular road follows a Poisson distribution with parameter λ then find the maximum likelihood estimate of parameter λ on the basis of the following data:

Number of Accidents	0	1	2	3	4	5	6
Frequency	10	12	12	9	5	3	1

Solution: Here, the number of weekly accidents occurring on a mile stretch of a particular road follows a Poisson distribution with parameter λ so probability mass function (pmf) of the Poisson distribution is given by

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

We find the theoretical maximum likelihood estimate of the parameter λ (refer to references). It was found that maximum likelihood estimate of parameter λ of Poisson distribution is sample mean.

Since we observed that the sample mean is the maximum likelihood estimate of parameter λ so we calculate the sample mean of the given data as

S. No.	Number of Accidents(X)	Frequency(f)	fX
1	0	10	0
2	1	12	12
3	2	12	24
4	3	9	27
5	4	5	20
6	5	3	15
7	6	1	6
		N = 52	$\sum fX = 104$

The formula for calculating mean is

$$\begin{aligned} \bar{X} &= \frac{1}{N} \sum fX \text{ where, N is the total number of accidents} \\ &= \frac{1}{52} \times 104 = 2 \end{aligned}$$

Hence, maximum likelihood estimate of λ is 2.

Example 3: For random sampling from normal population $N(\mu, \sigma^2)$, find the maximum likelihood estimators for μ and σ^2 .

Estimation Parameters

$\hat{\theta}$ - is used to represent the estimate / estimator of parameter and read as cap i.e.

$\hat{\theta}$ - means it is the estimate of parameter θ and read as theta cap.

Solution: Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from normal population $N(\mu, \sigma^2)$, whose probability density function is given by

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}; -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

The likelihood function for parameters μ and σ^2 are computed in the references.

The maximum likelihood estimates for μ and σ^2 are $\bar{x}$ and s'^2 respectively. Hence, ML estimators for μ and σ^2 are $\bar{X}$ and S'^2 respectively.

Note 2: Since throughout the course we are using capital letters for estimators, therefore in the last line of the above example we use capital letters for maximum likelihood estimators for μ and σ^2 .

Properties of Maximum Likelihood (ML) Estimators

The following are the properties of maximum likelihood estimators:

1. A ML estimator is not necessarily unique.
2. A ML estimator is not necessarily unbiased.
3. A ML estimator may not be consistent in a rare case.
4. If a sufficient statistic exists, it is a function of the ML estimators.
5. If $T = t(X_1, X_2, \dots, X_n)$ is a ML estimator of θ and $\gamma(\theta)$ is a one to one function of θ , then $\gamma(T)$ is a ML estimator of $\gamma(\theta)$. This is known as invariance property of ML estimator.
6. When ML estimator exists, then it is most efficient in the group of such estimators.

9.6 INTERVAL ESTIMATION

When we find two values with the help of sample observations and constitute an interval such that it contains the true value of the parameter with a certain probability, then it is known as an interval estimate of the parameter. This technique of estimation is known as “Interval Estimation”.

In this section, we will define:

- Confidence Interval and Confidence Coefficient
- One-sided Confidence Intervals

in the following two sub-sections.

Confidence Interval and Confidence Coefficient

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from a population whose probability density (mass) function is $f(x, \theta)$. Let $T_1 = t_1(X_1, X_2, \dots, X_n)$ and $T_2 = t_2(X_1, X_2, \dots, X_n)$ (where $T_1 \leq T_2$) be two statistics such that the probability that the random interval $[T_1, T_2]$ includes the true value of population parameter θ is $(1 - \alpha)$, that is,

$$P[T_1 \leq \theta \leq T_2] = 1 - \alpha$$

as shown in the Fig.9.1, where, α does not depend on θ .

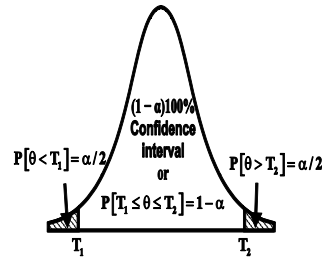


Fig. 9.1: Confidence Interval

Then the random interval $[T_1, T_2]$ is known as $(1-\alpha)$ 100% confidence interval for unknown population parameter θ and $(1-\alpha)$ is known as confidence coefficient or confidence level. The above probability statement may be explained with the help of an example:

Suppose we say that the probability that the random interval contains the true value of parameter θ is 0.95 then by this statement we simply mean that if 100 samples of the same size, say, 'n' are drawn from the given population $f(x, \theta)$ and the random interval $[T_1, T_2]$ is computed for each sample, then 95 times out of 100 intervals, the random interval $[T_1, T_2]$ contains the true value of the parameter θ . Hence, the higher the probability $(1-\alpha)$, more the confidence we will have that the random interval $[T_1, T_2]$ will include the true value of θ . The statistics T_1 and T_2 are known as lower and upper confidence limits or confidence bounds or fiducial limits, respectively for θ . And the interval is known as two-sided confidence interval. The length of the confidence interval is defined as

$$L = \text{Upper confidence limit} - \text{Lower confidence limit}$$

$$\text{i.e. } L = T_2 - T_1$$

The confidence interval may also be one-sided so we define one-sided confidence intervals next.

One-Sided Confidence Intervals

In some situations, we may be interested in finding an upper bound or lower bound but not both for population parameter with a given confidence. For example, one may be interested to obtain a bound such that he /she is 95% confident that the average life of the electric bulbs of a company is no less than one years. In such cases, we construct one-sided confidence intervals.

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from the population having probability density(mass) function $f(x, \theta)$ and also let T_1 be a statistic such that

$$P[\theta \geq T_1] = 1 - \alpha$$

as shown in Fig. 9.2, then statistic T_1 is called lower confidence bound for parameter θ with confidence coefficient $(1-\alpha)$ and $[T_1, \infty)$ is called a lower one-sided $(1-\alpha)$ 100% confidence interval for parameter θ .

Similarly, let T_2 be a statistic such that

$$P[\theta \leq T_2] = 1 - \alpha$$

as shown in Fig. 9.3, then statistic T_2 is called upper confidence bound for parameter θ with confidence coefficient $(1-\alpha)$ and $(-\infty, T_2]$ is called an upper one-sided $(1-\alpha)$ 100% confidence interval for parameter θ .

Note 3: Generally, one-sided confidence intervals are rarely used so we focus on two-sided confidence interval in this course. **Generally, confidence interval means two-sided confidence interval unless or otherwise it is stated as one-sided.**

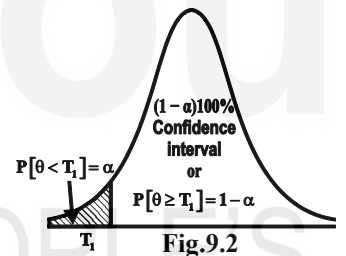


Fig.9.2

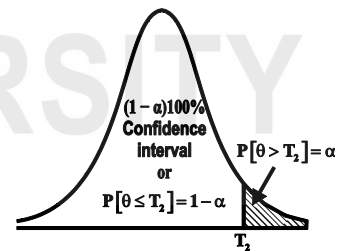


Fig. 9.3

Check Your Progress 2

- 1) What is the density function of the binomial population? What is the maximum likelihood estimator for p in a binomial population?

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.....

- 2) List any five properties of maximum likelihood estimators.

.....
.....
.....

- 3) Find the length of the following confidence intervals:

- (i) $P[-1.65 \leq \mu \leq 3.0] = 0.95$ (ii) $P[-1.68 \leq \mu \leq 2.70] = 0.95$
(iii) $P[-1.70 \leq \mu \leq 2.54] = 0.95$ (iv) $P[-1.96 \leq \mu \leq 1.96] = 0.95$

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- 4) Find the lower and upper confidence limits and also the confidence coefficient of the following confidence intervals:

- (i) $P[0 \leq \theta \leq 1.5] = 0.90$ (ii) $P[-1 \leq \theta \leq 2] = 0.95$
(iii) $P[-2 \leq \theta \leq 2] = 0.98$ (iv) $P[-2.5 \leq \mu \leq 2.5] = 0.99$

.....
.....
.....

9.7 CONFIDENCE INTERVAL FOR POPULATION MEAN

There are so many problems in real life where it becomes necessary to obtain the confidence interval of the population mean. For example, an investigator may interest to find the interval estimate of the average income of the people living in a particular geographical area, a product manager may want to find the interval estimate of the average life of electric bulbs manufactured by a company, a pathologist may want to obtain the interval estimate of the mean time required to complete a certain analysis, etc.

For describing confidence interval for population mean, let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from a normal population having mean μ and variance σ^2 . We can determine confidence interval for population mean μ under the following two cases:

1. When population variance σ^2 is known

2. When population variance σ^2 is unknown.

These two cases for one population are discussed in Sections 9.7.1 and 9.7.2, and for two populations in Sections 9.7.3 and 9.7.4.

9.7.1 Confidence Interval for Population Mean when Population Variance is Known

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from normal population $N(\mu, \sigma^2)$ when σ^2 is known, that is, σ^2 has a specified value, say, σ_0^2 .

To find out the confidence interval for population mean, first of all, we search the statistic for estimating μ whose distribution is completely known. Generally, we use the value of statistic (sample mean) $\bar{X}$ to estimate the population mean μ and also it is a sufficient statistic for parameter θ . Therefore, we use $\bar{X}$ to make the pivotal quantity.

We know that when the parent population is normal $N(\mu, \sigma^2)$, then sampling distribution of the sample mean $\bar{X}$ is normally distributed with mean μ and variance σ^2/n , that is, if

$$X_i \sim N(\mu, \sigma^2)$$

then

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

and the variate

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

follows the normal distribution with mean 0 and variance unity. Therefore, the probability density function of Z is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}; -\infty < z < \infty$$

This can be used to compute the $(1-\alpha)$ 100% confidence interval of the population mean as:

$$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] \quad \dots (1)$$

and corresponding limits are given by

$$\bar{X} \mp z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \dots (2)$$

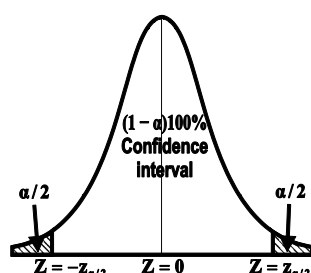


Fig. 9.4: Confidence Interval of Population Mean (Variance Known)

Note 4: In interval estimation generally, we have to find the 90%, 95%, 98% and 99% confidence intervals, therefore, the corresponding values of $z_{\alpha/2}$ are summaries in Table 9.2 and we will use these values directly when needed.

Table 9.2: Communally Used Values of Standard Normal Variate Z

$1 - \alpha$	0.90	0.95	0.98	0.99
α	0.10	0.05	0.02	0.01
$\alpha/2$	0.05	0.025	0.01	0.005
$z_{\alpha/2}$	1.645	1.96	2.33	2.58

For example, if we want to find the 99% confidence interval (two-sided) for μ then

$$1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$$

For $\alpha = 0.01$, the value of $z_{\alpha/2} = z_{0.005}$ is 2.58 therefore, 99% confidence interval for μ is given by

$$\left[\bar{X} - 2.58 \frac{\sigma}{\sqrt{n}}, \bar{X} + 2.58 \frac{\sigma}{\sqrt{n}} \right]$$

The application of the above discussion can be seen in the following example.

Example 4: The mean life of the tyres manufactured by a company follows a normal distribution with standard deviation 3200 km. A sample of 250 tyres is taken and it is found that the average life of the tyres is 50000 km with a standard deviation of 3500 km. Establish the 99% confidence interval within which the mean life of tyres of the company is expected to lie.

Solution: Here, we are given that

$$n = 250, \sigma = 3200, \bar{X} = 50000, S = 3500$$

Since population standard deviation i.e. population variance σ^2 is known, therefore, we use $(1-\alpha)$ 100% confidence limits for population mean when population variance is known which are given by

$$\bar{X} \mp z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \dots(3)$$

where, $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z and for 99% confidence interval, we have $1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$. For $\alpha = 0.01$ we have, $z_{\alpha/2} = z_{0.005} = 2.58$.

Therefore, the 99% confidence limits are

$$\bar{X} \mp 2.58 \frac{\sigma}{\sqrt{n}}$$

By putting the values of n , $\bar{X}$ and σ , the 99% confidence limits are

$$\begin{aligned} & 50000 \mp 2.58 \times \frac{3200}{\sqrt{250}} \\ & 50000 \mp 522.20 = 49477.80 \text{ and } 50522.20 \end{aligned}$$

Hence, 99% confidence interval within which the mean life of tyres of the company is expected to lie is

$$[49477.80, 50522.20]$$

9.7.2 Confidence Interval for Population Mean when Population Variance is Unknown

In the cases, described in the previous sub-section we assume that variance σ^2 of the normal population is known but in general, it is not known and in such a

situation the only alternative left is to estimate the unknown σ^2 . The value of sample variance (S^2) is used to estimate the σ^2 where,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2$$

In this situation, we know that the variate

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}$$

follows t-distribution with $(n-1)$ df. Therefore, the probability density function of statistic t is given by

$$f(t) = \frac{1}{B\left(\frac{n-1}{2}, \frac{1}{2}\right) \sqrt{n-1}} \left(1 + \frac{t^2}{n-1}\right)^{-\frac{n}{2}}; \quad -\infty < t < \infty$$

This can be used to find $(1-\alpha)$ 100% confidence interval for population mean of the normal population when variance σ^2 is unknown as:

$$\left[\bar{X} - t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}}, \quad \bar{X} + t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}} \right] \quad \dots (4)$$

and corresponding limits are given by

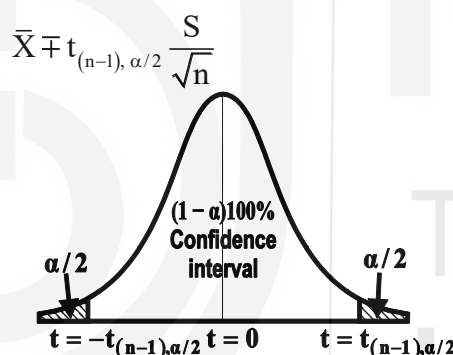


Fig. 9.5: Confidence Interval using t-Distribution (Variance Unknown)

Note 5: For different confidence intervals and different degrees of freedom, the values of $t_{(n-1), \alpha/2}$ are different. Therefore, for given values of α and n we read the tabulated value of t-statistic from the table of t-distribution (t-table) given in references.

For example, if we want to find the 95% confidence interval for μ when $n = 8$, then we have

$$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$$

From the t-table, for $\alpha = 0.05$ and $v = n - 1 = 7$, we have the value of t as:

$$t_{(n-1), \alpha/2} = t_{(7), 0.025} = 2.365.$$

You may observe in the t-table of the References that when the sample size is greater than 30 ($n > 30$) then all values of variate t are not given in this table so for convenience when n is sufficiently large (> 30) then we know that almost all the distributions are very closely approximated by a normal distribution. Thus, in this case t-distribution is also approximated a normal distribution. So the variate

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim N(0, 1) \quad \dots(5)$$

also follows the normal distribution with mean 0 and variance unity. Therefore, when population variance is unknown and the sample size is large then the $(1-\alpha)$ 100% confidence interval for population mean may be obtained by using the same procedure as we have followed in the case when σ^2 is known by taking S^2 in place of σ^2 which is given as

$$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right] \quad \dots (6)$$

and corresponding limits are given by

$$\bar{X} \mp z_{\alpha/2} \frac{S}{\sqrt{n}} \quad \dots (7)$$

The following example will explain the application of t-distribution.

Example 5: It is known that the average weight of students of a Study Centre of IGNOU follows a normal distribution. To estimate the average weight, a sample of 10 students is taken from this Study Centre and measured their weights (in kg) which are given below:

48, 50, 62, 75, 80, 60, 70, 56, 52, 77

Compute the 95% confidence interval for the average weight of students of the Study Centre of IGNOU.

Solution: Since population variance is unknown, therefore, $(1-\alpha)$ 100% confidence limits for the average weight of students of Study Centre are given by

$$\bar{X} \mp t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}}$$

$$\text{where, } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Calculation for $\bar{X}$ and S:

S. No.	Weight (X)	$(X - \bar{X})$	$(X - \bar{X})^2$
1	48	-15	225
2	50	-13	169
3	62	-1	1
4	75	12	144
5	80	17	289
6	60	-3	9
7	70	7	49
8	56	-7	49
9	52	-11	121
10	77	14	196
Sum	$\sum X = 630$		$\sum (X - \bar{X})^2 = 1252$

From the above calculation, we have

$$\bar{X} = \frac{1}{n} \sum X = \frac{1}{10} \times 630 = 63$$

$$S^2 = \frac{1}{n-1} \sum (X - \bar{X})^2 = \frac{1}{9} \times 1252 = 139.11$$

$$\Rightarrow S = \sqrt{139.11} = 11.79$$

For 95% confidence interval, we have $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$. Also from t-table, we have, $t_{(n-1), \alpha/2} = t_{(9), 0.025} = 2.262$.

Thus, the 95% confidence limits are

$$\begin{aligned} \bar{X} \mp t_{(n-1), 0.025} \frac{S}{\sqrt{n}} &= 63 \mp 2.262 \times \frac{11.79}{\sqrt{10}} \\ &= 63 \mp 8.43 = 54.57 \text{ and } 71.43 \end{aligned}$$

Hence, the required 95% confidence interval for the average weight of students of the Study Centre of IGNOU is given by

$$[54.57, 71.43]$$

Example 6: The mean life of 100 electric bulbs produced by a company is 2550 hours with a standard deviation 54 hours. Find 95% confidence limits for population mean life of the electric bulbs produced by the company.

Solution: Here, we are given that

$$n = 100, \bar{X} = 2550, S = 54$$

Since population variance is unknown and the sample size is large (> 30), therefore, we can use $(1 - \alpha)$ 100 % confidence limits for population mean which are given by

$$\bar{X} \mp z_{\alpha/2} \frac{S}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z. For 95% confidence limits, we have $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and for $\alpha = 0.05$, we have $z_{\alpha/2} = z_{0.025} = 1.96$.

Thus, the 95% confidence limits for mean life of electric bulbs are

$$\begin{aligned} \bar{X} \mp 1.96 \frac{S}{\sqrt{n}} &= 2550 \mp 1.96 \frac{54}{\sqrt{100}} \\ &= 2550 \mp 1.96 \times 5.4 \\ &= 2550 \mp 10.58 = 2539.42 \text{ and } 2560.58 \end{aligned}$$

9.7.3 Confidence Interval for Difference of Two Population Means when Samples are Independent and Population Variances are Known

Let $X_1, X_2, \dots, X_{n_1}$ be a random sample of size n_1 taken from normal population-I $N(\mu_1, \sigma_1^2)$ and $Y_1, Y_2, \dots, Y_{n_2}$ be another independent random sample of size n_2 taken from normal population-II $N(\mu_2, \sigma_2^2)$. Let $\bar{X}$ and $\bar{Y}$ be the sample means

of random samples $X_1, X_2, \dots, X_{n_1}$ and $Y_1, Y_2, \dots, Y_{n_2}$ respectively then we know that sample means $\bar{X}$ and $\bar{Y}$ are normally distributed, that is,

$$\bar{X} \sim N(\mu_1, \sigma_1^2/n_1) \text{ and } \bar{Y} \sim N(\mu_2, \sigma_2^2/n_2)$$

Also, by the property of normal distribution, we have

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

And the variate

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

is normally distributed with mean 0 and variance unity. Since the distribution of Z is independent of the parameters so it can be taken as a pivotal quantity.

The required $(1 - \alpha)$ 100% confidence interval for $(\mu_1 - \mu_2)$ is

$$\left[(\bar{X} - \bar{Y}) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, (\bar{X} - \bar{Y}) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right] \quad \dots (8)$$

and corresponding limits are given by

$$(\bar{X} - \bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad \dots (9)$$

Note 6: When $\bar{Y}$ is greater than $\bar{X}$ then we take $(\bar{Y} - \bar{X})$ in place of $(\bar{X} - \bar{Y})$.

The following example will make you user friendly with the way how the above concepts can be used in a practical situation:

Example 7: A company manufacturing two types of bulbs and product manager tested a random sample of 50 bulbs of type I and 60 bulbs of type II and found the following information:

	Mean life (hrs.)	Population Standard Deviation (hrs.)
Type I	1300	41
Type II	1280	46

Obtain the 95% confidence interval for the difference of average life of two types of bulbs assuming that the distributions of average lives of two types of bulbs follow the normal distribution.

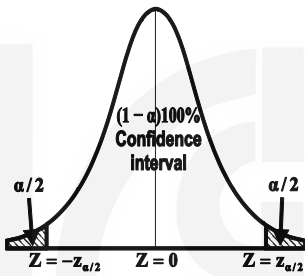
Solution: Here, we are given that

$$n_1 = 50, \quad \bar{X} = 1300, \quad \sigma_1 = 41$$

$$n_2 = 60, \quad \bar{Y} = 1280, \quad \sigma_2 = 46$$

Since population standard deviations, i.e. population variances of both the populations, are known so we use $(1 - \alpha)$ 100% confidence limits for the

Fig. 9.6



difference of population mean when population variances are known which are given by

$$(\bar{X} - \bar{Y}) \mp z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

For 95% confidence interval, $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$. Also $z_{\alpha/2} = 1.96$.

Thus, the 95% confidence limits for the difference of average lives of two types of bulbs are

$$\begin{aligned} (\bar{X} - \bar{Y}) \mp z_{0.025} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} &= (1300 - 1280) \mp 1.96 \sqrt{\frac{(41)^2}{50} + \frac{(46)^2}{60}} \\ &= 20 \mp 1.96 \sqrt{33.62 + 35.27} \\ &= 20 \mp 1.96 \times 8.3 \\ &= 20 \mp 16.27 \\ &= 3.73 \text{ and } 36.27 \end{aligned}$$

Hence, the required 95% confidence interval is

$$[3.73, 36.27]$$

9.7.4 Confidence Interval for Difference of Two Population Means when Population Variances are Unknown

Let $X_1, X_2, \dots, X_{n_1}$ be a random sample of size n_1 taken from normal population-I with mean μ_1 and unknown variance σ_1^2 and also let $Y_1, Y_2, \dots, Y_{n_2}$ be another independent random sample of size n_2 taken from a second normal population with mean μ_1 and unknown variance σ_2^2 . Then, keeping the scope of the course in mind, let us discuss the following two cases:

Case I: When $\sigma_1^2 = \sigma_2^2 = \sigma^2$

In this case, σ^2 is estimated by pooled sample variance S_p^2 where,

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} [(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2]$$

$$\text{where, } S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2 \text{ and } S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

This can also be written as

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum_{i=1}^{n_1} (X_i - \bar{X})^2 + \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2 \right]$$

In this situation, the variate

$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1+n_2-2)}$$

follows t-distribution with $(n_1 + n_2 - 2)$ degrees of freedom. Since the distribution of variate t is independent of the parameters, therefore, it can be taken as a pivotal quantity.

Hence, the required $(1-\alpha)$ 100% confidence interval is

$$\left[(\bar{X} - \bar{Y}) - t_{(n_1+n_2-2), \alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, (\bar{X} - \bar{Y}) + t_{(n_1+n_2-2), \alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right] \quad \dots (10)$$

and the corresponding limits are given by

$$(\bar{X} - \bar{Y}) \pm t_{(n_1+n_2-2), \alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \quad \dots (11)$$

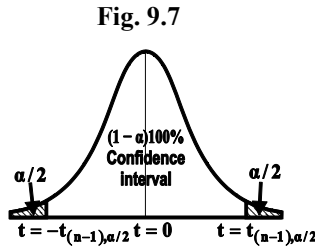


Fig. 9.7

Note 7: When $\bar{Y}$ is greater than $\bar{X}$ then we take $(\bar{Y} - \bar{X})$ in place of $(\bar{X} - \bar{Y})$.

Case II: When $\sigma_1^2 \neq \sigma_2^2$

In this case, σ_1^2 & σ_2^2 are estimated by the values of their estimators S_1^2 & S_2^2 respectively, where

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2 \quad \& \quad S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

In this situation, the pivotal quantity is difficult to obtain. But we know by the central limit theorem that for large sample sizes n_1 and n_2 (> 30) the variate

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim N(0, 1)$$

follows normal distribution with mean 0 and variance unity, so $(1-\alpha)$ 100% confidence interval for difference of population means may be obtained by the procedure as same as we have followed in the case when σ_1^2 & σ_2^2 are known by taking S_1^2 & S_2^2 in place of σ_1^2 & σ_2^2 respectively. Then $(1-\alpha)$ 100% confidence interval for $(\mu_1 - \mu_2)$ is given by

$$\left[(\bar{X} - \bar{Y}) - z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}, (\bar{X} - \bar{Y}) + z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \right] \quad \dots (12)$$

and corresponding limits are given by

$$(\bar{X} - \bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \quad \dots (13)$$

Note 8: When $\bar{Y}$ is greater than $\bar{X}$ then we take $(\bar{Y} - \bar{X})$ in place of $(\bar{X} - \bar{Y})$.

The following examples will make you user friendly with the way how above concepts can be used in a practical situation:

Example 8: A random sample of 120 colonies was taken from city A and the average population per colony was found to be 540 with a standard deviation 18. Another sample of 150 colonies taken from another city B gave an average population 460 per colony with a standard deviation 15. Find 95% confidence interval for the difference of two population averages by assuming that both average populations follow the normal distributions.

Solution: Here, we are given that

$$n_1 = 120, \quad \bar{X} = 540, \quad S_1 = 18$$

$$n_2 = 150, \quad \bar{Y} = 460, \quad S_2 = 15$$

Since population variances are unknown, therefore, we use $(1-\alpha)$ 100% confidence limits for the difference of population mean when population variances are unknown which are given by

$$(\bar{X} - \bar{Y}) \mp z_{0.025} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

For 95% confidence interval, $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$. Also $z_{\alpha/2} = 1.96$.

So, 95% confidence limits for the difference of two population averages are given by

$$\begin{aligned} (540 - 460) \mp 1.96 \sqrt{\frac{(18)^2}{120} + \frac{(15)^2}{150}} \\ = 80 \mp 1.96 \sqrt{\frac{324}{120} + \frac{225}{150}} \\ = 80 \mp 1.96 \sqrt{2.70 + 1.50} \\ = 80 \mp 1.96 \times 2.05 \\ = 80 \mp 4.02 \\ = 75.98 \text{ and } 84.02 \end{aligned}$$

Hence, the required 95% confidence interval for $(\mu_1 - \mu_2)$ is

$$[75.98, 84.02]$$

9.8 CONFIDENCE INTERVAL FOR POPULATION PROPORTION

In last section, we have discussed the confidence interval for population mean. But in many real-world situations, in business and other areas, the data are collected in the form of counts or the collected data classified into two categories or groups according to an attribute or characteristic under study. Generally, such types of data are considered in terms of the proportion of elements/individuals/units/items that possess or not possess a given characteristic or attribute. For example, the proportion of female in the

population, proportion of diabetes patients in a hospital, proportion of Science books in a library, proportion of defective articles in a lot, etc.

In such situations, we deal population proportion instead of population mean and one may want to obtain the confidence interval for population proportion. For example, a sociologist may want to know the confidence interval for proportion of female in the population of a state. A doctor may want to know the confidence interval for proportion of diabetes patients in a hospital, a product manager may want to know the confidence interval for proportion of defective articles in a lot, etc.

Generally, population proportion is estimated by sample proportion.

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from a population with population proportion P . Also, let X denotes the number of observations or elements that possess a certain attribute (successes) out of n observations of the sample then sample proportion p can be defined as

$$p = \frac{X}{n} \leq 1$$

The mean and variance of the sampling distribution of sample proportion are

$$E(p) = P \text{ and } \text{Var}(p) = \frac{PQ}{n}$$

where, $Q = 1 - P$.

But sample proportion is generally considered for a large sample so if the sample size is sufficiently large, such that $np > 5$ and $nq > 5$ then by central limit theorem, the sampling distribution of sample proportion p is approximately normally distributed with mean P and variance PQ/n .

Hence, $(1-\alpha)$ 100% confidence interval for population proportion is given by

$$\left[p - z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}, p + z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \right] \dots (14)$$

Therefore, corresponding confidence limits are

$$p \mp z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \dots (15)$$

This also defines the *Confidence Interval for the Bernoulli distribution*.

The following example will explain the application of the above discussion.

Example 9: A sample of 200 voters is chosen at random from all voters in a given city. 60% of them were in favour of a particular candidate. If a large number of voters cast their votes then find 99% and 95% confidence intervals for the proportion of voters in favour of a particular candidate.

Solution: Here, we are given

$$n = 200, p = 0.60$$

First, we check the condition of normality as

$$\therefore np = 200 \times 0.60 = 120 > 5 \text{ and } nq = 200 \times (1 - 0.60) = 200 \times 0.40 = 80 > 5$$

so $(1-\alpha)$ 100% confidence limits for the proportion are given by

$$p \mp z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

For 99% confidence interval, we have $1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$. For $\alpha = 0.01$, we have $z_{0.005} = 2.58$ and for $\alpha = 0.05$, $z_{0.025} = 1.96$.

Therefore, 99% confidence limits of voters in favour of a particular candidate are

$$\begin{aligned} p \pm z_{0.005} \sqrt{\frac{p(1-p)}{n}} &= 0.60 \pm 2.58 \times \sqrt{\frac{0.60 \times 0.40}{200}} \\ &= 0.60 \pm 2.58 \times 0.03 \\ &= 0.60 \pm 0.08 = 0.52 \text{ and } 0.68 \end{aligned}$$

Hence, the required 99% confidence interval for the proportion of voters in favour of a particular candidate is given by

$$[0.52, 0.68]$$

Similarly, 95% confidence limits are given by

$$\begin{aligned} p \pm z_{0.025} \sqrt{\frac{p(1-p)}{n}} &= 0.60 \pm 1.96 \times 0.03 \\ &= 0.60 \pm 0.06 = 0.54 \text{ and } 0.66 \end{aligned}$$

Hence, 95% confidence interval for the proportion of voters in favour of a particular candidate is given by

$$[0.54, 0.66]$$

Check Your Progress 3

1. Certain refined oil is packed in tins holding 15 kg each. The filling machine maintains this but has a standard deviation 0.30 kg. A sample of 200 tins is taken from the production line. If the sample mean is 15.25 kg then find the 95% confidence interval for the average weight of oil tins.
.....
.....
2. Sample mean of weights (in kg) of 150 students of IGNOU is found to be 65 kg with standard deviation 12 kg. Find the 95% confidence limits in which the average weight of all students of IGNOU expected to lie.
.....
.....
3. A sample of the height of 2500 Nepalis has a mean of 68.50 inches and a standard deviation of 2.52 inches, while the sample of the height of 1600 Indians has a mean 70.25 inches and a standard deviation 2.58 inches. Find a 90% interval estimate for the difference of mean heights of both the countries by assuming that the height of the persons of both the countries is normally distributed.
.....
.....
.....
.....

4. A random sample of 400 apples was taken from a large consignment and 80 were found to be bad. Obtain the 99% confidence limits for the proportion of bad apples in the consignment.

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9.9 DETERMINATION OF SAMPLE SIZE

To estimate some population parameter we have to draw a random sample. A natural question that may arise in your mind: “how large should my sample be?” And this is a very important question that is commonly asked. From a statistical point of view, the best answer to this question is “take as large a sample as you can afford. That is, if possible ‘sample’ the entire population that means study all units of the population under study because by taking all units of the population we will have all the information about the parameter and we will know the exact value of the population parameter which is better than any estimate of that parameter. Generally, this is impractical to take the entire population to be sampled due to economic constraints, time constraints and other limitations. So the answer, “take as large a sample as you can afford is best if we ignore all costs because, as you have studied in this unit, the larger the sample size, the smaller the standard error of the statistic, which means less uncertainty.

When the resources in terms of money and time are limited then the question is “how to find the minimum sample size which will satisfy some precision requirements”. In such cases, we should require first the answers to the following three questions about his / her requirement about the survey:

1. How close do you want your sample estimate to be to the unknown parameter? That means what should be the allowable difference between the sample estimate and the true value of the population parameter. This difference is known as sampling error or margin of error and represented by E .
2. The next question is, what do you want the confidence level to be so that the difference between the estimate and the parameter is less than or equal to E ? That is, 90%, 95%, 99%, etc.
3. The last question is what is the population variance or population proportion as may be the case?

When we have the answers of these three questions then we will get an answer of the minimum required sample size.

In this section, we will describe the procedure of determining of sample size for estimating population mean and population proportion.

Determination of Minimum Sample Size for Estimating Population Mean

For determining the minimum sample size for estimating the population mean we use a confidence interval. Since the confidence interval for the population mean depends upon the nature of the population and population variance (σ^2) is known or unknown, the following cases may arise:

Case I: Population is normal and population variance (σ^2) is known

If σ^2 is known and form of the population is normal then we know that $(1 - \alpha)100\%$ confidence interval for population mean is given by

$$P\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

$$\text{Also } P\left[-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha \quad \dots (16)$$

Since the normal distribution is symmetric, we can concentrate on the right-hand equality, that is:

$$\bar{X} - \mu \leq z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \dots (17)$$

This inequality implies that the largest value that the difference $\bar{X} - \mu$ can assume is $z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$.

Also the difference between the estimator (sample mean $\bar{X}$) and the population parameter (population mean μ) is called the sampling error, so

$$E = \bar{X} - \mu = z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \dots (18)$$

Solving this equation for n, we have

$$n = \frac{z_{\alpha/2}^2 \sigma^2}{E^2} \quad \dots (19)$$

When population size N is finite and sampling is to be done without replacement then finite population correction $\sqrt{\frac{N-n}{N-1}}$ is required so equation (20) becomes

$$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

which gives

$$n = \frac{N z_{\alpha/2}^2 \sigma^2}{E^2 (N-1) + z_{\alpha/2}^2 \sigma^2} \quad \dots (20)$$

If the finite population correction is ignored then equation (20) is reduced to equation (19).

Case II: Population is non-normal and population variance (σ^2) is known

If the population is not assumed to be normal and the population variance σ^2 is known then by central limit theorem we know that sampling distribution of mean is approximately normally distributed as sample size increases. So the above method can be used determining minimum sample size. Once the required sample size is obtained, we can check to see it that sample size greater than 30 and if it does, we may be confident that our method of solution was appropriate.

Case III: Population is normal or non-normal and population variance (σ^2) is unknown

In this case, we use value of sample variance (S^2) to estimate the population variance but S^2 is also calculated from a sample and we have not yet taken a sample. So in this case, determination of sample size is not directly obtained.

The most frequent methods for estimating σ^2 are as follows:

1. A pilot or preliminary sample may be drawn from the population under study and the variance computed from this sample may be used as an estimate of σ^2 .
2. The variance of previous or similar studies may be used to estimate σ^2 .

3. If the population is assumed to be normal then we may use the fact that the range is approximately equal to six times of standard deviation i.e. $\sigma = R/6$. This approximate require only knowledge of the largest and smallest value of the variable under study because range may be defined as

$$R = \text{largest value} - \text{smallest value}$$

Determination of Minimum Sample Size for Estimating Population Proportion

The method of determination of the minimum sample size for estimating population proportion is similar to that described in estimating the population mean. So the formula for minimum sample size is given by

$$n = \frac{z_{\alpha/2}^2 P(1-P)}{E^2} \quad \dots (21)$$

where P is the population proportion and $E = p - P$ is the sampling error or margin of error.

When the population is finite of size N and sampling is to be done without replacement then finite population correction $\sqrt{\frac{N-n}{N-1}}$ is required so

$$n = \frac{N z_{\alpha/2}^2 P(1-P)}{E^2 (N-1) + z_{\alpha/2}^2 P(1-P)} \quad \dots (22)$$

Here, the sample size is dependent upon the population proportion and it is generally unknown so the most frequent methods for estimating P are as follows:

1. A pilot or preliminary sample may be drawn from the population under study and the sample proportion computed from this sample may be used as an estimate of P .
2. The proportion of previous or similar studies may be used to estimate P .
3. Absence of any information, we may use $P = 0.5$.

Now, it is time to do some examples based on the determination of sample size.

Example 10: A hospital administrator wishes to estimate the mean weight of babies born in her hospital. How large a sample of birth records should be taken if she wants a 99 per cent confidence that the estimate is within the range of 0.4 pounds? Assume that a reasonable estimate of σ is 0.5 pounds.

Solution: Here, we are given that

$$E = \text{margin of error} = 0.4, \text{ confidence level} = 0.99 \text{ and } \sigma = 0.5$$

Also for 99% confidence, $1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$ and $\alpha/2 = 0.005$, we have

$$z_{\alpha/2} = z_{0.005} = 2.58.$$

The hospital administrator wishes to obtain the minimum sample size for estimating the mean weight of babies born in her hospital so we have

$$n = \frac{z_{\alpha/2}^2 \sigma^2}{E^2} = \frac{(2.58)^2 (0.5)^2}{(0.4)^2} = 10.40 \sim 11$$

Hence, the hospital administrator should obtain a random sample of at least 11 babies.

Example 11: The manufacturers of a car want to estimate the proportion of people who are interested in a certain model. The company wants to know the population proportion, P , to within 0.05 with 95% confidence. Current company's records indicate that the proportion P is around 0.20. What is the minimum required sample size for this survey?

Solution: Here, we are given that

$$E = \text{margin of error} = 0.05, \text{ confidence level} = 0.95 \text{ and } P = 0.20$$

Also for 95% confidence, $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$, we have:

$$z_{\alpha/2} = z_{0.025} = 1.96.$$

The manufacturers of a car interested to obtain the minimum sample size for estimating the population proportion so the required formula is given below

$$n = \frac{z_{\alpha/2}^2 P(1-P)}{E^2} = \frac{(1.96)^2 (0.20)(0.80)}{(0.05)^2} = 245.86 \sim 246$$

Hence, the company should require a random sample of at least 246 people.

Check Your Progress 4

- 1) A survey is planned to determine the average annual family medical expenses of employees of a large company. The management of the company wishes to be 95% confident that the sample average is correct to within ± 100 Rs of the true average family expenses. A pilot study indicates that the standard deviation can be estimated as 400 Rs. How large a sample size is necessary?

.....

- 2) The manager of a bank in a small city would like to determine the proportion of the depositors per week. The manager wants to be 95% confident of being correct to within ± 0.10 of the true proportion of depositors per week. A guess is that the parentage of such depositors is about 8%, what sample size is needed?

.....

We now end this unit by giving a summary of what we have covered in it.

9.10 SUMMARY

In this unit, we have covered the following points:

1. The parameter space and the basic characteristics of an estimator.
2. Unbiasedness, consistency, efficiency and sufficiency of an estimator
3. The point estimation.
4. The method of maximum likelihood and its properties.
5. The interval estimation.
6. The method of obtaining confidence interval for population mean of a normal population when variance is known and unknown

7. The method of construction of confidence interval for difference of means of two normal populations when variances are known and unknown.
8. The concept of the shortest interval.
9. Determination of sample size.

9.11 SOLUTIONS / ANSWERS

Check Your Progress 1

- 1) Refer to Section 9.3.
- 2)
 - (i) Here, investigator estimates the average income Rs. 1.5 lakh i.e. he/she estimate average income with a single value, therefore, the investigator used the point estimation technique.
 - (ii) The product manager estimates the average life of electric bulbs with the help of two values 800 hours and 1000 hours, therefore, the product manager used the interval estimation technique.
 - (iii) A pathologist estimates the mean time required completing a certain analysis with the help of two values 30 minutes and 45 minutes, therefore, he/she used the interval estimation technique.

- 3) Refer to Section 9.4.

Check Your Progress 2

- 1) Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from $B(n, p)$ whose pmf is given by

$$P[X = x] = {}^nC_x p^x q^{n-x}; \quad x = 0, 1, \dots, n \text{ \& } q = 1 - p$$

ML estimator for parameter p is $\bar{X}/n$.

- 2) Refer to Section 9.5.
- 3) We know that the length of confidence interval $[T_1, T_2]$ is given by

$$L = T_2 - T_1$$

Therefore, in our case we have

- (i) $L = 3.0 - (-1.65) = 4.65$
 - (ii) $L = 2.70 - (-1.68) = 4.38$
 - (iii) $L = 2.54 - (-1.70) = 4.24$
 - (iv) $L = 1.96 - (-1.96) = 3.92$
- 4) We know that if we have confidence interval for parameter θ is

$$P[T_1 \leq \theta \leq T_2] = 1 - \alpha$$

then

$$\text{Lower confidence limit (LCL)} = T_1$$

$$\text{Upper confidence limit (UCL)} = T_2$$

$$\text{Confidence coefficient (CC)} = 1 - \alpha$$

Therefore, in our cases, we have

- (i) LCL = 0, UCL = 1.5 & CC = 0.90
- (ii) LCL = -1, UCL = 2 & CC = 0.95
- (iii) LCL = -2, UCL = 2 & CC = 0.98
- (iv) LCL = -2.5, UCL = 2.5 & CC = 0.99

Check Your Progress 3

- 1) Here, we are given that

$$n = 200, \sigma = 0.30, \bar{X} = 15.25$$

Since population standard deviation or variance is known so we use. (1- α) 100% confidence limits for population mean when population variance is known which are given by

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Where $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z. For 95% confidence interval, we have $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and for $\alpha = 0.05$, we have, $z_{\alpha/2} = z_{0.025} = 1.96$.

Thus, the 95% confidence limits for the average weight of oil tins are

$$\begin{aligned} \bar{X} \pm 1.96 \frac{\sigma}{\sqrt{n}} &= 15.25 \pm 1.96 \times \frac{0.30}{\sqrt{200}} \\ &= 15.25 \pm 0.04 = 15.21 \text{ and } 15.29 \end{aligned}$$

- 2) Here, we are given that

$$n = 150, \bar{X} = 65, S = 12$$

Since population variance is unknown and the sample size is large sample,

so we use (1- α) 100% confidence limits for population mean when population variance is unknown which are given by

$$\bar{X} \pm z_{\alpha/2} \frac{S}{\sqrt{n}}$$

For 95% confidence interval, we have $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and for $\alpha = 0.05$, we have, $z_{\alpha/2} = z_{0.025} = 1.96$.

Thus, the 95% confidence limits in which the average weight of all students of a Study centre of IGNOU expected to lie are given by

$$\begin{aligned} \bar{X} \pm 1.96 \frac{S}{\sqrt{n}} &= 65 \pm 1.96 \times \frac{12}{\sqrt{150}} \\ &= 65 \pm 1.92 = 63.08 \text{ and } 66.92 \end{aligned}$$

Hence, required confidence limits are

$$63.08 \text{ and } 66.92.$$

$$n = 400, X = 80$$

$$p = \frac{X}{n} = \frac{80}{400} = \frac{1}{5} = 0.20$$

$\therefore np = 400 \times 0.20 = 80 > 5$ and $nq = 400 \times (1 - 0.20) = 400 \times 0.80 = 320 > 5$ so $(1-\alpha)100\%$ confidence limits for the proportion are given by

$$p \mp z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

For $\alpha = 0.01$, we have, $z_{\alpha/2} = z_{0.005} = 2.58$.

Therefore, 99% confidence limits are

$$\begin{aligned} & 0.20 \mp 2.58 \sqrt{\frac{0.20 \times 0.80}{400}} \\ & = 0.2 \mp 2.58 \times 0.02 \\ & = 0.2 \mp 0.05 = 0.15 \text{ and } 0.25 \end{aligned}$$

For 99% confidence interval
 $1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$
and $\alpha/2 = 0.005$.

3) Here, we are given that

$$n_1 = 2500, \quad \bar{X} = 68.50, \quad S_1 = 2.52$$

$$n_2 = 1600, \quad \bar{Y} = 70.25, \quad S_2 = 2.58$$

Since population variances are unknown, therefore, we use $(1-\alpha) 100\%$ confidence limits for the difference of two population means when population variances are unknown which are given by

$$(\bar{X} - \bar{Y}) \mp z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

For 90% confidence interval, $1 - \alpha = 0.90 \Rightarrow \alpha = 0.10$ and $\alpha/2 = 0.05$. Also $z_{\alpha/2} = 1.645$.

Since in this case, $\bar{Y} > \bar{X}$ therefore, we take $(\bar{Y} - \bar{X})$ in place of $(\bar{X} - \bar{Y})$.

Thus, 90% confidence limits are given by

$$\begin{aligned} & (\bar{Y} - \bar{X}) \mp z_{0.05} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \\ & = (70.25 - 68.50) \mp 1.645 \times \sqrt{\frac{(2.52)^2}{2500} + \frac{(2.58)^2}{1600}} \\ & = 1.75 \mp 1.645 \times \sqrt{\frac{6.35}{2500} + \frac{6.66}{1600}} \\ & = 1.75 \mp 1.645 \times \sqrt{0.0025 + 0.0042} \\ & = 1.75 \mp 1.645 \times 0.08 \\ & = 1.62 \text{ and } 1.88 \end{aligned}$$

Hence, the required 90% confidence interval is

$$[1.62, 1.88]$$

4) We have

$$n = 400, X = 80$$

$$p = \frac{X}{n} = \frac{80}{400} = \frac{1}{5} = 0.20$$

$\therefore np = 400 \times 0.20 = 80 > 5$ and $nq = 400 \times (1 - 0.20) = 400 \times 0.80 = 320 > 5$ so $(1-\alpha)100\%$ confidence limits for the proportion are given by

$$p \mp z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

For $\alpha = 0.01$, we have, $z_{\alpha/2} = z_{0.005} = 2.58$.

Therefore, 99% confidence limits are

For 99% confidence interval

$1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$
and $\alpha/2 = 0.005$.

$$\begin{aligned} &0.20 \mp 2.58 \sqrt{\frac{0.20 \times 0.80}{400}} \\ &= 0.2 \mp 2.58 \times 0.02 \\ &= 0.2 \mp 0.05 = 0.15 \text{ and } 0.25 \end{aligned}$$

Check Your Progress 4

1) Here, we are given that

$E = \text{margin of error} = 100$, confidence level = 0.95 and $\sigma = 400$

Also for 95% confidence, $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$, we have $z_{\alpha/2} = z_{0.025} = 1.96$.

The management of the company wishes to obtain the minimum sample size for estimating the average annual family medical expenses of employees of the company so the required formula is given below

$$n = \frac{z_{\alpha/2}^2 \sigma^2}{E^2} = \frac{(1.96)^2 (400)^2}{(100)^2} = 61.47 \sim 62$$

Hence, the management of the company should require a random sample of at least 62 employees.

2) Here, we are given that

$E = \text{margin of error} = 0.1$, confidence level = 0.95 and $P = 0.08$

Also for 95% confidence, $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$, we have $z_{\alpha/2} = z_{0.025} = 1.96$.

The manager wants to obtain the minimum sample size for determining the proportion so the required formula is given below

$$n = \frac{z_{\alpha/2}^2 P(1-P)}{E^2} = \frac{(1.96)^2 (0.08)(0.92)}{(0.1)^2} = 28.27 \sim 29$$

Hence, the manager should require a random sample of at least 29 depositors.

9.12 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 2: Estimation UNIT 5: Introduction to Estimation (for Sections 9.2 and 9.3 and related portions in Sections 9.0 , 9.1, 9.10, 9.11 and Check Your Progress questions)
2. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 2: Estimation UNIT 6: Point Estimation (for Sections 9.4 and 9.5 and related portions in Sections 9.0 , 9.1, 9.10, 9.11 and Check Your Progress questions)
3. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 2: Estimation UNIT 7: Interval Estimation for One Population (for Sections 9.6 to 9.8 and related portions in Sections 9.0 , 9.1, 9.10, 9.11 and Check Your Progress questions)
4. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 2: Estimation UNIT 8: Interval Estimation for Two Population (for Section 9.7 and related portions in Sections 9.0 , 9.1, 9.10, 9.11 and Check Your Progress questions)

Reference Book

5. “Introduction to Probability and Statistics for Engineers and Scientists”, 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004

UNIT 10 HYPOTHESIS TESTING

Structure

- 10.0 Introduction
- 10.1 Objectives
- 10.2 Hypothesis
- 10.3 Key Concepts for Hypothesis Testing
 - 10.3.1 Critical Region
 - 10.3.2 Type-I and Type-II Errors
 - 10.3.3 Level of Significance
 - 10.3.4 One-Tailed and Two-Tailed Tests
 - 10.3.5 Concept of p-Value
- 10.4 General Procedure of Testing a Hypothesis
- 10.5 Testing of Hypothesis for Population Mean using Z-Test
- 10.6 Testing of Hypothesis for Difference of Two Population Means using Z-Test
- 10.7 Testing of Hypothesis for Population Proportions using Z-Test
- 10.8 Testing of Hypothesis for Population Mean using t-Test
- 10.9 Summary
- 10.10 Solutions /Answers
- 10.11 References

10.0 INTRODUCTION

In the previous units of this course, we have discussed one part of statistical inference, that is, estimation and we have learnt how we estimate the unknown population parameter(s) by using **point estimation** and **interval estimation**. In this block, we will focus on the second part of statistical inference which is known as **testing of hypothesis**.

In our day-to-day life, we see different commercials and advertisements on television, newspapers, magazines, etc. such as

- (i) The refrigerator of a certain brand saves up to 20% electric bill,
- (ii) The motorcycle of a certain brand gives 60 km/liter mileage,
- (iii) A detergent of a certain brand produces the cleanest wash,
- (iv) Ninety-nine out of hundred dentists recommend brand A toothpaste for their patients to save the teeth against cavity, etc.

Now, the question may arise in our mind “can such types of claims be verified statistically?” Fortunately, in many cases, the answer is “yes”.

The technique of testing such type of claims or statements or assumptions is known as testing of hypothesis. The truth or falsity of a claim or statement is never known unless we examine the entire population. But practically it is not possible in most situations so we take a random sample from the population under study and use the information contained in this sample to take the decision whether a claim is true or false.

This Unit covers in detail the concepts of hypothesis testing. It discusses the hypothesis testing of the population mean and population proportions using the Z-test. Further, it discusses the testing of hypotheses using the t-test. This unit also covers a topic on testing of hypothesis for two population means.

10.1 OBJECTIVES

After reading this unit, you should be able to:

- define a hypothesis;
- formulate the null and alternative hypotheses;
- explain what we mean by type-I and type-II errors;
- explore the concept of critical region and level of significance;
- define one-tailed and two-tailed tests;
- concept of p-value;
- applying the Z-test for testing the hypothesis about the population mean and difference of two population means;
- applying the Z-test for testing the hypothesis about the population proportions; and
- describe testing of hypothesis for the population mean using t-test.

10.2 HYPOTHESIS

In our day-to-day life, we see different commercials and advertisements in television, newspapers, magazines, etc. and if someone may be interested to test such type of claims or statement then we come across the problem of testing of hypothesis. For example,

- (i) a customer of motorcycle wants to test whether the claim about the motorcycle of the certain brand gives the average mileage 60 km/liter is true or false,
- (ii) the businessman of banana wants to test whether the average weight of a banana of Kerala is more than 200 gm,
- (iii) a doctor wants to test whether a new medicine is more effective for controlling high blood pressure than old medicine,
- (iv) an economist wants to test whether the variability in incomes differ in two populations,
- (v) a psychologist wants to test whether the proportion of literates between two groups of people is the same, etc.

In all the cases discussed above, the decision-maker is interested in making inference about the population parameter(s). However, he/she is not interested in estimating the value of the parameter(s) but he/she is interested in testing a claim or statement or assumption about the value of population parameter(s). Such a claim or statement is postulated in terms of hypothesis.

In statistics, a hypothesis is a statement or a claim or an assumption about the value of a population parameter (e.g., mean, median, variance, proportion, etc.).

Similarly, in the case of two or more populations, a hypothesis is a comparative statement or a claim or an assumption about the values of population parameters. (e.g., means of two populations are equal, variance of one population is greater than other, etc.). The plural of a hypothesis is hypotheses.

In hypothesis testing problems, first of all, we should be identifying the claim or statement or assumption or hypothesis to be tested and write it in the words. Once

the claim has been identified then we write it in symbolical form if possible. As in the above examples,

- (i) Customer of the motorcycle may write the claim or postulate the hypothesis “the motorcycle of the certain brand gives the average mileage 60 km/liter.” Here, the concern is the **average** mileage of the motorcycle so let μ represents the average mileage then our hypothesis becomes $\mu = 60$ km/ liter.
- (ii) Similarly, the businessman of banana may write the statement or postulate the hypothesis “the average weight of a banana of Kerala is greater than 200 gm.” So our hypothesis becomes $\mu > 200$ gm.
- (iii) The doctor may write the claim or postulate the hypothesis “ the new medicine is more effective for controlling high blood pressure than old medicine.” Here, we are concerned with the **average** effect of the medicines so let μ_1 and μ_2 represent the average effect of new and old medicines respectively on controlling blood pressure, then our hypothesis becomes $\mu_1 > \mu_2$.
- (iv) The economist may write the statement or postulate the hypothesis “ the variability in incomes differ in two populations.” Here, our concern is the **variability** in income so let σ_1^2 and σ_2^2 represent the variability in incomes in two populations respectively then our hypothesis becomes $\sigma_1^2 \neq \sigma_2^2$.
- (v) The psychologist may write the statement or postulate the hypothesis “the proportion of literates between two groups of people is same.” Here, we are concerning the **proportion** of literates so let P_1 and P_2 represent the proportions of literates of two groups of people respectively then our hypothesis becomes $P_1 = P_2$ or $P_1 - P_2 = 0$.

The hypothesis is classified according to its nature and usage, as we will discuss next.

Null and Alternative Hypotheses

As we have discussed in the last page that in hypothesis testing problems, first of all, we identify the claim or statement to be tested and write it in symbolical form. After that, we write the complement or opposite of the claim or statement in symbolical form. In our example of motorcycle, the claim is $\mu = 60$ km/liter then its complement is $\mu \neq 60$ km/liter. In (ii) the claim is $\mu > 200$ gm then its complement is $\mu \leq 200$ gm. If the claim is $\mu < 200$ gm then its complement is $\mu \geq 200$ gm. The claim and its complement are formed in such a way that they cover all possibilities of the value of the population parameter.

Once the claim and its complement have been established then we decide of these two which is the null hypothesis and which is the alternative hypothesis. The thumb rule is that the statement containing equality is the null hypothesis. That is, a hypothesis which contains symbols $=$ or $\leq$ or $\geq$ is taken as a null hypothesis and a hypothesis which does not contain equality i.e. contains $\neq$ or $<$ or $>$ is taken as an alternative hypothesis. The null hypothesis is denoted by H_0 and the alternative hypothesis is denoted by H_1 or H_A .

In our example of motorcycle, the claim is $\mu = 60$ km/liter and its complement is $\mu \neq 60$ km/liter. Since claim $\mu = 60$ km/liter contains equality sign so we take it as a null hypothesis and complement $\mu \neq 60$ km/liter as an alternative hypothesis, that is,

$$H_0: \mu = 60 \text{ km/liter and } H_1: \mu \neq 60 \text{ km/liter}$$

In our second example of banana, the claim is $\mu > 200$ gm and its complement is $\mu \leq 200$ gm. Since complement $\mu \leq 200$ gm contains equality sign so we take complement as a null hypothesis and claim $\mu > 200$ gm as an alternative hypothesis, that is,

$$H_0: \mu \leq 200 \text{ gm and } H_1: \mu > 200 \text{ gm}$$

Formally, these hypotheses are defined as

The hypothesis which we wish to test is called the null hypothesis.

According to Prof. R.A. Fisher,

“A null hypothesis is a hypothesis which is tested for possible rejection under the assumption that it is true.”

The hypothesis which complements to the null hypothesis is called the alternative hypothesis.

Note 1: Some authors use equality sign (=) in the null hypothesis instead of $\geq$ and $\leq$ signs.

The alternative hypothesis has two types:

- (i) Two-sided (tailed) alternative hypothesis
- (ii) One-sided (tailed) alternative hypothesis

If an alternative hypothesis gives the alternate of the null hypothesis in both directions (less than and greater than) of the value of the parameter specified in null hypothesis then it is known as two-sided alternative hypothesis and if it gives an alternate only in one direction (less than or greater than) only then it is known as one-sided alternative hypothesis. For example, if our alternative hypothesis is $H_1: \theta \neq 60$ then it is a two-sided alternative hypothesis because it means that the value of parameter θ is greater than or less than 60. Similarly, if $H_1: \theta > 60$ then it is a right-sided alternative hypothesis because it means that the value of parameter θ is greater than 60 and if $H_1: \theta < 60$ then it is a left-sided alternative hypothesis because it means that the value of parameter θ is less than 60.

In the testing procedure, we assume that the null hypothesis is true until there is sufficient evidence to prove that it is false. Generally, a hypothesis is tested with the help of a sample so the evidence about the claim comes from a sample. If there is enough sample evidence to suggest that the null hypothesis is false then we reject the null hypothesis and support the alternative hypothesis. If the sample fails to provide us sufficient evidence against the null hypothesis we are not saying that the null hypothesis is true because here, we take the decision on the basis of a random sample which is a small part of the population. To say that a null hypothesis is true, we must study all observations of the population under study. For example, if someone wants to test that the person of India has two online accounts then to prove that this is true we must check all the persons of India whereas to prove that it is false we require a person he/ she has one online account or no online account. So we can only say that there is not enough evidence against the null hypothesis.

Note 2: When we assume that a null hypothesis is true then we are actually assuming that the population parameter is equal to the value in the claim. In our example of motorcycle, we assume that $\mu = 60$ km/liter whether the null hypothesis is $\mu = 60$ km/liter or $\mu \leq 60$ km/liter or $\mu \geq 60$ km/liter.

10.3 KEY CONCEPTS FOR HYPOTHESIS TESTING

After describing the hypothesis, let us discuss some basic concepts needed for hypothesis testing.

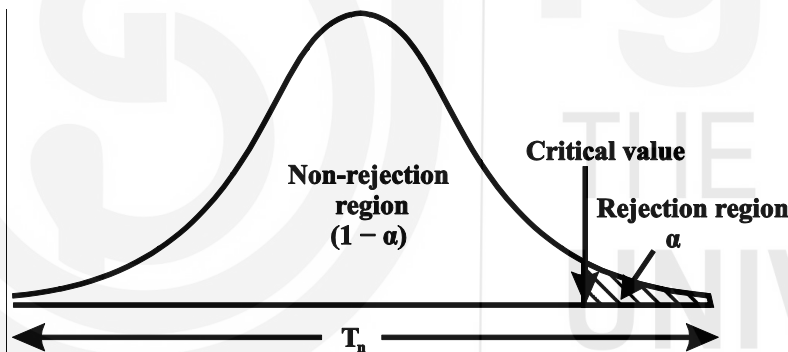
10.3.1 Critical Region

In order to test a hypothesis, the entire sample space is partitioned into two disjoint sub-spaces, say ω and $S - \omega = \bar{\omega}$. If calculated value of the test statistic lies in ω , then we reject the null hypothesis and if it lies in $\bar{\omega}$, then we do not reject the null hypothesis. The region ω is called a “**rejection region or critical region**” and the region $\bar{\omega}$ is called a “**non-rejection region**”. Therefore, we can say that

“A region in the sample space in which if the calculated value of the test statistic lies, we reject the null hypothesis then it is called critical region or rejection region.”

The rejection (critical) region lies in one-tail or two-tails on the probability curve of the sampling distribution of the test statistic its depends upon the alternative hypothesis. Therefore, three cases arise:

Case I: If the alternative hypothesis is right-sided such as $H_1: \theta > \theta_0$ or $H_1: \theta_1 > \theta_2$ then the entire critical or rejection region of size α lies on the right tail of the probability curve of the sampling distribution of the test statistic as shown in Fig. 10.1.



Critical value is a value or values that separate the region of rejection from the non-rejection region.

Fig. 10.1: Rejection Region for Right-sided Hypothesis

Case II: If the alternative hypothesis is left-sided such as $H_1: \theta < \theta_0$ or $H_1: \theta_1 < \theta_2$ then the entire critical or rejection region of size α lies on the left tail of the probability curve of the sampling distribution of the test statistic as shown in Fig. 10.2.

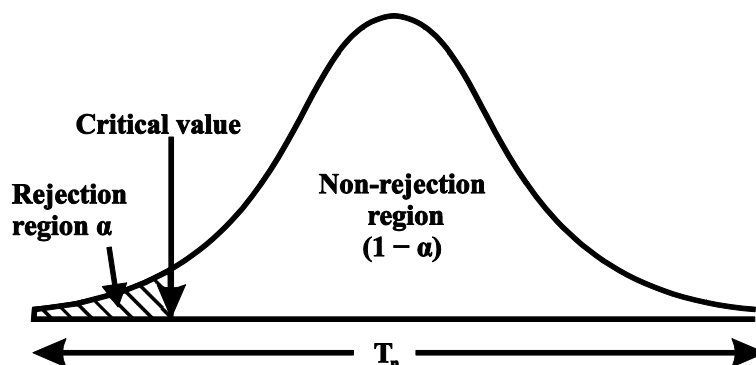


Fig. 10.2: Rejection Region for Left-sided Hypothesis

Case III: If the alternative hypothesis is two-sided such as $H_1: \theta \neq \theta_0$ or $H_1: \theta_1 \neq \theta_2$ then critical or rejection regions of size $\alpha/2$ lies on both tails

of the probability curve of the sampling distribution of the test statistic as shown in Fig. 10.3.

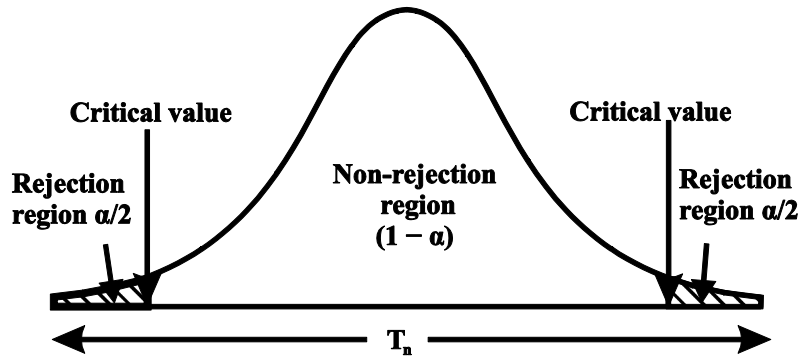


Fig. 10.3: Rejection Region for Two-sided Hypothesis

10.3.2 Type-I and Type-II Errors

In the last sub-section, we have discussed a rule that if the value of test statistic falls in rejection (critical) region then we reject the null hypothesis and if it falls in the non-rejection region then we do not reject the null hypothesis. A test statistic is calculated on the basis of observed sample observations. But a sample is a small part of the population about which decision is to be taken. A random sample may or may not be a good representative of the population.

A faulty sample misleads the inference (or conclusion) relating to the null hypothesis. For example, an engineer infers that a packet of screws is sub-standard when actually it is not. It is an error caused due to poor or inappropriate (faulty) sample. Similarly, a packet of screws may be inferred to be good when actually it is sub-standard. So we can commit two kinds of errors in testing a hypothesis which are summarised in Table 10.1 which is given below:

Table 10.1: Type of Errors

Decision	H_0 True	H_1 True
Reject H_0	Type-I Error	Correct Decision
Do not reject H_0	Correct Decision	Type-II Error

Let us take a situation where a patient suffering from high fever reaches to a doctor. And suppose the doctor formulates the null and alternative hypotheses as

H_0 : The patient is a malaria patient

H_1 : The patient is not a malaria patient

Then the following cases arise:

Case I: Suppose that the hypothesis H_0 is really true, that is, the patient actually a malaria patient and after observation, pathological and clinical examination, the doctor rejects H_0 , that is, he/ she declares him/ her a non-malaria-patient. It is not a correct decision and he/ she commits an error in the decision known as type-I error.

Case II: Suppose that the hypothesis H_0 is actually false, that is, the patient actually a non-malaria patient and after observation, the doctor rejects H_0 , that is, he/ she declares him/ her a non-malaria-patient. It is a correct decision.

Case III: Suppose that the hypothesis H_0 is really true, that is, the patient actually a malaria patient and after observation, the doctor does not reject H_0 , that is, he/ she declares him/ her a malaria-patient. It is a correct decision.

Case IV: Suppose that the hypothesis H_0 is actually false, that is, the patient actually a non-malaria patient and after observation, the doctor does not reject H_0 , that is, he/she declares him/her a malaria-patient. It is not a correct decision and he/she commits an error in the decision known as type-II error.

10.3.3 Level of Significance

So far in this unit, we have discussed the hypothesis, types of hypothesis, critical region and types of errors. In this section, we shall discuss a very useful concept “**level of significance**”, which plays an important role in decision making while testing a hypothesis.

The probability of type-I error is known as level of significance of a test. It is also called the **size of the test** or **size of the critical region**, denoted by α . Generally, it is pre-fixed as 5% or 1% level ($\alpha = 0.05$ or 0.01). As we have discussed in Section 10.3 that if calculated value of the test statistic lies in rejection (critical) region, then we reject the null hypothesis and if it lies in non-rejection region, then we do not reject the null hypothesis. Also, we note that when H_0 is rejected then automatically the alternative hypothesis H_1 is accepted. Now, one point of our discussion is how to decide critical value(s) or cut-off value(s) for a known test statistic.

If distribution of test statistic could be expressed into some well-known distributions like Z , χ^2 , t , F etc. then our problem will be solved and using the probability distribution of test statistic, we can find the cut-off value(s) that provides us critical area equal to 5% (or 1%).

Another viewpoint about the level of significance relates to the trueness of the conclusion. If H_0 does not reject at a level, say, $\alpha = 0.05$ (5% level) then a person will be confident that “concluding statement about H_0 ” is true with 95% assurance. But even then it may false with 5% chance. There is no cent-per cent assurance about the trueness of a statement made for H_0 .

As an example, if among 100 scientists, each draws a random sample and use the same test statistic to test the same hypothesis H_0 conducting the same experiment, then 95 of them will reach to the same conclusion about H_0 . But still, 5 of them may differ (i.e. against the earlier conclusion).

Similar argument can be made for, say, $\alpha = 0.01$ (=1%). It is like when H_0 is rejected at $\alpha = 0.01$ by a scientist, then out of 100 similar researchers who work together at the same time for the same problem, but with different random samples, 99 of them would reach to the same conclusion, however, one may differ.

10.3.4 One-Tailed and Two-Tailed Test

We have seen that rejection (critical) region lies at one-tail or two-tails on the probability curve of sampling distribution of the test statistic, depending on the form of alternative hypothesis. Similarly, the test of testing the null hypothesis also depends on the alternative hypothesis.

A test of testing the null hypothesis is said to be a two-tailed test if the alternative hypothesis is two-tailed whereas if the alternative hypothesis is one-tailed then a test of testing the null hypothesis is said to be a one-tailed test.

For example, if our null and alternative hypotheses are

$$H_0 : \theta = \theta_0 \text{ and } H_1 : \theta \neq \theta_0$$

then the test for testing the null hypothesis is two-tailed because the alternative hypothesis is two-tailed that means, the parameter θ can take value greater than θ_0 or less than θ_0 .

If the null and alternative hypotheses are

$$H_0 : \theta \leq \theta_0 \text{ and } H_1 : \theta > \theta_0$$

then the test for testing the null hypothesis is right-tailed because the alternative hypothesis is right-tailed.

Similarly, if the null and alternative hypotheses are

$$H_0 : \theta \geq \theta_0 \text{ and } H_1 : \theta < \theta_0$$

then the test for testing the null hypothesis is left-tailed because the alternative hypothesis is left-tailed.

The above discussion can be summarised in Table 10.2.

Table 10.2: Null and Alternative Hypotheses and Corresponding One-tailed and Two-tailed Tests

Null Hypothesis	Alternative Hypothesis	Types of Critical Region / Test
$H_0 : \theta = \theta_0$	$H_1 : \theta \neq \theta_0$	Two-tailed test having critical regions under both tails.
$H_0 : \theta \leq \theta_0$	$H_1 : \theta > \theta_0$	Right-tailed test having critical region under right tail only.
$H_0 : \theta \geq \theta_0$	$H_1 : \theta < \theta_0$	Left- tailed test having critical region under left tail only.

Let us do one example based on the type of tests.

Example 1: A company has replaced its original technology of producing electric bulbs by CFL technology. The company manager wants to compare the average life of bulbs manufactured by original technology and new technology CFL. Write appropriate null and alternate hypotheses. Also, say about one-tailed and two-tailed tests.

Solution: Suppose the average lives of original and CFL technology bulbs are denoted by μ_1 and μ_2 respectively.

If the company manager is interested just to know whether any significant difference exists in the average lifetime of two types of bulbs then null and alternative hypotheses will be:

$$H_0: \mu_1 = \mu_2 \quad [\text{average lives of two types of bulbs are same}]$$

$$H_1: \mu_1 \neq \mu_2 \quad [\text{average lives of two types of bulbs are different}]$$

Since the alternative hypothesis is two-tailed therefore corresponding test will be two-tailed.

If company manager is interested just to know whether average life of CFL is greater than original technology bulbs then our null and alternative hypotheses will be

$$H_0: \mu_1 \geq \mu_2$$

$$H_1: \mu_1 < \mu_2 \quad \left[\begin{array}{l} \text{average life of CFL technology bulbs} \\ \text{is greater than average life of original technology} \end{array} \right]$$

Since the alternative hypothesis is left-tailed therefore the corresponding test will be left-tailed test.

10.3.5 Concept of p-Value

Use of p-value is becoming more and more popular because of the following two reasons:

- most of the statistical software provides p-value rather than the critical value.
- p-value provides more information compared to critical value as far as rejection or does not rejection H_0 .

The first point listed above needs no explanation. But the second point lies in the heart of p-value and needs to explain more clearly. Moving in this direction, we note that in scientific applications one is not only interested simply in rejecting or not rejecting the null hypothesis but he/she is also interested to assess how strong the data has the evidence to reject H_0 . For example, for testing a hypothesis where we test the null hypothesis

$$H_0: \theta \leq 50 \text{ gm against } H_1: \theta > 50 \text{ gm}$$

To test the null hypothesis, we calculated the value of the test statistic as 2.78 and the critical value (z_α) at $\alpha = 0.01$ was $z_\alpha = 2.33$.

Since the calculated value of test statistic (= 2.78) is greater than the critical (tabulated) value (= 2.33), therefore, we reject the null hypothesis at 1% level of significance.

Now, if we reject the null hypothesis at this level (1%) surely we have to reject it at higher level because at $\alpha = 0.05$, $z_\alpha = 1.645$ and at $\alpha = 0.10$, $z_\alpha = 1.28$. However, the calculated value of test statistic is much higher than 1.645 and 1.28, therefore, the question arises “could the null hypothesis also be rejected at values of α smaller than 0.01?” The answer is “yes” and we can compute the smallest level of significance (α) at which a null hypothesis can be rejected. This smallest level of significance (α) is known as “p-value”.

The p-value is the smallest value of the level of significance (α) at which a null hypothesis can be rejected using the obtained value of the test statistic and can be defined as:

The p-value is the probability of obtaining a test statistic equal to or more extreme (in the direction of supporting H_1) than the actual value obtained when null hypothesis is true.

The p-value also depends on the type of test. If a test is one-tailed then the p-value is defined as:

For right-tailed test:

$$\text{p-value} = P[\text{Test Statistic (T)} \geq \text{observed value of the test statistic}]$$

For left-tailed test:

$$\text{p-value} = P[\text{Test Statistic (T)} \leq \text{observed value of the test statistic}]$$

If a test is two-tailed then the p-value is defined as:

For two-tailed test:

$$\text{p-value} = 2P[T \geq |\text{observed value of the test statistic}|]$$

The procedure of taking the decision about the null hypothesis on the basis of p-value:

To take the decision about the null hypothesis based on p-value, the p-value is compared with the level of significance (α) and if p-value is equal or less than

α then we reject the null hypothesis and if the p-value is greater than α we do not reject the null hypothesis.

The p-value for various tests can be obtained with the help of the tables given in the Appendix at the end of this course. But unless we are dealing with the standard normal distribution, the exact p-value is not obtained with the tables as mentioned above. But if we test our hypothesis with the help of computer packages, then these types of computer packages or software present the p-value as part of the output for each hypothesis testing procedure.

Check Your Progress 1

- 1) A company manufactures car tyres. The company claims that the average life of its tyres is 50000 miles. To test the claim of the company, formulate the null and alternative hypotheses.

.....
.....

- 2) Write the null and alternative hypotheses in cases (iii), (iv) and (v) of example given in Section 10.2.

.....
.....

- 3) If $H_0: \theta = 60$ and $H_1: \theta \neq 60$ then critical region lies in one-tail or two-tails.

.....
.....

- 4) If the probability of type-I error is 0.05 then what is the level of significance?

.....
.....

- 5) If we have null and alternative hypotheses as

$$H_0: \theta = \theta_0 \text{ and } H_1: \theta \neq \theta_0$$

then the corresponding test will be

- (i) left-tailed test (ii) right-tailed test (iii) both-tailed test

Write the correct option.

.....
.....

- 6) The test whether one-tailed or two-tailed depends on

- (i) Null hypothesis (H_0) (ii) Alternative hypothesis (H_1)

- (iii) Neither H_0 nor H_1 (iv) Both H_0 and H_1

.....
.....

10.4 GENERAL PROCEDURE OF TESTING A HYPOTHESIS

Testing of hypothesis is a huge demanded statistical tool by many discipline and professionals. It is a step by step procedure as you will see in the next three units through a large number of examples. The aim of this section just gives you the flavor of that sequence which involves the following steps:

Step I: First of all, we have to set up the null hypothesis H_0 and alternative hypothesis H_1 . Suppose, we want to test the hypothetical/ claimed/ assumed value θ_0 of parameter θ . So we can take the null and alternative hypotheses as

$$H_0 : \theta = \theta_0 \text{ and } H_1 : \theta \neq \theta_0 \quad [\text{for two-tailed test}]$$

$$\text{or} \quad \left. \begin{array}{l} H_0 : \theta \leq \theta_0 \text{ and } H_1 : \theta > \theta_0 \\ H_0 : \theta \geq \theta_0 \text{ and } H_1 : \theta < \theta_0 \end{array} \right\} \quad [\text{for one-tailed test}]$$

In case of comparing the same parameter of two populations of interest, say, θ_1 and θ_2 , then our null and alternative hypotheses would be

$$H_0 : \theta_1 = \theta_2 \text{ and } H_1 : \theta_1 \neq \theta_2 \quad [\text{for two-tailed test}]$$

$$\text{or} \quad \left. \begin{array}{l} H_0 : \theta_1 \leq \theta_2 \text{ and } H_1 : \theta_1 > \theta_2 \\ H_0 : \theta_1 \geq \theta_2 \text{ and } H_1 : \theta_1 < \theta_2 \end{array} \right\} \quad [\text{for one-tailed test}]$$

Step II: After setting the null and alternative hypotheses, we establish criteria for rejection or non-rejection of the null hypothesis, that is, decide the level of significance (α), at which we want to test our hypothesis. Generally, it is taken as 5% or 1% ($\alpha = 0.05$ or 0.01).

Step III: The third step is to choose an appropriate test statistic under H_0 for testing the null hypothesis, as given below:

$$\text{Test statistic} = \frac{\text{Statistic} - \text{Value of the parameter under } H_0}{\text{Standard error of statistic}}$$

After that, specify the sampling distribution of the test statistic preferably in the standard form like Z (standard normal), χ^2 , t, F or any other well-known in the literature.

Step IV: Calculate the value of the test statistic described in Step III on the basis of observed sample observations.

Step V: Obtain the critical (or cut-off) value(s) in the sampling distribution of the test statistic and construct rejection (critical) region of size α . Generally, critical values for various levels of significance are put in the form of a table for various standard sampling distributions of test statistic such as Z-table, χ^2 -table, t-table, etc. (See Appendix at the end of this course).

Step VI: After that, compare the calculated value of test statistic obtained from Step IV, with the critical value(s) obtained in Step V and locates the position of the calculated test statistic, that is, it lies in rejection region or non-rejection region.

Step VII: In testing of hypothesis ultimately we have to reach at a conclusion. It is done as explained below:

- (i) If the calculated value of test statistic lies in the rejection region at α level of significance then we reject the null hypothesis. It means that the sample data provide us sufficient evidence against the null hypothesis and there is a significant difference between hypothesized value and the observed value of the parameter.

- (ii) If the calculated value of test statistic lies in the non-rejection region at α level of significance then we do not reject the null hypothesis. It means that the sample data fails to provide us sufficient evidence against the null hypothesis and the difference between hypothesized value and the observed value of the parameter due to fluctuation of the sample.

Note 3: Nowadays, the decision about the null hypothesis is taken with the help of p-value. The concept of p-value is very important because computer packages and statistical software all provide p-value.

Now, with the help of an example, we explain the above general procedure.

Example 2: Suppose it is found that the average weight of potatoes was 50 gm and the standard deviation was 5.1 gm nearly 5 years ago. We want to test that due to advancement in agricultural technology, the average weight of potato has been increased. To test this, a random sample of 50 potatoes is taken and calculate the sample mean ($\bar{X}$) as 52 gm. Describe the procedure to carry out this test.

Solution: Here, we are given that

Specified value of population mean = $\mu_0 = 50$ gm,

Population standard deviation = $\sigma = 5.1$ gm,

Sample size = $n = 50$,

Sample mean = $\bar{X} = 52$ gm

To carry out the above test, we have to follow up the following steps:

Step I: First of all, we set up null and alternative hypotheses. Here, we want to test that the average weight of potato is increased. So our claim is “average weight of potato has increased” i.e. $\mu > 50$ and its complement is $\mu \leq 50$. Since complement contains equality sign so we can take the complement as the null hypothesis and claim as the alternative hypothesis, that is,

$$H_0: \mu \leq 50 \text{ gm and } H_1: \mu > 50 \text{ gm [Here, } \theta = \mu]$$

Since the alternative hypothesis is right-tailed, so our test is right-tailed.

Step II: After setting the null and alternative hypotheses, we fix the level of significance α . Suppose, $\alpha = 0.01$ ($= 1\%$ level).

Step III: Define a test statistic to test the null hypothesis as

$$\text{Test statistic} = \frac{\text{Statistic} - \text{Value of the parameter under } H_0}{\text{Standard error of statistic}}$$

$$T = \frac{\bar{X} - 50}{\sigma / \sqrt{n}}$$

Since the sample size is large ($n = 50 > 30$) so by central limit theorem the sampling distribution of test statistic approximately follows standard normal distribution, i.e. $T \sim N(0,1)$

Step IV: Calculate the value of test statistic on the basis of sample observations as

$$T = \frac{52 - 50}{5.1 / \sqrt{50}} = \frac{2}{0.72} = 2.78$$

Step V: Now, we find the critical value. The critical value or cut-off value for standard normal distribution is given in **Table I (Z-table)** in the

Appendix of this course. So from this table, the critical value for right-tailed test at $\alpha = 0.01$ is $z_\alpha = 2.33$.

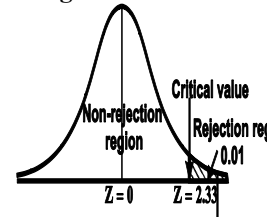
Step VI: Now, to take the decision about the null hypothesis, we compare the calculated value of the test statistic with the critical value.

Since the calculated value of test statistic ($= 2.78$) is greater than the critical value ($= 2.33$), that means the calculated value of test statistic lies in rejection region at 1% level of significance as shown in Fig. 10.4. So we reject the null hypothesis and support the alternative hypothesis. Since the alternative hypothesis is our claim, so we support the claim.

Thus, we conclude that the average weight of potato has increased.

Hypothesis Testing

Fig. 10.4



10.5 TESTING OF HYPOTHESIS FOR POPULATION MEAN USING Z-TEST

In the previous section, we have discussed the general procedure for hypothesis testing. Now we are discussing the Z-test for testing the hypothesis or claim about the population mean when the sample size is large. Let the population under study has mean μ and variance σ^2 , where μ is unknown and σ^2 may be known or unknown. We will consider both cases in this unit. For testing a hypothesis about population mean, we draw a random sample $X_1, X_2, \dots, X_n$ of size $n > 30$ from this population. As we know that for drawing the inference about the population mean, we generally use the sample mean and for the test statistic, we require the mean and standard error of sampling distribution of the statistic (mean). Here, we are considering a large sample so we know by the central limit theorem that sample mean is asymptotically normally distributed with mean μ and variance σ^2/n whether parent population is **normal or non-normal**. That is, if $\bar{X}$ is the sample mean of the random sample then

$$E(\bar{X}) = \mu, \text{ Var}(\bar{X}) = \frac{\sigma^2}{n} \quad \dots (1)$$

But we know that standard error = $\sqrt{\text{Variance}}$

$$\therefore SE(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}} \quad \dots (2)$$

Now, follow the same procedure as we have discussed in the previous section, that is, first of all, we have to set up null and alternative hypotheses. Since here we want to test the hypothesis about the population mean so we can take the null and alternative hypotheses as

$$\begin{aligned} & H_0 : \mu = \mu_0 \text{ and } H_1 : \mu \neq \mu_0 \text{ [for two-tailed test]} \left[\begin{array}{l} \text{Here, } \theta = \mu \text{ and } \theta_0 = \mu_0 \\ \text{if we compare it with} \\ \text{general procedure.} \end{array} \right] \\ \text{or} & \left. \begin{array}{l} H_0 : \mu \leq \mu_0 \text{ and } H_1 : \mu > \mu_0 \\ H_0 : \mu \geq \mu_0 \text{ and } H_1 : \mu < \mu_0 \end{array} \right\} \text{ [for one-tailed test]} \end{aligned}$$

When we assume that the null hypothesis is true then we are actually assuming that the population parameter is equal to the value in the null hypothesis. is $\mu = 60$ or $\mu \leq 60$ or $\mu \geq 60$.

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{\bar{X} - E(\bar{X})}{SE(\bar{X})}$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \quad \left[\begin{array}{l} \text{Using equations (1) and (2) and} \\ \text{under } H_0 \text{ we assume that } \mu = \mu_0. \end{array} \right]$$

The sampling distribution of the test statistic depends upon σ^2 that it is known or unknown. Therefore, two cases arise:

Case I: When σ^2 is known

In this case, the test statistic follows the normal distribution with mean 0 and variance unity when the sample size is large as the population under study is normal or non-normal. If the sample size is small then the test statistic Z follows the normal distribution only when the population under study is normal. Thus,

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Case II: When σ^2 is unknown

In this case, we estimate σ^2 by the value of sample variance (S^2) where,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Then the test statistic becomes $\frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ and follows the t-distribution

with $(n-1)$ degrees of freedom (df) as the sample size is large or small, provided the population under study is normal. But when the population under study is not normal and the sample size is large then this test statistic approximately follows a normal distribution with mean 0 and variance unity, that is,

$$Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim N(0, 1)$$

After that, we calculate the value of the test statistic as may be the case (σ^2 is known or unknown) and compare it with the critical value given in **Tables of Z-value given in the Appendix at the end of this course**, at the prefixed level of significance α . Take the decision about the null hypothesis as described in the previous section.

From the above discussion of testing of hypothesis about the population mean, we note the following point:

- (i) When σ^2 is known then we apply the Z-test as the population under study is normal or non-normal for the large sample. But when the sample size is small then we apply the Z-test only when the population under study is normal.
- (ii) When σ^2 is unknown then we apply the t-test only when the population under study is normal as the sample size is large or small. But when the assumption of normality is not fulfilled and the sample size is large then we can apply the Z-test.
- (iii) When the sample is small and σ^2 is known or unknown and the form of the population is not known then we apply the non-parametric test, which are not part of this course.

The following examples will help you to understand the procedure more clearly.

Example 3: A light bulb company claims that the 100-watt light bulb it sells has an average life of 1200 hours with a standard deviation of 100 hours. For testing the claim 50 new bulbs were selected randomly and allowed to burn out. The average lifetime of these bulbs was found to be 1180 hours. Is the company's claim true at 5% level of significance?

Solution: Here, we are given that

Specified value of population mean = $\mu_0 = 1200$ hours,

Population standard deviation = $\sigma = 100$ hours,

Sample size = $n = 50$

Sample mean = $\bar{X} = 1180$ hours.

In this example, the population parameter being tested is the population mean i.e. the average life of a bulb (μ) and we want to test the company's claim that the average life of a bulb is 1200 hours. So our claim is $\mu = 1200$ and its complement is $\mu \neq 1200$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. So

$$H_0 : \mu = \mu_0 = 1200 \text{ [average life of the bulbs is 1200 hours]}$$

$$H_1 : \mu \neq 1200 \text{ [average life of the bulbs is not 1200 hours]}$$

Also, the alternative hypothesis is two-tailed, so the test is two-tailed.

Here, we want to test the hypothesis regarding mean when population SD (variance) is known and sample size $n = 50 (> 30)$ is large. So we will go for Z-test.

Thus, for testing the null hypothesis the test statistic is given by

$$\begin{aligned} Z &= \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \\ &= \frac{1180 - 1200}{100 / \sqrt{50}} = \frac{-20}{14.14} = -1.41 \end{aligned}$$

The critical (tabulated) values for two-tailed test at 5% level of significance are $\pm Z_{\alpha/2} = \pm Z_{0.025} = \pm 1.96$.

Since the calculated value of the test statistic $Z (= -1.41)$ is greater than the critical value ($= -1.96$) and less than the critical value ($= 1.96$), that means it lies in the non-rejection region as shown in Fig. 10.5, so we do not reject the null hypothesis. Since the null hypothesis is the claim so we support the claim at 5% level of significance.

Decision according to p-value:

The test is two-tailed, therefore,

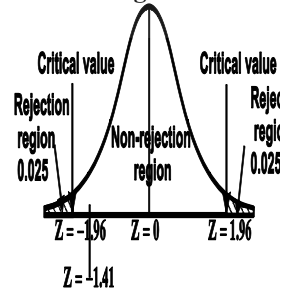
$$p\text{-value} = 2P[Z \geq |z|] = 2P[Z \geq 1.41]$$

$$= 2[0.5 - P[0 \leq Z \leq 1.41]] = 2(0.5 - 0.4207) = 0.1586$$

Since p-value ($= 0.1586$) is greater than $\alpha (= 0.05)$ so we do not reject the null hypothesis at 5% level of significance.

Decision according to confidence interval:

Fig. 10.5



Here, the test is two-tailed, therefore, we construct a two-sided confidence interval for the population mean.

Since population standard deviation is known, therefore, we can use $(1-\alpha)$ 100 % confidence interval for the population mean when the population variance is known which is given by

$$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

Here, $\alpha = 0.05$, we have $z_{\alpha/2} = z_{0.025} = 1.96$.

Thus, 95% confidence interval for the average life of a bulb is given by

$$\left[1180 - 1.96 \frac{100}{\sqrt{50}}, 1180 + 1.96 \frac{100}{\sqrt{50}} \right]$$

or $[1180 - 27.71, 1180 + 27.71]$

or $[1152.29, 1207.71]$

Since 95% confidence interval for the average life of a bulb contains the value of the parameter specified by the null hypothesis, that is, $\mu = \mu_0 = 1200$ so we do not reject the null hypothesis.

Thus, we conclude that the sample does not provide us sufficient evidence against the claim so we may assume that the company's claim that the average life of a bulb is 1200 hours is true.

Note 4: Here, we note that the decisions about the null hypothesis based on three approaches (critical value or classical p-value and confidence interval) are same. The learners are advised to make the decision about the claim or statement by using only one of the three approaches in the examination. Here, we used all these approaches only to give you an idea how they can be used in a given problem.

Example 4: A manufacturer of ballpoint pens claims that a certain pen manufactured by him has a mean writing life at least 460 A-4 size pages. A purchasing agent selects a sample of 100 pens and put them on the test. The mean writing-life of the sample found 453 A-4 size pages with standard deviation 25 A-4 size pages. Should the purchasing agent reject the manufacturer's claim at 1% level of significance?

Solution: Here, we are given that

Specified value of population mean $= \mu_0 = 460$,

Sample size $= n = 100$,

Sample mean $= \bar{X} = 453$,

Sample standard deviation $= S = 25$

Here, we want to test the manufacturer's claim that the mean writing-life (μ) of the pen is at least 460 A-4 size pages. So our claim is $\mu \geq 460$ and its complement is $\mu < 460$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. So

$$H_0 : \mu \geq \mu_0 = 460 \text{ and } H_1 : \mu < 460$$

Also, the alternative hypothesis is left-tailed so the test is left-tailed.

Here, we want to test the hypothesis regarding population mean when population SD is unknown. So we should use the t-test if the writing life of the pen follows a normal distribution. But it is not the case. Since the sample size is $n = 100$ ($n > 30$) large so we go for the Z-test. The test statistic of the Z-test is given by

$$Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

$$= \frac{453 - 460}{25/\sqrt{100}} = \frac{-7}{2.5} = -2.8$$

From Table I of the Appendix given at the end of this course, the value for the left-tailed Z test at 1% level of significance is $z_\alpha = -2.33$.

Since the calculated value of the test statistic Z ($= -2.8$) is less than the critical value ($= -2.33$), that means the calculated value of test statistic Z lies in the rejection region as shown in Fig. 10.6, so we reject the null hypothesis. Since the null hypothesis is the claim so we reject the manufacturer's claim at 1% level of significance.

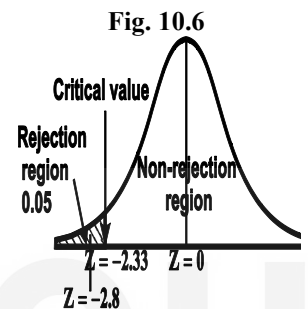
Decision according to p-value:

The test is left-tailed, therefore,

$$\begin{aligned} \text{p-value} &= P[Z \leq z] = P[Z \leq -2.8] = P[Z \geq 2.8] \quad \left[\because Z \text{ is symmetrical about } Z = 0 \text{ line} \right] \\ &= 0.5 - P[0 \leq Z \leq 2.8] = 0.5 - 0.4974 = 0.0026 \end{aligned}$$

Since the p-value ($= 0.0026$) is less than α ($= 0.01$) so we reject the null hypothesis at 1% level of significance.

Therefore, we conclude that the sample provides us sufficient evidence against the claim so the purchasing agent rejects the manufacturer's claim at 1% level of significance.



10.6 TESTING OF HYPOTHESIS FOR DIFFERENCE OF TWO POPULATION MEANS USING Z-TEST

In the previous section, we have learnt about the testing of hypothesis about the population mean. But there are so many situations where we want to test the hypothesis about difference of two population means or two population means. For example, two manufacturing companies of bulbs produce the same type of bulbs and one may be interested to test that one is better than the other, an investigator may want to test the equality of the average incomes of the people living in two cities, etc. Therefore, we require an appropriate test for testing the hypothesis about the difference of two population means.

Let there be two populations, say, population-I and population-II under study. Also, let μ_1, μ_2 and σ_1^2, σ_2^2 denote the means and variances of population-I and population-II respectively where both μ_1 and μ_2 are unknown but σ_1^2 and σ_2^2 may be known or unknown. We will consider all

possible cases here. For testing the hypothesis about the difference of two population means, we draw a random sample of large size n_1 from population-I and a random sample of large size n_2 from population-II. Let $\bar{X}$ and $\bar{Y}$ be the means of the samples selected from population-I and II, respectively.

These two populations may or may not be normal but according to the central limit theorem, the sampling distribution of difference of two large sample means asymptotically normally distributed with mean $(\mu_1 - \mu_2)$ and variance $(\sigma_1^2/n_1 + \sigma_2^2/n_2)$ as described earlier in this course.

Thus,

$$E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2 \quad \dots (3)$$

and

$$\text{Var}(\bar{X} - \bar{Y}) = \text{Var}(\bar{X}) + \text{Var}(\bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

But we know that standard error = $\sqrt{\text{Variance}}$

$$\therefore SE(\bar{X} - \bar{Y}) = \sqrt{\text{Var}(\bar{X} - \bar{Y})} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad \dots (4)$$

Now, follow the same procedure as we have discussed in previous section, that is, first of all, we have to set up null and alternative hypotheses. Here, we want to test the hypothesis about the difference of two population means so we can take the null hypothesis as

$$H_0 : \mu_1 = \mu_2 \text{ (no difference in means)}$$

$$\text{or } H_0 : \mu_1 - \mu_2 = 0 \text{ (difference in two means is 0)}$$

and the alternative hypothesis as

$$H_1 : \mu_1 \neq \mu_2 \quad [\text{for two-tailed test}]$$

$$\text{or } \left. \begin{array}{l} H_0 : \mu_1 \leq \mu_2 \text{ and } H_1 : \mu_1 > \mu_2 \\ H_0 : \mu_1 \geq \mu_2 \text{ and } H_1 : \mu_1 < \mu_2 \end{array} \right\} [\text{for one-tailed test}]$$

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{(\bar{X} - \bar{Y}) - E(\bar{X} - \bar{Y})}{SE(\bar{X} - \bar{Y})}$$

$$\text{or } Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad [\text{using equations (3) and (4)}]$$

Since under the null hypothesis, we assume that $\mu_1 = \mu_2$, therefore, we have

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Now, the sampling distribution of the test statistic depends upon σ_1^2 and σ_2^2 that both are known or unknown. Therefore, four cases arise:

Case I: When σ_1^2 & σ_2^2 are known and $\sigma_1^2 = \sigma_2^2 = \sigma^2$

In this case, the test statistic follows a normal distribution with mean 0 and variance unity when the sample sizes are large as both the populations under study are normal or non-normal. But when sample sizes are small then the test statistic Z follows normal distribution only when populations under study are normal, that is,

$$Z = \frac{\bar{X} - \bar{Y}}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0,1)$$

Case II: When σ_1^2 & σ_2^2 are known and $\sigma_1^2 \neq \sigma_2^2$

In this case, the test statistic also follows the normal distribution as described in case I, that is,

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

Case III: When σ_1^2 & σ_2^2 are unknown and $\sigma_1^2 = \sigma_2^2 = \sigma^2$

In this case, σ^2 is estimated by the value of pooled sample variance S_p^2 where,

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} [n_1 S_1^2 + n_2 S_2^2]$$

and

$$S_1^2 = \frac{1}{(n_1 - 1)} \sum_{i=1}^{n_1} (X_i - \bar{X})^2 \text{ and } S_2^2 = \frac{1}{(n_2 - 1)} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

and the test statistic follows t-distribution with $(n_1 + n_2 - 2)$ degrees of freedom as the sample sizes are large or small provided populations under study follow a normal distribution. But when the populations are under study are not normal and sample sizes n_1 and n_2 are large (> 30) then by central limit theorem, the test statistic approximately normally distributed with mean 0 and variance unity, that is,

$$Z = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0,1)$$

Case IV: When σ_1^2 & σ_2^2 are unknown and $\sigma_1^2 \neq \sigma_2^2$

In this case, σ_1^2 & σ_2^2 are estimated by the values of the sample variances S_1^2 & S_2^2 respectively and the exact distribution of the test statistic is difficult to derive. But when sample sizes n_1 and n_2 are large (> 30) then according to the central limit theorem, the test statistic is approximately normally distributed with mean 0 and variance unity, that is,

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim N(0, 1)$$

After that, we calculate the value of the test statistic and compare it with the critical value given in the Appendix at a prefixed level of significance α . Take the decision about the null hypothesis as described in the last section.

From the above discussion of testing of hypothesis about population mean, we note the following point:

- (i) When σ_1^2 & σ_2^2 are known then we apply the Z-test as both the population under study are normal or non-normal for the large sample. But when sample sizes are small then we apply the Z-test only when populations under study are normal.
- (ii) When σ_1^2 & σ_2^2 are unknown then we apply the t-test only when the populations under study are normal as sample sizes are large or small. But when the assumption of normality is not fulfilled and sample sizes are large then we can apply the Z-test.
- (iii) When samples are small and σ_1^2 & σ_2^2 are known or unknown and the form of the population is not known then we apply the non-parametric tests which are beyond the scope of this course.

Let us do some examples based on the above test.

Example 5: In two samples of women from Punjab and Tamil Nadu, the mean height of 1000 and 2000 women are 67.6 and 68.0 inches respectively. If the population standard deviation of Punjab and Tamil Nadu are same and equal to 5.5 inches then, can the mean heights of Punjab and Tamil Nadu women be regarded as same at 1% level of significance?

Solution: We are given

$$n_1 = 1000, n_2 = 2000, \bar{X} = 67.6, \bar{Y} = 68.0 \text{ and } \sigma_1 = \sigma_2 = \sigma = 5.5$$

Here, we wish to test that the mean height of Punjab and Tamil Nadu women is same. If μ_1 and μ_2 denote the mean heights of Punjab and Tamil Nadu women respectively then our claim is $\mu_1 = \mu_2$ and its complement is $\mu_1 \neq \mu_2$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 \neq \mu_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed.

Here, we want to test the hypothesis regarding two population means. The standard deviations of both populations are known and sample sizes are large, so we should go for Z-test.

So, for testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\begin{aligned}
 &= \frac{67.6 - 68.0}{\sqrt{\left(\frac{(5.5)^2}{1000} + \frac{(5.5)^2}{2000}\right)}} = \frac{-0.4}{5.5 \sqrt{\left(\frac{1}{1000} + \frac{1}{2000}\right)}} \\
 &= \frac{-0.4}{5.5 \times 0.0387} = -1.88
 \end{aligned}$$

The critical (tabulated) values for the two-tailed test at 1% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 2.58$.

Since the calculated value of Z ($= -1.88$) is greater than the critical value ($= -2.58$) and less than the critical value ($= 2.58$), that means it lies in the non-rejection region as shown in Fig. 10.7, so we do not reject the null hypothesis i.e. we fail to reject the claim.

Decision according to p-value:

The test is two-tailed, therefore,

$$\begin{aligned}
 \text{p-value} &= 2P[Z \geq |z|] = 2P[Z \geq 1.88] \\
 &= 2[0.5 - P[0 \leq Z \leq 1.88]] = 2(0.5 - 0.4699) = 0.0602
 \end{aligned}$$

Since the p-value ($= 0.0602$) is greater than $\alpha (= 0.01)$ so we do not reject the null hypothesis at 1% level of significance.

Thus, we conclude that the samples do not provide us sufficient evidence against the claim so we may assume that the average height of women of Punjab and Tamil Nadu is the same.

Example 6: A university conducts both face to face and distance mode classes for a particular course intended to be identical. A sample of 50 students of face to face mode yields examination results mean and SD respectively as:

$$\bar{X} = 80.4, \quad S_1 = 12.8$$

and another sample of 100 distance-mode students yields mean and SD of their examination results in the same course respectively as:

$$\bar{Y} = 74.3, \quad S_2 = 20.5$$

Are both educational methods statistically equal at 5% level?

Solution: Here, we are given that

$$n_1 = 50, \quad \bar{X} = 80.4, \quad S_1 = 12.8;$$

$$n_2 = 100, \quad \bar{Y} = 74.3, \quad S_2 = 20.5$$

We wish to test that both educational methods are statistically equal. If μ_1 and μ_2 denote the average marks of face to face and distance mode students respectively then our claim is $\mu_1 = \mu_2$ and its complement is $\mu_1 \neq \mu_2$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 \neq \mu_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed.

We want to test the null hypothesis regarding two population means when standard deviations of both populations are unknown. So we should go for

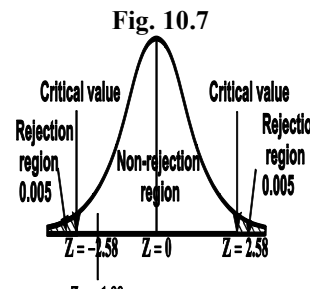
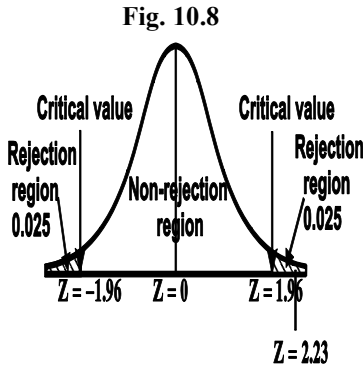


Fig. 10.7

the t-test if the population of difference is known to be normal. But it is not the case. Since sample sizes are large (n_1 , and $n_2 > 30$) so we go for Z-test.

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{80.4 - 74.3}{\sqrt{\frac{(12.8)^2}{50} + \frac{(20.5)^2}{100}}} = \frac{6.1}{\sqrt{3.28 + 4.20}} = 2.23$$



The critical (tabulated) values for the two-tailed test at 5% level of significance are $\pm Z_{\alpha/2} = \pm Z_{0.025} = \pm 1.96$.

Since the calculated value of Z (= 2.23) is greater than the critical values (= ± 1.96), that means it lies in the rejection region as shown in Fig. 10.8, so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Decision according to p-value:

The test is two-tailed, therefore,

$$\begin{aligned} \text{p-value} &= 2P[Z \geq |z|] = 2P[Z \geq 2.23] \\ &= 2[0.5 - P[0 \leq Z \leq 2.23]] = 2(0.5 - 0.4871) = 0.0258 \end{aligned}$$

Since the p-value (= 0.0258) is less than α (= 0.05) so we reject the null hypothesis at 5% level of significance.

Thus, we conclude that the samples provide us sufficient evidence against the claim so both methods of education, i.e. face-to-face and distance-mode, provided by that University are not statistically equal.

10.7 TESTING OF HYPOTHESIS FOR POPULATION PROPORTION USING Z-TEST

In the last section, we have discussed the procedure of testing of hypothesis for the population mean when the sample size is large and the data is quantitative. But in many real-world situations, in business and other areas where collected data are in the form of counts or the collected data are classified into two categories or groups according to an attribute or a characteristic. For example, the people living in a colony may be classified into two groups (male and female) with respect to the characteristic gender, the patients in a hospital may be classified into two groups as cancer and non-cancer patients, articles may be classified as defective and non-defective, etc. Here, collected data are available in dichotomous or binary outcomes which is a special case of nominal scale and the data is categorised into two mutually exclusive and exhaustive classes generally known as success and failure outcomes. For example, the characteristic gender can be measured as a success if female and a failure if male or vice versa. So in such situations, proportion is a suitable measure to apply.

In such situations, we require a test for testing a hypothesis about the population proportion.

For this purpose, let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from a population with population proportion P . Also let X denotes the number of observations or elements that possess a certain attribute (number of successes) out of n observations of the sample then sample proportion p can be defined as

$$p = \frac{X}{n} \leq 1$$

As we have seen in an earlier unit of this course that the mean and variance of the sampling distribution of sample proportion are:

$$E(p) = P \text{ and } \text{Var}(p) = \frac{PQ}{n}$$

where $Q = 1 - P$.

Now, two cases arise:

Case I: When the sample size is not sufficiently large i.e. either of the conditions $np > 5$ or $nq > 5$ does not meet, then we use the exact binomial test. But the exact binomial test is beyond the scope of this course.

Case II: When sample size is sufficiently large, such that $np > 5$ and $nq > 5$ then by central limit theorem, the sampling distribution of sample proportion p is approximately normally distributed with mean and variance as

$$E(p) = P \text{ and } \text{Var}(p) = \frac{PQ}{n} \quad \dots (5)$$

But we know that standard error = $\sqrt{\text{Variance}}$

$$\therefore \text{SE}(p) = \sqrt{\frac{PQ}{n}} \quad \dots (6)$$

Now, follow the same procedure as we have discussed in the last Section. First of all, we set up null and alternative hypotheses. Since here we want to test the hypothesis about a specified value P_0 of the population proportion so we can take the null and alternative hypotheses as

$$H_0 : P = P_0 \text{ and } H_1 : P \neq P_0 \text{ [for two-tailed test]} \left[\begin{array}{l} \text{Here, } \theta = P \text{ and } \theta_0 = P_0 \\ \text{if we compare it with} \\ \text{general procedure.} \end{array} \right]$$

$$\text{or } \left. \begin{array}{l} H_0 : P \leq P_0 \text{ and } H_1 : P > P_0 \\ H_0 : P \geq P_0 \text{ and } H_1 : P < P_0 \end{array} \right\} \text{ [for one-tailed test]}$$

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{p - E(p)}{\text{SE}(p)}$$

$$Z = \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \sim N(0, 1) \text{ under } H_0 \text{ [using equations (5) and (6)]}$$

After that, we calculate the value of the test statistic and compare it with the critical value(s) at a prefixed level of significance α . We take the decision about the null hypothesis as described in the last Section.

Let us do some examples of testing of hypothesis about population proportion.

Example 7: A machine produces a large number of items out of which 25% are found to be defective. To check this, the company manager takes a random sample of 100 items and found 35 items defective. Is there evidence of more deterioration of quality at 5% level of significance?

Solution: The company manager wants to check that his machine produces 25% defective items. Here, attribute under study is defectiveness. And we define our success and failure as getting a defective or non-defective item.

Let P = Population proportion of defectives items = 0.25 ($= P_0$)

p = Observed proportion of defectives items in the sample = $35/100 = 0.35$

Here, we want to test that the machine produces more defective items, that is, the proportion of defective items (P) greater than 0.25. So our claim is $P > 0.25$ and its complement is $P \leq 0.25$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. So

$$H_0 : P \leq P_0 = 0.25 \text{ and } H_1 : P > 0.25$$

Since the alternative hypothesis is right-tailed so the test is right-tailed.

Before proceeding further, first, we have to check whether the condition of normality meets or not.

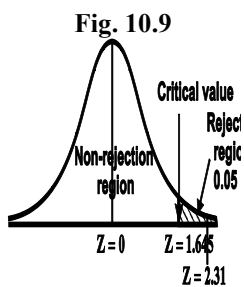
$$\therefore np = 100 \times 0.35 = 35 > 5$$

$$nq = 100 \times (1 - 0.35) = 100 \times 0.65 = 65 > 5$$

We see that condition of normality meets, so we can go for Z-test.

So, for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \\ &= \frac{0.35 - 0.25}{\sqrt{\frac{0.25 \times 0.75}{100}}} = \frac{0.10}{0.0433} = 2.31 \end{aligned}$$



Since test is right-tailed so the critical value at 5% level of significance is $Z_{\alpha} = Z_{0.05} = 1.645$.

Since the calculated value of the test statistic Z ($= 2.31$) is greater than the critical value ($= 1.645$), that means it lies in the rejection region as shown in Fig. 10.9. So we reject the null hypothesis and support the alternative hypothesis i.e. we support the claim at 5% level of significance.

Decision according to p-value:

The test is right-tailed, therefore,

$$\begin{aligned}
 \text{p-value} &= P[Z \geq z] = P[Z \geq 2.31] \\
 &= 0.5 - P[0 < Z < 2.31] = 0.5 - 0.4896 \\
 &= 0.0104
 \end{aligned}$$

Since the p-value (= 0.0104) is less than α (= 0.05) so we reject the null hypothesis at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that deterioration in quality exists at 5% level of significance.

Example 8: A die is thrown 9000 times and a draw of 2 or 5 is observed 3100 times. Can we regard that die is unbiased at 5% level of significance?

Solution: Let getting a 2 or 5 be our success, and getting a number other than 2 or 5 to be a failure. In usual notions, we have

$$n = 9000, X = \text{number of successes} = 3100, p = 3100/9000 = 0.3444$$

Here, we want to test that the die is unbiased and we know that if die is unbiased then the proportion or probability of getting 2 or 5 is

$$\begin{aligned}
 P &= \text{Probability of getting a 2 or 5} \\
 &= \text{Probability of getting 2} + \text{Probability of getting 5} \\
 &= \frac{1}{6} + \frac{1}{6} = \frac{1}{3} = 0.3333
 \end{aligned}$$

So our claim is $P = 0.3333$ and its complement is $P \neq 0.3333$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : P = P_0 = 0.3333 \quad \text{and} \quad H_1 : P \neq 0.3333$$

Since the alternative hypothesis is two-tailed so the test is two-tailed.

Before proceeding further, first, we have to check whether the condition of normality meets or not.

$$\therefore np = 9000 \times 0.3444 = 3099.6 > 5$$

$$nq = 9000 \times (1 - 0.3444) = 9000 \times 0.6556 = 5900.4 > 5$$

We see that condition of normality meets, so we can go for Z-test.

So, for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned}
 Z &= \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \\
 &= \frac{0.3444 - 0.3333}{\sqrt{\frac{0.3333 \times 0.6667}{9000}}} = \frac{0.0111}{0.005} = 2.22
 \end{aligned}$$

Since the test is a two-tailed test so the critical values at 5% level of significance are $\pm Z_{\alpha/2} = \pm Z_{0.025} = \pm 1.96$.

Since the calculated value of Z (= 2.22) is greater than the critical value (= 1.96), that means it lies in the rejection region, so we reject the null hypothesis i.e. we reject our claim.

Decision according to p-value:

Since the test is two-tailed so

$$\begin{aligned} \text{p-value} &= 2P[Z \geq |z|] = 2P[Z \geq 2.22] \\ &= 2[0.5 - P[0 < Z < 2.22]] = 2(0.5 - 0.4868) = 0.0264 \end{aligned}$$

Since the p-value (= 0.0264) is less than α (= 0.05), so we reject the null hypothesis at 5% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so die cannot be considered as unbiased.

Check Your Progress 2

- 1) A sample of 900 bolts has a mean length 3.4 cm. Is the sample regarded to be taken from a large population of bolts with mean length 3.25 cm and standard deviation 2.61 cm at 5% level of significance?

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.....

- 2) Two brands of electric bulbs are quoted at the same price. A buyer has tested a random sample of 200 bulbs of each brand and found the following information:

	Mean Life (hrs.)	SD(hrs.)
Brand A	1300	41
Brand B	1280	46

Is there any significant difference in the mean duration of their lives of two brands of electric bulbs at 1% level of significance?

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.....

- 3) Out of 200 patients who are given a particular injection 180 survived. Test the hypothesis that the survival rate is more than 80% at 5% level of significance?

.....
.....

10.8 TESTING OF HYPOTHESIS FOR POPULATION MEAN USING t-TEST

In the last Section, we have discussed Z-test for testing the hypothesis about the population mean when population variance σ^2 is known and unknown.

Recall from these, we have already pointed out that one basic difference between the Z-test and the t-test is that the Z-test is used when population SD is known whether the sample size is large or small and t-test is used when population SD is unknown whether the sample size is small or large. But in the case of large sample size, the Z-test is appropriate for the t-test. But in practice, the standard deviation of the population is not known and the sample size is small so in this situation, we use the t-test provided population under study is normal.

Assumptions

Virtually every test has some assumptions which must be met before the application of the test. This t-test needs the following assumptions to work:

- (i) The characteristic under study follows a normal distribution. In other words, populations from which a random sample is drawn should be normal with respect to the characteristic of interest.
- (ii) Sample observations are random and independent.
- (iii) Population variance σ^2 is unknown.

For describing this test, let $X_1, X_2, \dots, X_n$ be a random sample of **small size n (< 30)** selected from a **normal population** with mean μ and unknown variance σ^2 .

Now, follow the same procedure as we have discussed in the previous section, that is, first of all, we set up the null and alternative hypotheses. Here, we want to test the claim about the specified value μ_0 of the population mean μ so we can take the null and alternative hypotheses as

$$\begin{aligned}
 &H_0 : \mu = \mu_0 \text{ and } H_1 : \mu \neq \mu_0 \quad \left[\text{for two-tailed test} \right] \quad \left[\begin{array}{l} \text{Here, } \theta = \mu \text{ and } \theta_0 = \mu_0 \\ \text{if we compare it with} \\ \text{general procedure.} \end{array} \right] \\
 \text{or} \quad &\left. \begin{aligned} H_0 : \mu \leq \mu_0 \text{ and } H_1 : \mu > \mu_0 \\ H_0 : \mu \geq \mu_0 \text{ and } H_1 : \mu < \mu_0 \end{aligned} \right\} \quad \left[\text{for one-tailed test} \right]
 \end{aligned}$$

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{(n-1)} \quad \text{under } H_0$$

where, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is the sample variance.

For computational simplicity, we may use the following formulae for $\bar{X}$, S^2 :

$$\bar{X} = a + \frac{1}{n} \sum d \text{ and } S^2 = \frac{1}{n-1} \left[\sum d^2 - \frac{(\sum d)^2}{n} \right]$$

where, $d = (X - a)$, 'a' being the assumed arbitrary value.

Here, the test statistic t follows t-distribution with $(n - 1)$ degrees of freedom.

After substituting values of $\bar{X}$, S and n , we get the calculated value of the test statistic t . Then we look for critical (or cut-off or tabulated) value(s) of the test statistic t from the t-table, given in the Appendix at the end of this course. On comparing calculated value and critical value(s), we take the decision about the null hypothesis as discussed in the previous section.

Let us do some examples of testing of hypothesis about the population mean using the t-test.

Example 9: A manufacturer claims that a special type of projector bulb has an average life 160 hours. To check this claim an investigator takes a sample of 20 such bulbs, puts on the test, and obtains an average life 167 hours with standard deviation 16 hours. Assuming that the lifetime of such bulbs follows a normal

distribution, does the investigator accept the manufacturer's claim at 5% level of significance?

Solution: Here, we are given that

$$\mu_0 = 160, \quad n = 20, \quad \bar{X} = 167 \quad \text{and} \quad S = 16$$

Here, we want to test the manufacturer's claims that a special type of projector bulb has an average life (μ) 160 hours. So claim is $\mu = 160$ and its complement is $\mu \neq 160$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu = \mu_0 = 160 \quad \text{and} \quad H_1 : \mu \neq 160$$

Since the alternative hypothesis is two-tailed so the test is two-tailed.

Here, we want to test the hypothesis regarding population mean when population SD is unknown. Also, the sample size is small $n = 20$ ($n < 30$) and the population under study is normal, so we can go for the t-test for testing the hypothesis about the population mean.

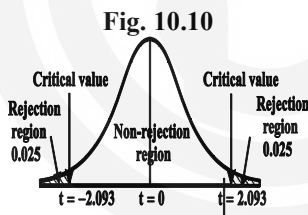
For testing the null hypothesis, the test statistic t is given by

$$\begin{aligned} t &= \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \\ &= \frac{167 - 160}{16/\sqrt{20}} = \frac{7}{3.58} = 1.96 \end{aligned}$$

The critical value of the test statistic t for various df and different level of significance α are given in Table V of the Appendix at the end of this course.

The critical (tabulated) values of the test statistic for two-tailed test corresponding $(n-1) = 19$ df at 5% level of significance are

$$\pm t_{(n-1), \alpha/2} = \pm t_{(19), 0.025} = \pm 2.093.$$



Since the calculated value of the test statistic t ($= 1.96$) is greater than the critical value ($= -2.093$) and is less than the critical value ($= 2.093$), that means the calculated value of the test statistic lies in the non-rejection region as shown in Fig. 10.10. So we do not reject the null hypothesis i.e. we support the manufacturer's claim at 5% level of significance.

Decision according to p-value:

Since the calculated value of the test statistic t is based on 19 df , therefore, we use row for 19 df in the t -table (Table V in Appendix) and move across this row to find the values in which the calculated t -value falls. Since the calculated t -value falls between 1.729 and 2.093 corresponding to one-tailed area $\alpha = 0.05$ and 0.025 respectively therefore p -value lies between 0.025 and 0.05, that is,

$$0.025 < p\text{-value} < 0.05$$

Since the test is two-tailed so

$$2 \times 0.025 = 0.05 < p\text{-value} < 0.10 = 0.05 \times 2$$

Since the p -value is greater than α ($= 0.05$) so we do not reject the null hypothesis at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the null hypothesis so we may assume that the manufacturer's claim is true so the investigator may accept the manufacturer's claim at 5% level of significance.

Example 10: The mean share price of companies in the Pharma sector is Rs.70. The share prices of all companies were changed from time to time. After a month, a sample of 10 Pharma companies was taken and their share prices were noted as below:

70, 76, 75, 69, 70, 72, 68, 65, 75, 72

Assuming that the distribution of the share prices follows a normal distribution, test whether the mean share price is still the same at 1% level of significance?

Solution: Here, we wish to test that the mean share price (μ) of companies in the Pharma sector is still Rs.70 besides all changes. So our claim is $\mu = 70$ and its complement is $\mu \neq 70$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu = \mu_0 = 70 \quad [\text{mean share price of companies is still Rs. 70}]$$

$$H_1 : \mu \neq \mu_0 = 70 \quad [\text{mean share price of companies is not still Rs. 70}]$$

Since the alternative hypothesis is two-tailed so the test is two-tailed.

Here, we want to test the hypothesis regarding population mean when population SD is unknown. Also, the sample size is small $n = 10$ ($n < 30$) and the population under study is normal, so we can go for the t-test for testing the hypothesis about the population mean.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{(n-1)} \quad \dots (1)$$

Calculation for $\bar{X}$ and S:

S. No.	Sample value (X)	Deviation $d = (X-a), a = 70$	d^2
1	70	0	0
2	76	6	36
3	75	5	25
4	69	-1	1
5	70	0	0
6	72	2	4
7	68	-2	4
8	65	-5	25
9	75	5	25
10	72	2	4
Total		12	124

The assumed value of a is 70.

From the above calculation, we have

$$\bar{X} = a + \frac{1}{n} \sum d = 70 + \frac{1}{10} \times 12 = 71.2$$

$$S^2 = \frac{1}{n-1} \left[\sum d^2 - \frac{(\sum d)^2}{n} \right]$$

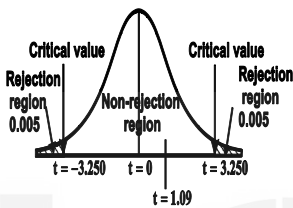
$$= \frac{1}{10-1} \left[124 - \frac{(12)^2}{10} \right] = \frac{1}{9} \left[124 - \frac{144}{10} \right] = 12.18$$

$$\Rightarrow S = \sqrt{12.18} = 3.49$$

Putting the values in equation (1), we have

$$t = \frac{71.2 - 70}{3.49 / \sqrt{10}} = \frac{1.2}{1.10} = 1.09$$

Fig. 10.11



The critical (tabulated) values of the test statistic for the two-tailed test corresponding $(n-1) = 9$ df at 1% level of significance are

$$\pm t_{(n-1), \alpha/2} = \pm t_{(9), 0.005} = \pm 3.250.$$

Since the calculated value of the test statistic $t (= 1.09)$ is less than the critical value $(= 3.250)$ and greater than the critical value $(= -3.250)$, that means the calculated value of t lies in the non-rejection region as shown in Fig. 10.11. So we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Decision according to p-value:

Since the calculated value of the test statistic t is based on 9 df, therefore, we use row for 9 df in the t -table (Table V in Appendix) and move across this row to find the values in which the calculated t -value falls. Since all values in this row are greater than the calculated t -value 1.09 and the smallest value is 1.383 corresponding to one-tailed area $\alpha = 0.10$, therefore, the p -value is greater than 0.10, that is,

$$p\text{-value} > 0.10$$

Since the test is two-tailed so

$$p\text{-value} > 2 \times 0.10 = 0.20$$

Since the p -value $(= 0.20)$ is greater than $\alpha (= 0.01)$ so we do not reject the null hypothesis at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so may assume that the mean share price is still Rs. 70.

Check Your Progress 3

- 1) A tyre manufacturer claims that the average life of a particular category of his tyre is 18000 km when used under normal driving conditions. A random sample of 16 tyres was tested. The mean and SD of life of the tyres in the sample were 20000 km and 6000 km respectively. Assuming that the life of the tyres is normally distributed, test the claim of the manufacturer at 1% level of significance using the appropriate test.

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- 2) It is known that the average weight of cadets of a centre follows a normal distribution. Weights of 10 randomly selected cadets from the same centre are as given below:

48, 50, 62, 75, 80, 60, 70, 56, 52, 77

Can we say that the average weight of all cadets of the centre from which the above sample was taken is equal to 60 kg at 5% level of significance?

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With this, we end this unit. Let us summarise what we have discussed in this unit.

10.9 SUMMARY

In this unit, we have covered the following points:

1. Statistical hypothesis, null hypothesis, alternative hypothesis.
2. Critical region and Type-I and Type-II errors.
3. One-tailed and two-tailed tests.
4. General procedure of testing a hypothesis.
5. Level of significance and Concept of p-value.
6. Applying the Z-test for testing the hypothesis about the population mean and the difference of two population means.
7. Applying the Z-test for testing the hypothesis about the population proportion.
8. Testing of hypothesis for population mean using t-test.

10.10 SOLUTIONS / ANSWERS

Check Your Progress 1

- 1) Here, we wish to test the claim of the company that the average life of its car tyres is 50000 miles so

Claim: $\mu = 50000$ miles and complement: $\mu \neq 50000$ miles

Since claim contains equality sign so we take claim as the null hypothesis and complement as the alternative hypothesis i.e.

$H_0: \mu = 50000$ miles

$H_1: \mu \neq 50000$ miles

- 2) (iii) Here, the doctor wants to test whether new medicine is more effective for controlling blood pressure than old medicine so

Claim: $\mu_1 > \mu_2$ and complement: $\mu_1 \leq \mu_2$

Since complement contains equality sign so we take complement as the null hypothesis and claim as the alternative hypothesis i.e.

$H_0: \mu_1 \leq \mu_2$ and $H_1: \mu_1 > \mu_2$

(iv) Here, the economist wants to test whether the variability in incomes differ in two populations so

Claim: $\sigma_1^2 \neq \sigma_2^2$ and complement: $\sigma_1^2 = \sigma_2^2$

Since complement contains equality sign so we take complement as the null hypothesis and claim as the alternative hypothesis i.e.

$$H_0: \sigma_1^2 = \sigma_2^2 \quad \text{and} \quad H_1: \sigma_1^2 \neq \sigma_2^2$$

(v) Here, psychologist wants to test whether the proportion of literates between two groups of people is same so

Claim: $P_1 = P_2$ and complement: $P_1 \neq P_2$

Since claim contains equality sign so we take claim as the null hypothesis and complement as the alternative hypothesis i.e.

$$H_0: P_1 = P_2 \quad \text{and} \quad H_1: P_1 \neq P_2$$

- 3) Since alternative hypothesis $H_1: \theta \neq 60$ is two-tailed so critical region lies in two-tails.
- 4) Since level of significance is the probability of type-I error so in this case level of significance is 0.05 or 5%.
- 5) Here, the alternative hypothesis is two-tailed therefore, the test will be two-tailed.
- 6) Whether the test of testing a hypothesis is one-tailed or two-tailed depends on the alternative hypothesis. So the correct option is (ii).

Check Your Progress 2

- 1) We are given that

$$n = 900, \bar{X} = 3.4 \text{ cm}, \mu_0 = 3.25 \text{ cm and } \sigma = 2.61 \text{ cm}$$

Here, we wish to test that the sample comes from a large population of bolts with mean (μ) 3.25cm. So our claim is $\mu = 3.25\text{cm}$ and its complement is $\mu \neq 3.25\text{cm}$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0: \mu = \mu_0 = 3.25 \text{ and } H_1: \mu \neq 3.25$$

Since the alternative hypothesis is two-tailed, so the test is two-tailed.

Here, we want to test the hypothesis regarding population mean when population SD is unknown, so we should use the t-test if the population of bolts known to be normal. But it is not the case. Since the sample size is large ($n > 30$) so we can go for the Z-test instead of the t-test as an approximate. So the test statistic is given by

$$\begin{aligned} Z &= \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \\ &= \frac{3.40 - 3.25}{2.61 / \sqrt{900}} = \frac{0.15}{0.087} = 1.72 \end{aligned}$$

The critical (tabulated) values for the two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since the calculated value of the test statistic $Z (= 1.72)$ is less than the critical value $(= 1.96)$ and greater than the critical value $(= -1.96)$, that means it lies in the non-rejection region, so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample does not provide us sufficient evidence against the claim so we may assume that the sample comes from the population of bolts with mean 3.25cm.

2) Given that

$$n_1 = 200, \quad \bar{X} = 1300, \quad S_1 = 41;$$

$$n_2 = 200, \quad \bar{Y} = 1280, \quad S_2 = 46$$

Here, we want to test that there is a significant difference in the mean duration of their lives of two brands of electric bulbs. If μ_1 and μ_2 denote the mean lives of two brands of electric bulbs respectively then our claim is $\mu_1 \neq \mu_2$ and its complement is $\mu_1 = \mu_2$. Since the complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 \neq \mu_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed.

We want to test the null hypothesis regarding the equality of two population means. The standard deviations of both populations are unknown so we should go for the t-test if the population of difference is known to be normal. But it is not the case. Since sample sizes are large (n_1 , and $n_2 > 30$) so we go for the Z-test.

So for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \\ &= \frac{1300 - 1280}{\sqrt{\frac{(41)^2}{200} + \frac{(46)^2}{200}}} = \frac{20}{\sqrt{8.41 + 10.58}} = \frac{20}{4.36} = 4.59 \end{aligned}$$

The critical (tabulated) values for the two-tailed test at 1% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 2.58$.

Since the calculated value of the test statistic $Z (= 4.59)$ is greater than the critical value $(= \pm 2.58)$, that means it lies in the rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. support the claim at 1% level of significance.

Thus, we conclude that the samples do not provide us sufficient evidence against the claim so there is a significant difference in the mean duration of their lives of two brands of electric bulbs.

3) We define our success and failure as a patient is survived and not survived. Here, we are given that

$$n = 200$$

X = Number of survived patients who are given a particular injection

$$= 180$$

p = Sample proportion of survived patients who are given a particular injection

$$= \frac{X}{n} = \frac{180}{200} = 0.9$$

$$P_0 = 80\% = \frac{80}{100} = 0.80 \Rightarrow Q_0 = 1 - P_0 = \frac{80}{100} = 0.20$$

Here, we want to test that the survival rate of the patients is more than 80%. If P denotes the proportion of survival patients then our claim is $P > 0.80$ and its complement is $P \leq 0.80$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : P \leq P_0 = 0.80 \text{ and } H_1 : P > 0.80$$

Since the alternative hypothesis is right-tailed so the test is right-tailed.

Before proceeding further, first, we have to check whether the conditions of normality meet or not.

$$\therefore np = 200 \times 0.9 = 180 > 5$$

$$nq = 200 \times (1 - 0.9) = 200 \times 0.1 = 20 > 5$$

We see that the conditions of normality meet, so we can go for the Z -test.

For testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \\ &= \frac{0.9 - 0.8}{\sqrt{\frac{(0.8)(0.2)}{200}}} = \frac{0.1}{0.0283} = 3.53 \end{aligned}$$

The critical (tabulated) value for the right-tailed test at 5% level of significance is $z_\alpha = z_{0.05} = 1.645$.

Since the calculated value of the test statistic Z ($= 3.53$) is greater than the critical value ($= 1.645$), that means it lies in the rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. support our claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the survival rate is greater than 80% in the population using that injection.

Check Your Progress 3

1) Here, we are given that

$$n = 16, \mu_0 = 18000, \bar{X} = 20000, S = 6000$$

Here, we want to test that the manufacturer's claim is true that the average life (μ) of tyres is 18000 km. So claim is $\mu = 18000$ and its complement is $\mu \neq 18000$. Since the claim contains the equality sign so

we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0: \mu = \mu_0 = 18000 \quad [\text{average life of tyres is } 18000 \text{ km}]$$

$$H_1: \mu \neq 18000 \quad [\text{average life of tyres is not } 18000 \text{ km}]$$

Here, population SD is unknown and the population under study is given to be normal. So we can go for the t-test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} = \frac{20000 - 18000}{6000/\sqrt{16}} = \frac{2000}{1500} = 1.33$$

The critical value of the test statistic t for the two-tailed test corresponding $(n-1) = 15$ df at 1% level of significance are

$$\pm t_{(15), 0.005} = \pm 2.947.$$

Since the calculated value of the test statistic t ($= 1.33$) is less than the critical (tabulated) value ($= 2.947$) and greater than critical value ($= -2.947$), that means the calculated value of the test statistic lies in the non-rejection region, so we do not reject the null hypothesis i.e. we support the manufacturer's claim at 1% level of significance.

Thus, we conclude that the sample fails to provide sufficient evidence against the claim so we may assume that manufacturer's claim is true.

- 2) Here, we want to test that the average weight (μ) of all cadets of the centre is 60 kg. So our claim is $\mu = 60$ and its complement is $\mu \neq 60$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu = \mu_0 = 60 [\text{average weight of all cadets is } 60 \text{ kg}]$$

$$H_1 : \mu \neq 60 \quad [\text{average weight of all cadets is not } 60 \text{ kg}]$$

Since the alternative hypothesis is two-tailed so the test is two-tailed.

Here, population SD is unknown and the population under study is given to be normal. So we can go for the t-test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{(n-1)} \quad \text{under } H_0 \quad \dots (5)$$

Before moving further, first, we have to calculate the value of $\bar{X}$ and S.

Calculation for $\bar{X}$ and S:

Sample value (X)	$(x - \bar{X})$	$(x - \bar{X})^2$
48	-15	225
50	-13	169
62	-1	1
75	12	144
80	17	289
60	-3	9
70	7	49
56	-7	49
52	-11	121

77	14	196
$\sum X = 630$		$\sum (X - \bar{X})^2 = 1252$

From the above calculation, we have

$$\bar{X} = \frac{1}{n} \sum X = \frac{1}{10} \times 630 = 63$$

$$S^2 = \frac{1}{n-1} \sum (X - \bar{X})^2 = \frac{1}{10-1} \times 1252 = 139.11$$

$$\Rightarrow S = \sqrt{139.11} = 11.79$$

Putting the values in equation (5), we have

$$t = \frac{63 - 60}{11.79 / \sqrt{10}} = \frac{3}{3.73} = 0.80$$

The critical values of the test statistic t for the two-tailed test corresponding $(n-1) = 9$ df at 5% level of significance are $\pm t_{(9), 0.025} = \pm 2.262$.

Since the calculated value of the test statistic t ($= 0.80$) is less than the critical value ($= 2.262$) and greater than the critical value ($= -2.262$), that means the calculated value of the test statistic t lies in the non-rejection region so we do not reject H_0 i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide sufficient evidence against the claim so we may assume that the average weight of all the cadets of the given centre is 60 kg.

10.11 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 3: Testing of Hypothesis UNIT 9: Concepts of Testing of Hypothesis (for Sections 10.2 to 10.4 and related portions in Sections 10.0 , 10.1, 10.9, 10.10 and Check Your Progress questions)
2. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 3: Testing of Hypothesis UNIT 10: Large Sample Tests (for Sections 10.5 to 10.7 and related portions in Sections 10.0 , 10.1, 10.9, 10.10 and Check Your Progress questions)
3. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 3: Testing of Hypothesis UNIT 11: Small Sample Tests (for Sections 10.8 and related portions in Sections 10.0 , 10.1, 10.9, 10.10 and Check Your Progress questions)

Reference Book

4. "Introduction to Probability and Statistics for Engineers and Scientists", 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004